Modern Academy In Maadi
For Engineering & Technology



DIGITAL SIGNAL PROCESSING

(ELC 422)

Fifth Year Communications

(Parts Selected)

By

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الرؤيـــــة

أن تصبح الأكاديمية الحديثة للهندسة والتكنولوجيا ضمن المعاهد الهندسية العليا الأكثر تميزاً على المستوى المحلى والإقليمى فى مجال العلوم الهندسية وتطبيقاتها.

الرســـالة

إعداد الكوادر الهندسية المتميزة والقادرة على المنافسة علمياً ومهنياً على المستوى المحلى والإقليمى لكى تفى بإحتياجات المجتمع فى قطاعات الدولة المختلفة عن طريق إكساب الدارسين المعارف والمفاهيم والمهارات التكنولوجية الحديثة والمتطورة بإستخدام البرامج المتنوعة فى إطار من القيم الثقافية والإجتماعية والأخلاقية.

ELC-422 - Digital signal Processing (DSP)- ILOS

1 1– Course Learning Objectives:

By the end of this course the students should demonstrate the knowledge and understanding of the basic concepts & theory of Signals, Systems, signal Processing and discrete transformations and digital filters. They should be able to design, calculate and analyze the performance of digital systems.

2 - Intended Learning Outcomes (ILOS)

a - Knowledge and understanding:

On successful completion of the course, the student should demonstrate knowledge and understanding of:

a1- the principles and concepts of digital signal processing (A8, A24).
a2- the concept of Analog-to-Digital and Digital-to-Analog Conversion (A2).
a3- the concept of Discrete Transformations (FS, FT, DFT, FFT and z-Transform) (A5).
a4- the concept of Digital Filters design (FIR & IIR), (A10).

b - Intellectual skills:

On successful completion of the course, the student should be able to.

b1- select appropriate transformation (FS, FT, DFT or FFT), (B1).
b2- solve engineering problems using the concepts of Z-Transform in DS Processing (B7, B14).
b3- logically analyze the digital systems (B11 & B15).
b4- join the different topics in this subject to design a good new digital system (B3, B14).

c - Professional and practical skills:

On successful completion of the course, the student should be able to:

c1- discriminate between different signal types (C2).
c2- use the gained Lab information in this subject to design numerous of digital signal processing systems, e.g., A/D, D/A converters and DSP processor (C2, C5, C15).
c3- design different digital filters (FIR&IIR), (C2, C5, C14).
c4- develop some DSP systems to attain high qualified system (C6).
c5- prepare and present works both in written & oral form (C12).

d - General and transferable skills:

On successful completion of the course, the student should be able to:

d1- search for information from references and internet (D7).
d2- communicate effectively and present data and results orally and in written form(D3).
d3- use ICT facilities in presentations (D4).

Course Contribution in the Program ILO's

ILO's		Program ILO's
A	Knowledge and understanding	A2, A5, A8, A10 & A24
B	Intellectual skills	B1, B3, B7, B11, B14 & B15
C	Professional and practical skills	C2, C5, C6, C12, C14 & C15
D	General and transferable skills	D3, D4 & D7

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INTRODUCTION

- **DSP:** stands for, Digital Signal Processing.
- **DSP** is distinguished from other areas in computer science by the unique type of data it uses: ***signals*.**
- **DSP** is one of the most powerful technologies that will shape science and engineering in the twenty-first century.
- In most cases, these signals originate as sensory data from the real world:
    * seismic vibrations,
    * visual images,
    * sound waves, etc.
- **DSP** is the:
    * mathematics,
    * algorithms, and
    * techniques used to manipulate these signals after they have been converted into a digital form.
- This includes a wide variety of goals, such as:
    * enhancement of visual images,
    * recognition and generation of speech,
    * compression of data for storage and transmission,
    * etc.

1- THE BREADTH AND DEPTH OF DSP

- Revolutionary changes have already been made in **a *broad* range** of fields (breadth or applications):
    * communications,
    * medical imaging,
    * radar & sonar,

* high fidelity music reproduction, and
* oil prospecting.

- Each of these areas has developed **a *deep* DSP** technology, with its own:

* algorithms,
* mathematics, and
* specialized techniques.

- This combination of breath and depth makes it impossible for any one individual to master all of the DSP technology that has been developed.

- DSP education involves two tasks:

a- Learning general concepts that apply to the field as a whole, and

b- Learning specialized techniques for your particular area of interest.

2- THE ROOTS OF DSP

- The roots of DSP are in the 1960s and 1970s when digital computers first became available.
- Computers were expensive during this era, and DSP was limited to only a few critical applications, such as: *radar & sonar*, *oil exploration*, *space exploration*, and *medical imaging*.

*** *Radar & sonar*,** where national security was at risk.

*** *Oil exploration*,** where large amounts of money could be made.

*** S*pace exploration*,** where the data are irreplaceable.

*** M*edical imaging*,** where lives could be saved.

- The PC revolution of the 1980s and 1990s caused DSP to explode with new applications.
- **DSP** reached the public in such products as:

***** mobile telephones,

***** compact disc players, and

***** electronic voice mail.

3- DSP APPLICATIONS:

- These applications summarized**:**

- Space:

+ Space photograph enhancement.
+ Data compression.
+ Intelligent sensory analysis by remote space probes.

- Medical:

+ Diagnostic imaging (CT , MRI , ultrasound , and others).
+ Electrocardiogram analysis.
+ Medical image storage/retrieval.

- Commercial:

+ Image and sound compression for multimedia presentation.
+ Movie special effects.
+ Video conference calling.

- Telephone:

+ Voice and data compression.
+ Echo reduction.
+ Signal multiplexing.
+ Filtering.

- Military:

+ Radar.
+ Sonar.
+ Ordnance guidance.
+ Secure communication.

- Industrial:

+ Oil and mineral prospecting.
+ Process monitoring & control.
+ Nondestructive testing.
+ CAD and design tools.

- Scientific:

+ Earthquake recording & analysis.

+ Data acquisition.

+ Special analysis.

+ Simulation and modeling.

- Figure 1 illustrates a few of these varied applications. These applications summarized as **(DSP Applications):**

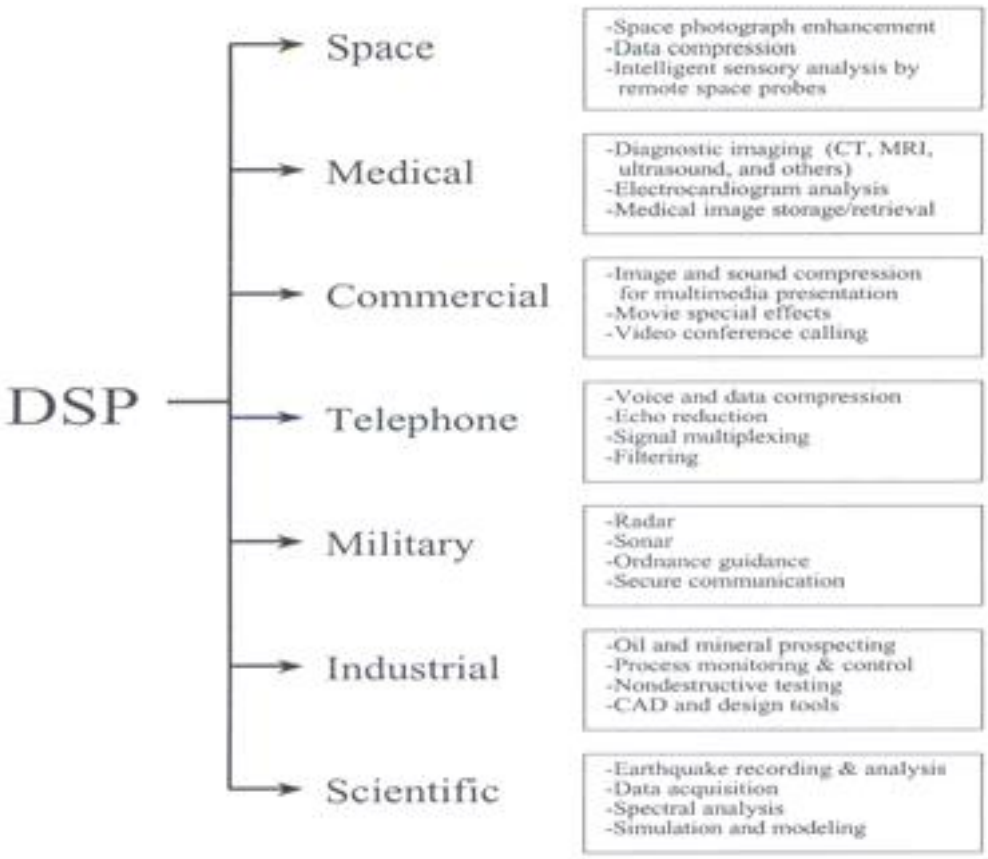


Fig. 1 A few of DSP applications.

- This technological revolution occurred from the top-down.
- In the early 1980s, DSP was taught as a *graduate* level course in electrical engineering.
- A decade later, DSP had become a standard part of the *undergraduate* curriculum.
- Today, DSP is a *basic skill* needed by scientists and engineers in many fields.

- **As an analogy,** DSP can be compared to a previous technological revolution: ***electronics***.
- If you want to specialize in DSP, these are the allied areas you also need to study:
    - Communication theory.
    - Probability and Statistics.
    - Decision Theory.
    - Analog Electronics.
    - Numerical Analysis.
    - Analog Signal Processing.
    - Digital Electronics.

- Figure 2 shows the borders between DSP and other technical disciplines.

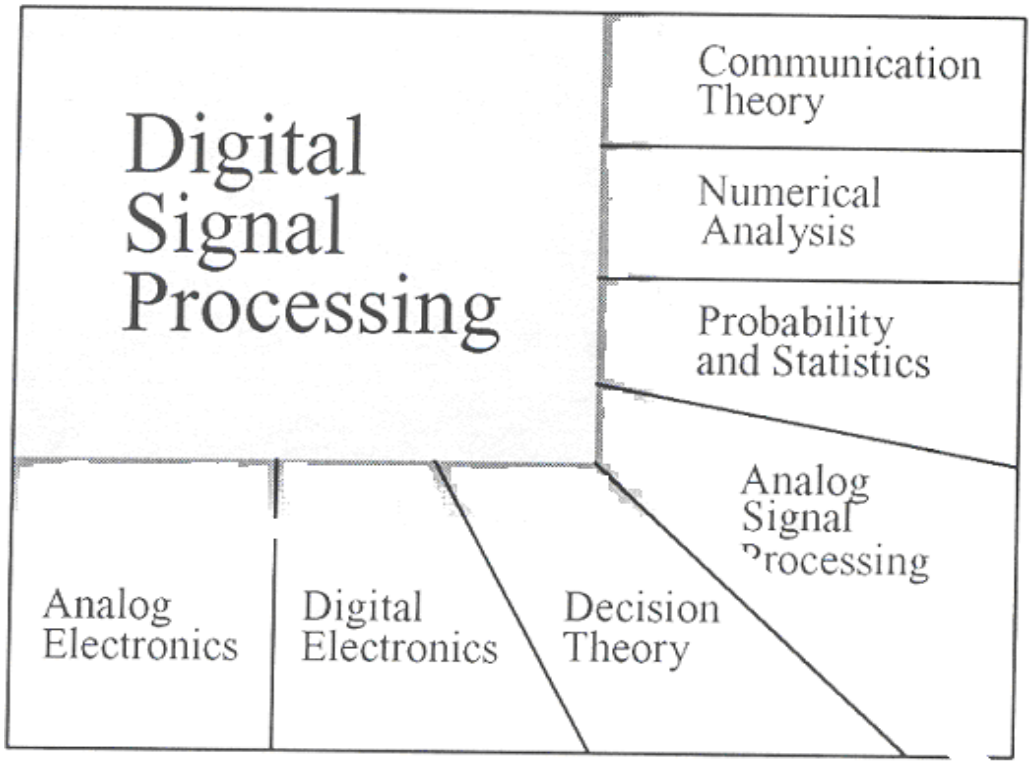


Fig. 2 DSP the allied areas.

CHAPTER 1

ANALOG AND DIGITAL SIGNAL PROCESSING

1.1 SIGNALS, SYSTEMS, AND SIGNAL PROCESSING

SIGNALS

A signal is defined as any physical quantity that varies with time, space, or any other independent variable or variables. Mathematically, we describe a signal as a function of one or more independent variables.

For example, the functions

$$s_1(t) = 5t$$
$$s_2(t) = 20t^2 \quad (1.1)$$

describe two signals, one that varies linearly with the independent variable time and second that varies quadratically with " t ".

As another example, consider the function

$$s(x, y) = 3x + 2xy + 10y^2 \quad (1.2)$$

This function describes a signal of two independent variables x and y that could represent the two spatial coordinates in a plane.

The signals described by (1.1) and (1.2) belong to a class of signals that are precisely defined by specifying the functional dependence on the independent variable. However, there are cases where such a functional relationship is unknown or too highly complicated to be of any practical use. For example, as speech signal cannot be described functionally by expressions such as (1.1).

In general, a segment of speech signal may be represented to a high degree of accuracy as a sum of several sinusoids of different amplitudes and frequencies, that is, as

$$\sum_{i=1}^{N} A_i(t) \sin [2 \pi f_i(t) t + \theta_i(t)] \qquad (1.3)$$

where $\{A_i(t)\}$, $f_i(t)$, and $\{\theta_i(t)\}$ are the sets of possibly time- varying amplitudes frequencies, and phases, respectively, of the sinusoids.

SYSTEMS

In fact, one way to interpret the information content or message conveyed by any short time segment of the speech signal is to measure the amplitudes, frequencies, an phases contained in the short time segment of the signal.

Another example of a natural signal is an electrocardiogram (ECG). Such a signal provides a doctor with information about the condition of the patient's heart. Speech and electrocardiogram signals are examples of information-bearing signals that evolve as functions of single independent variable, namely, time. An example of a signal that is a function of two independent variables is an image signal. The independent variables in this case are the spatial coordinates.

Associated with natural signals are the means by which such signals are generated. For example, speech signals are generated by forcing air through the vocal cords. Images are obtained by exposing a photographic film to a scene or an object. Thus signal generation is usually associated with a system that responds to a stimulus or force. In a speech signal, the system consists of the vocal cords and the vocal tract, also called the vocal cavity. The stimulus in combination with the system is called a signal source. Thus we have speech sources, images sources, and various other types of signal sources.

A system may also be defined as a physical device that performs an operation on a signal. For example, a filter used to reduce the noise and interference corrupting a desired information-bearing signal is called a system. In this case the filter performs some operation(s) on the signal, which has the effect of reducing (filtering) the noise and interference from the desired information-bearing signal.

SIGNAL PROCESSING

When we pass a signal through a system, as in filtering, we say that we have processed the signal. In this case the processing of the signal involves filtering the noise and interference from the desired signal. In general, the system is characterized by the type of operation that it performs on the signal. For example, if the operation is linear, the system is called linear. If the operation on the signal is nonlinear, the system is said to be nonlinear, and so forth. Such operations are usually referred to as signal processing.

For our purposes, it is convenient to broaden the definition of a system to include not only physical devices, but also software realizations of operations on a signal. In digital processing of signals on a digital computer, the operations performed on a signal consist of a number of mathematical operations as specified by a software program. In this case, the program represents an implementation of the system in software. Thus we have a system that is realized on a digital computer by means of a sequence of mathematical operations; that is, we have a digital signal processing system realized in software. For example, a digital computer can be programmed to perform digital filtering.

Alternatively, the digital processing on the signal may be performed by digital hardware (logic circuits) configured to perform the desired specified operations. In such a realization, we have a physical device that performs the specified operations. In a broader sense, a digital system can be implemented as a combination of digital hardware and software, each of which performs its own set of specified operations.

1.2 BASIC ELEMENTS OF ANALOG AND DIGITAL SIGNAL PROCESSING SYSTEMS

Most of the signals encountered in science and engineering are analog in nature that is, the signals are functions of a continuous variable, such as time or space, and usually take on values in a continuous range. Such signals may be processed directly by appropriate analog systems (such as filters or frequency analyzers) or some desired information. In such a case we say that the signal has been processed directly in its analog form, as illustrated in Fig. 1.1 Both the input signal and the output signal are in analog form.



Fig. 1.1 Analog signal processing.

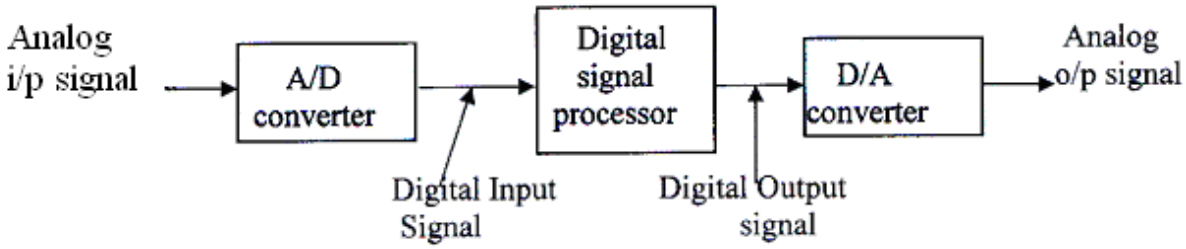


Fig. 1.2 Block diagram of a digital signal processing system.

Digital signal processing provides an alternative method for processing the analog signal, as illustrated in Fig. 1.2. To perform the processing digitally, there is a need for an interface between the analog signal and the digital processor. This interface is called an analog-to- digital (A/D) converter. The output of the A/D converter is a digital signal that is appropriate as an input to the digital processor. The digital signal processor may be a large programmable digital computer or a small microprocessor programmed to perform

the desired operations on the input signal. It may also be a hardwired digital processor configured to perform a specified set of operations on the input signal. Programmable machines provide the flexibility to change the signal processing operations through a change in the software, whereas hardwired machines are difficult to reconfigure. Consequently, programmable signal processors are in very common use. On the other hand, when signal processing operations are well defined, a hardwired implementation of the operations can be optimized, resulting in a cheaper signal processor and, usually, one that runs faster than its programmable counterpart. In applications where the digital output from the digital signal processor is to be given to the user in analog from, such as in speech communications, we must provide another interface from the digital domain to the analog domain. Such an interface is called a digital-to-analog (D/A) converter. Thus the signal is provided to the user in analog form, as illustrated in the block diagram of Fig. 1.2. However, there are other practical applications involving signal analysis, where the desired information is conveyed in digital form and no D/A converter is required. For example, in the digital processing of radar signals, the information extracted from the radar signal, such as the position of the aircraft and its speed, may simply be printed on paper. There is no need for a D/A converter in this case.

1.3 ADVANTAGES OF DIGITAL VERSUS ANALOG SIGNAL PROCESSING

There are many reasons why digital signal processing of an analog signal may be preferable to process the signal directly in the analog domain, as mentioned briefly earlier. First, a digital programmable system allows flexibility in reconfiguring the digital signal processing operations simply by changing the program. Reconfiguration of an analog system usually implies a redesign of the hardware followed by testing and verification to see that it operates properly. Accuracy considerations also play an important role in determining the form of the signal processor. Tolerances in analog circuit components make it extremely difficult for the system designer to control the accuracy of an analog signal processing system. On the other hand, a digital system provides much better control of accuracy requirements. Such requirements, in turn, result

in specifying the accuracy requirements in the A/D converter and the digital signal processor, in terms of word length, floating-point versus fixed-point arithmetic, and similar factors. Digital signals are easily stored on magnetic media (tape or disk) without deterioration or loss of signal fidelity beyond that introduced in the A/D conversion. As a consequence, the signals become transportable and can be processed off-line in a remote laboratory. The digital signal processing method also allows for the implementation of more sophisticated signal processing algorithms. It is usually very difficult to perform precise mathematical operations on signals in analog form but these same operations can be routinely implemented on a digital computer using software. In some cases a digital implementation of the signal processing system is cheaper than its analog counterpart. The lower cost may be due to the fact that the digital hardware is cheaper, or perhaps it is a result of the flexibility for modifications provided by the digital implementation.

Digital Signal Processing Limitations

As already indicated, however, digital implementation has its limitations. One practical limitation is the speed of operation of A/D converters and digital signal processors. We shall see that signals having extremely wide bandwidths require fast-sampling-rate A/D converters and fast digital signal processors. Hence there are analog signals with large bandwidths for which a digital processing approach is beyond the state of the art of digital hardware.

1.4 CLASSIFICATION OF SIGNALS

The methods we use in processing a signal or in analyzing the response of a system to a signal depend heavily on the characteristic attributes of the specific signal. There are techniques that apply only to specific families of signals. Consequently, any investigation in signal processing should start with a classification of the signals involved in the specific application.

1.4.1 Continuous-Time versus Discrete-Time Signals

Signals can be further classified into four different categories depending on the characteristics of the time (independent) variable and the values they take continuous-time signals or analog signals are defined for every value of time and they take on values in the continuous interval (a, b), where "a" can be -"∞"and "b" can be "∞". Mathematically, these signals can be described by functions of a continuous variable.

The signals:

$$x_1(t) = \cos(\pi t),\ x_2(t) = e^{-|t|}, \qquad -\infty < t < \infty \tag{1.4}$$

are example of analog signals. Discrete-time signals are defined only at certain specific values of time. These time instants need not be equidistant, but in practice they are usually taken at equally spaced intervals for computational convenience and mathematical tractability.

The signal:

$$x(t_n) = e^{-|t_n|},\ n = 0, \pm 1, \pm 2, \ldots \tag{1.5}$$

provides an example of a discrete-time signal. If we use the index "n" of the discrete-time instants as the independent variable, the signal value becomes a function of an integer variable (i.e., a sequence of numbers). Thus a discrete-time signal can be represented mathematically by a sequence of real or complex numbers. To emphasize the discrete-time nature of a signal, we shall denote such a signal as x(n) instead of x(t) . If the time instants t_n are equally spaced (i.e., $t_n = nT$), the notation x(nT) is also used. For example, the sequence

$$X(n) = \begin{cases} 0.8^n, & \text{if } n \geq 0 \\ 0, & \text{otherwise} \end{cases} \tag{1.6}$$

is a discrete-time signal, which is represented graphically as in Fig. 1.3

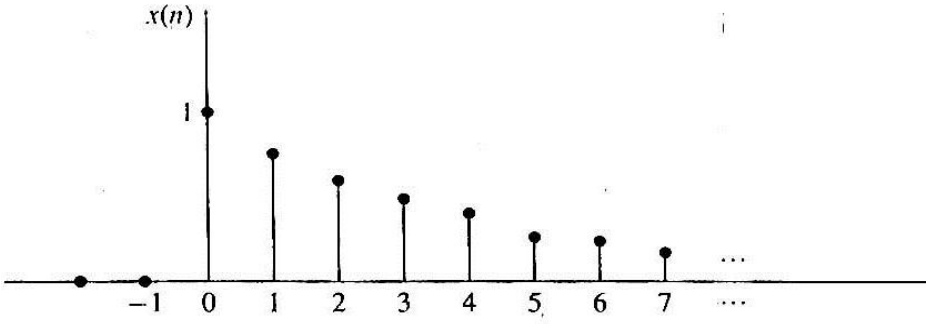


Fig 1.3 Representation of the discrete time signal $x(n) = 0.8^n$ for $n>0$ and $x(n) = 0$ for $n<0$.

1.4.2 Continuous-Valued Versus Discrete-Valued Signals

The values of a continuous-time or discrete-time signal can be continuous or discrete. If a signal takes on all possible values on a finite or an infinite range, it is to be continuous-valued signal. Alternatively, if the signal takes on values from a finite set of possible values, it is said to be a discrete-valued signal. Usually, these values are equidistant and hence can be expressed as an integer multiple of the distance between two successive values. A discrete-time signal having a set of discrete values is called a digital signal. Figure 1.4 shows a digital signal that takes on one of four possible values. In order for a signal to be processed digitally, it must be discrete in time and its values must be discrete (i.e., it must be a digital signal). If the signal to be processed is in analog form, it is converted to a digital signal by sampling the analog signal at discrete instants in time, obtaining a discrete-time signal, and then by quantizing its values to a set of discrete values, as described later in this chapter. The process of converting a continuous-valued signal into a discrete-valued signal, called quantization, is basically an approximation process. It may be accomplished simply by rounding or truncation. For example, if the allowable signal values in the digital signal are integers, say 0 through 15, the continuous-value signal is quantized into these integer values. Thus the signal value 8.58 will be approximated by the value 8 if the quantization process is performed by truncation or by 9 if the quantization process is performed by rounding to the nearest integer. An explanation of the analog-to-digital conversion process is given later in this chapter.

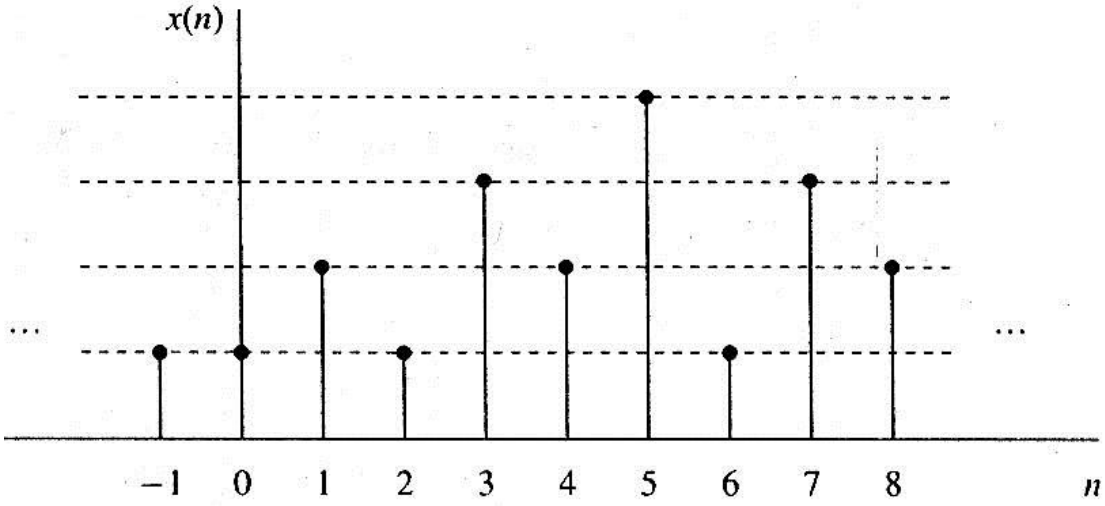


Fig. 1.4 Digital signal with four different amplitude values.

1.4.3 Deterministic Versus Random Signals

The mathematical analysis and processing of signals requires the availability of a mathematical description for the signal itself. This mathematical description, often referred to as the signal model, leads to another important classification of signals. Any signal that can be uniquely described by an explicit mathematical expression, a table of data, or a well-defined rule is called deterministic. This term is used to emphasize the fact that all past, present, and future values of the signal are known precisely, without any uncertainty. In many practical applications, however, there are signals that either cannot be described to any reasonable degree of accuracy by explicit mathematical formulas, or such description is too complicated to be of any practical use. The lack of such a relationship implies that such signals evolve in time in an unpredictable manner. We refer to these signals as random. The analysis and description of random signals are performed using statistical techniques instead of explicit formulas. The mathematical framework for the theoretical analysis of random signals is provided by the theory of probability and stochastic processes.

1.4.4 Multichannel and Multidimensional Signals

As explained in section 1.1, a signal is described by a function of one or more independent variables. The value of the function (i.e., the dependent variable) can be a real-valued scalar quantity, a complex-valued quantity, or perhaps a vector. For example, the signal

$$s_1(t) = A \sin 3\pi t \tag{1.7}$$

is a real-valued signal.

However, the signal

$$s_2(t) = Ae^{j3\pi t} = A \cos 3\pi t + j A \sin 3\pi t \tag{1.8}$$

is complex valued.

In some applications, signals are generated by multiple sources or multiple sensors. Such signals, in turn, can be represented in vector form.

If $S_k^{(t)}$, k = 1,2,3, denotes the electrical signal from the kth sensor as a function of time, the set of k = 3 signals can be represented by a vector S_3 (t), where

$$S_3(t) = \begin{bmatrix} s_1(t) \\ s_2(t) \\ s_3(t) \end{bmatrix} \tag{1.9}$$

we refer to such a vector of signals as a multi-channel signal.

Multidimensional Signals:

Let us now turn our attention to the independent variable (s). If the signal is a function of a single independent variable, the signal is called a one-dimensional signal. On the other hand, a signal is called M-dimensional if its value is a function of M independent variables. In our course, we deal mainly with single-channel; one-dimensional real-or complex-valued signals and we refer to them simply as signals. In mathematical terms these signals are described by a function of a single independent variable. Although the independent variable need not be time, it is common practice to use “t” as the independent variable.

1.5 CONCEPT OF FREQUENCY IN CONTINUOUS AND DISCRETE-TIME SIGNALS

The concept of frequency is familiar to students in engineering and the sciences. This concept is basic in, for example, the design of a radio receiver, a high-fidelity system, or a spectral filter for color photography. From physics we know that frequency is closely related to a specific type of periodic motion called harmonic oscillation, which is described by sinusoidal functions. The concept of frequency is directly related to the concept of time. Actually, it has the dimension of inverse time. Thus we should expect that the nature of time (continuous or discrete) would affect the nature of the frequency accordingly.

1.5.1 Continuous-Time Sinusoidal Signals

A simple harmonic oscillation is mathematically described by the following continuous-time sinusoidal signal:

$$x_a(t) = A\cos(\omega t + \theta) \qquad -\infty < t < \infty \tag{1.10}$$

shown in Fig. 1.5 . The subscript "a" used with x(t) denotes an analog signal. This signal is completely characterized by three parameters: A is the amplitude of the sinusoid, ω is the frequency in radians per second (rad/s), and θ is the phase in radians. Instead of "ω", we often use the frequency F in cycles per second or hertz (H_z), where

$$\omega = 2\pi f \tag{1.11}$$

In terms of "f", (1.10) can be written as

$$x_a(t) = A\cos(2\pi ft + \theta), \quad -\infty < t < \infty \tag{1.12}$$

We will use both forms, (1.10) and (1.12), in representing sinusoidal signals.

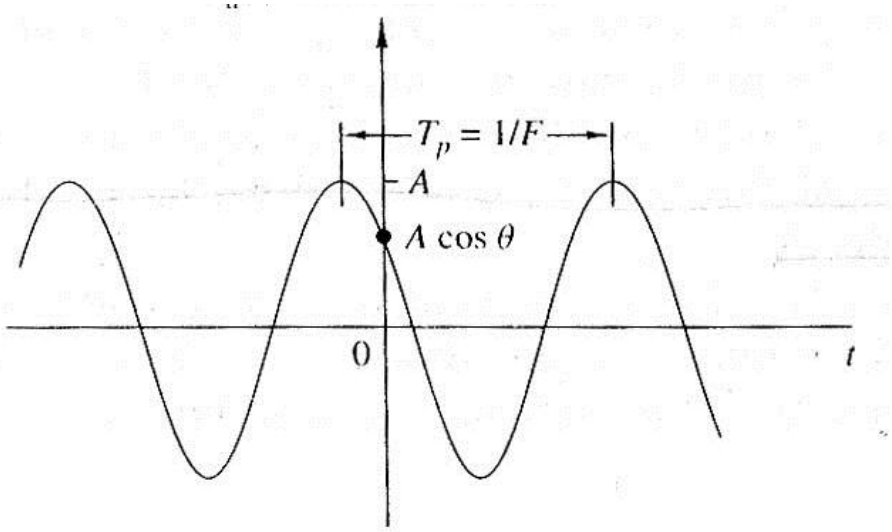


Fig. 1.5 Example of an analog sinusoidal signal Xa(t)=A cos(2 πft+θ).

The analog sinusoidal signal in (1.7) is characterized by the following properties:

(a) For every fixed value of the frequency f, $x_a(t)$ is periodic. Indeed, easily be shown, using elementary trigonometry, that

$$x_a (t + T_p) = x_a(t) \qquad (1.13)$$

where $T_p = 1/f$ is the fundamental period of the sinusoidal signal

(b) Continuous-time sinusoidal signals with distinct (different) frequencies are themselves distinct.

(c) Increasing the frequency "f" results in an increase in the rate of oscillation of the signal, in the sense that more periods are included in a given time interval.

We observe that for f = 0, the value $T_p = \infty$ is consistent with the fundamental relation $f = 1/T_p$. Due to continuity of the time variable t, we can increase the frequency "f", without limit, with a corresponding increase in the rate of oscillation.

The relationships we have described for sinusoidal signals carry over to the class of complex exponential signals

$$x_a(t) = A\, e^{j(\omega t + \theta)} \tag{1.14}$$

This can easily be seen by expressing these signals in terms of sinusoids using the Euler identity

$$e^{\pm j\phi} = \cos\phi \pm j \sin\phi \tag{1.15}$$

By definition, frequency is an inherently positive physical quantity. This is obvious if we interpret frequency as the number of cycles per unit time in a periodic signal. However, in many cases, only for mathematical convenience, we need to introduce negative frequencies. To see this we recall that the sinusoidal signal (1-5) may be expressed as:

$$x_a(t) = A\cos(\omega t + \theta) = \frac{A}{2} e^{j(\omega t + \theta)} + \frac{A}{2} e^{-j(\omega t + \theta)} \tag{1.16}$$

which follows from (1.10).

Note that a sinusoidal signal can be obtained by adding two equal-amplitude complex-conjugate exponential signals, sometimes called phasors, illustrated in Fig. 1.6.
As time progresses the phasors rotate in opposite directions with angular frequencies $\pm\omega$ radians per second.

Since a positive frequency corresponds to counterclockwise uniform angular motion, a negative frequency simply corresponds to clock wise angular motion. For mathematical convenience, we use both negative and positive frequencies. Hence the frequency range for analog sinusoids is $-\infty < f < \infty$

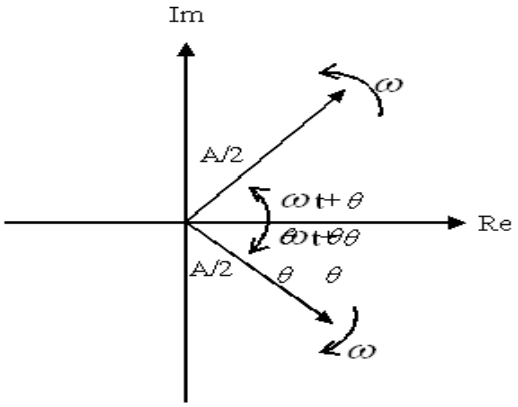


Fig. 1.6 Representation of a cosine functions by a pair of complex-conjugate exponentials (phasors).

1.5.2 Discrete-Time Sinusoidal Signals

A discrete-time sinusoidal signal may be expressed as

$$x(n) = A\cos(\Omega n + \theta),\ -\infty < n < \infty \qquad (1.17)$$

where n is an integer variable, called the sample number, A is the amplitude of the sinusoid, Ω is the frequency in radians per sample, and θ is the phase in radians.

If instead of Ω we use the frequency variable F defined by

$$\Omega = 2\pi F \qquad (1.18)$$

the relation (1.12) becomes

$$x(n) = A\cos(2\pi Fn + \theta),\ -\infty < n < \infty \qquad (1.19)$$

The frequency F has dimensions of cycles per sample. In section 1.4, where we consider the sampling of analog sinusoids, we relate the frequency variable F of a discrete – time sinusoid to the frequency f in cycles per second for the analog sinusoid. For the moment we consider the discrete-time sinusoid in (1.12) independently of the continuous-time sinusoid given in (1.5). Figure 1.7 shows a sinusoid with frequency $\Omega = \pi/6$ radians per sample (F= $\frac{1}{12}$ cycles per sample) and phase $\theta = \pi/3$.

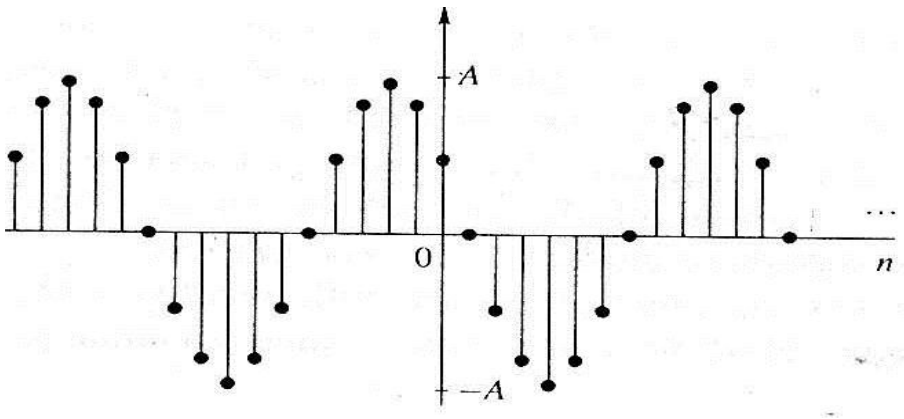


Fig. 1.7 Example of a discrete-time sinusoidal signal
(x(n) = A cos (Ω n+θ); $\Omega = \pi/6$ and $\theta = \pi/3$).

The discrete-time sinusoids are characterized by the following properties:

(a) A discrete-time sinusoid is periodic only if its frequency F is a rational number.

By definition, a discrete-time signal x(n) is periodic with period N (N> 0) if and only if

$$x(n + N) = x(n) \quad \text{for all } n \tag{1.20}$$

The smallest value of N for which (2.14) is true is called the fundamental period.

Proof:

For a sinusoid with frequency F_0 to be periodic, we should have

$$\cos [2\pi F_0 (N + n) + \theta] = \cos (2\pi F_0 n + \theta)$$

This relation is true if and only if there exists an integer k such that

$$2\pi F_0 N = 2k\pi$$

or , equivalently,

$$F_0 = \frac{k}{N} \tag{1.21}$$

According to (1.16), a discrete-time sinusoidal signal is periodic only if its frequency F_0 can be expressed as the ratio of two integers (i.e., F_0 is rational). To determine the fundamental period N of a periodic sinusoid, we express its frequency F_0 as in (1.16) and cancel common factors so that k and N are relatively prime.

Then the fundamental period of the sinusoid is equal to N. Observe that a small change in frequency can result in a large change in the period. For example, if F_1 = 31/60 implies that N_1 = 60, whereas F_2 = 30/60 results in N_2 = 2.

(b) Discrete-time sinusoids whose frequencies are separated by an integer multiple of 2π are identical.

Proof:

Let us consider the sinusoid $\cos(\Omega_0 n+\theta)$. It easily follows that:

$$\cos[(\Omega_0 + 2\pi)n + \theta] = \cos(\Omega_0 n + 2\pi n + \theta) = \cos(\Omega_0 n + \theta) \quad (1.22)$$

As a result, all sinusoidal sequences

$$x_k(n) = A\cos(\Omega_k n + \theta), \quad k = 0,1,2,\ldots\ldots \quad (1.23)$$

where

$$\Omega_k = \Omega_0 + 2k\pi, \quad -\pi \le \Omega_0 \le \pi \quad (1.24)$$

are indistinguishable (i.e., identical). Any sequence resulting from a sinusoid with a frequency $|\Omega| > \pi$, or $|F| > \frac{1}{2}$, is identical to a sequence obtained from a sinusoidal signal with frequency $|\Omega| < \pi$. Because of this similarity, we call the sinusoid having the frequency $|\Omega| > \pi$ an alias of a corresponding sinusoid with frequency $|\Omega| < \pi$. Thus we regard frequencies in the range $-\pi \le \Omega \le \pi$, or $-\frac{1}{2} \le F \le \frac{1}{2}$ as unique and all frequencies $|\Omega| > \pi$, or $|F| > \frac{1}{2}$, as aliases.

You should notice the difference between discrete-time sinusoids and continuous-time sinusoids, where the latter result in distinct signals for "ω" or "f" in the entire rang $-\infty < \omega < \infty$ or $-\infty < f < \infty$.

(c) The highest rate of oscillation in a discrete-time sinusoid is attained when $\Omega = \pi$ or ($\Omega = -\pi$) or, equivalently, F= $\frac{1}{2}$ (or F = $-\frac{1}{2}$).

To illustrate this property, let us investigate the characteristics of the sinusoidal signal sequence

$$x(n) = \cos \Omega_0 n \qquad (1.25)$$

when the frequency varies from 0 to π.

To simplify the argument, we take values of Ω_0 = 0, $\pi/8$, $\pi/4$, $\pi/2$ and π corresponding to F = 0, $\frac{1}{16}, \frac{1}{8}, \frac{1}{4}, \frac{1}{2}$, which result in periodic sequences having periods N = ∞, 16, 8, 4, 2, as depicted in Fig. 1.8.

We note that the period of the sinusoid decreases as the frequency increases. In fact, we can see that the rate of oscillation increases as the frequency increases.

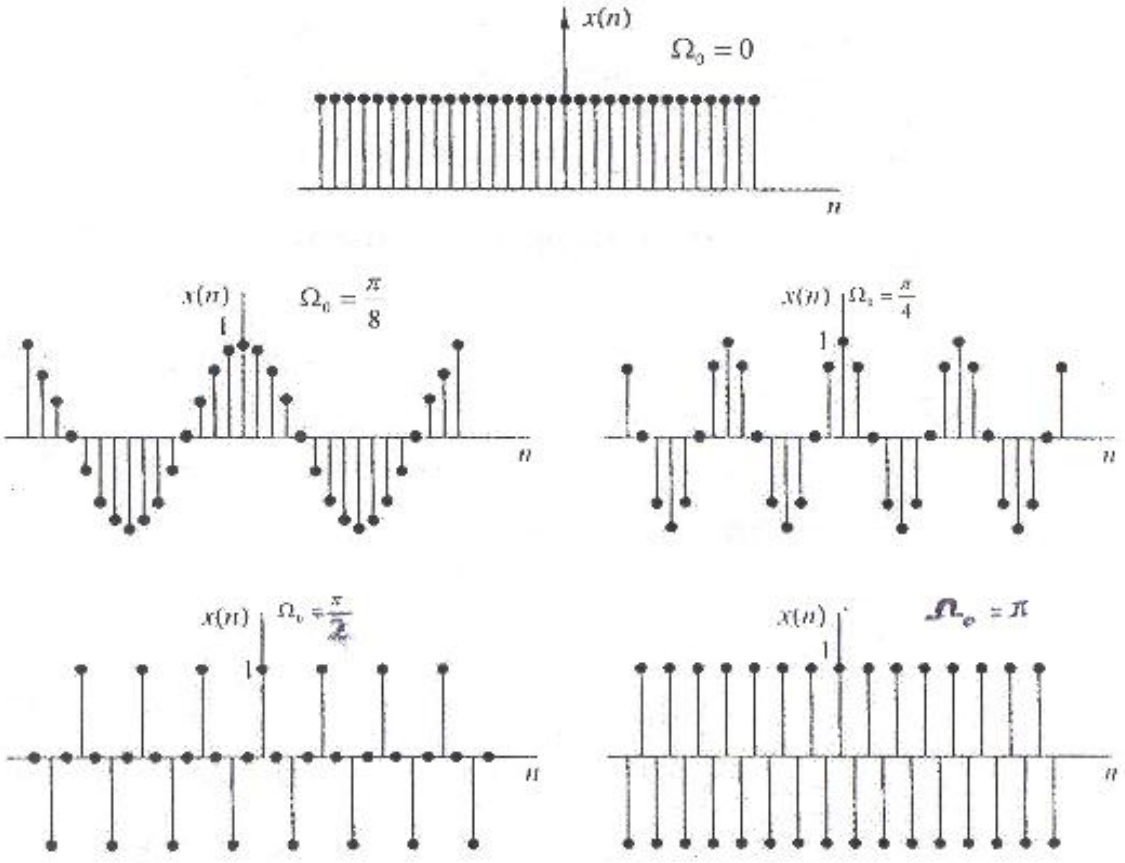


Fig. 1.8 Signal x(n) = cos $\Omega_0 n$ for various values of the frequency Ω_0

1.5.3 Harmonically Related Complex Exponentials

Sinusoidal signals and complex exponentials play a major role in the analysis of signals and systems. In some cases we deal with sets of periodic complex exponentials with fundamental frequencies that are multiples of a single positive frequency. Although we confine our discussion to complex exponentials, the same properties clearly hold for sinusoidal signals. We consider harmonically related complex exponentials in both continuous time and discrete time.

1.5.3.1 Continuous-time exponentials:

The basic signals for continuous-time, harmonically related exponentials are

$$s_k(t) = e^{jk\omega_0 t} = e^{j2\pi k f_0 t} \quad k = 0, \pm 1, \pm 2, \ldots \tag{1.26}$$

We note that for each value of k, $s_k(t)$ is periodic with fundamental period $1/(kf_0) = T_P/k$ or fundamental frequency kf_0.

Since a signal that is periodic with period T_P/k is also periodic with period $k(T_P/k) = T_P$ for any positive integer k, we see that all of the $s_k(t)$ have a common period of T_P. Furthermore, f_0 is allowed to take any value and all members of the set are distinct, in the sense that if $k_1 \neq k_2$, then $s_{k1}(t) \neq s_{k2}(t)$.

From the basic signals in (1.23) we can construct a linear combination of harmonically related complex exponentials of the form

$$x_a(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega 0t} = \sum_{k=-\infty}^{\infty} c_k s_k(t) \tag{1.27}$$

where c_k, k = 0, ± 1, ± 2, ….. are arbitrary complex constants.

The signal $x_a(t)$ is periodic with fundamental period $T_p = 1/f_0$, and its representation in terms of (1.24) is called the Fourier series expansion for $x_a(t)$. The complex-valued constants are the Fourier series coefficients and the signal $s_k(t)$ is called the kth harmonic of $x_a(t)$.

1.5.3.2 Discrete-time exponentials:

Since a discrete-time complex exponential is periodic if its relative frequency is a rational number, we choose $F_0 = 1/N$ and we define the sets of harmonically related complex exponentials by

$$s_k(n) = e^{jk\Omega_0 n} = e^{j2\pi kF_0 n}, \quad k = 0, \pm 1, \pm 2, \ldots. \tag{1.28}$$

In contrast to the continuous-time case, we note that

$$s_{k+N}(n) = e^{j2\pi n(k+N)/N} = e^{j2\pi n} s_k(n) = s_k(n) \tag{1.29}$$

This means that, consistent with (1.15), there are only N distinct periodic complex exponentials in the set described by (1.25).

Furthermore, all members of the set have a common period of N samples. Clearly, we can choose any consecutive N complex exponentials, say from $k = n_0$ to $k = n_0+N-1$ to form a harmonically related set with fundamental frequency $F_0 = 1/N$.

Most often, for convenience, we choose the set that corresponds to $n_0 = 0$, that is, the set

$$s_k(n) = e^{j2\pi kn/N}, \quad k = 0, 1, 2, \ldots.., N-1 \tag{1.30}$$

As in case of continuous-time signals, it is obvious that the linear combination.

$$x(n) = \sum_{k=0}^{N-1} c_k s_k(n) = \sum_{k=0}^{N-1} c_k e^{j2\pi kn/N} \qquad (1.31)$$

results in a periodic signal with fundamental period N. Equation (1.31), is the Fourier series representation for a periodic discrete-time sequence with Fourier coefficients $\{c_k\}$. The sequence $s_k(n)$ is called the kth harmonic of x(n).

1.6 ANALOG TO DIGITAL CONVERSION

Most signals of practical interest, such as speech, biological signals, seismic signals, radar signals, sonar signals, and various communications signals such as a audio and video signals, are analog. To process analog signals by digital means, it is first necessary to convert them into digital form, that is, to convert them to a sequence of number having finite precision. This procedure is called analog-to-digital (A/D) conversion, and the corresponding devices are called A/D conversion as a three-step process. This process is illustrated in Fig. 1.9

(1) Sampling: This is the conversion of a continuous-time signal into a discrete-time signal obtained by taking "samples" of the continuous-time signal at discrete-time instants. Thus, if $x_a(t)$ is the input to the sampler, the output is $x_a(nT) \equiv x(n)$, where T is called the sampling interval.

(2) Quantization: This is the conversion of a discrete-time continuous-valued signal into a discrete-time, discrete-valued (digital) signal. The value of each signal sample is represented by a value selected from a finite set of possible values. The difference between the un-quantized sample x(n) and the quantized output $x_q(n)$ is called the quantization error.

(3) Coding: In the coding process, each discrete value $x_q(n)$ is represented by a b-bit binary sequence.

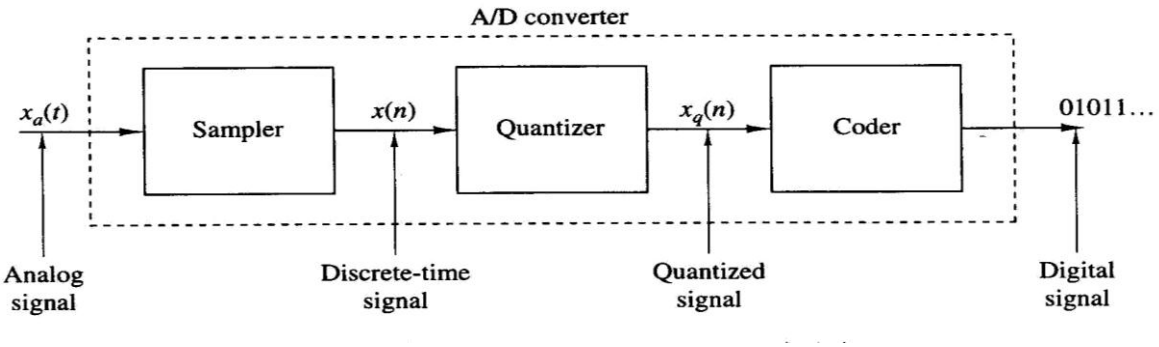


Fig. 1.9 Basic parts of an analog-to-digital (A/D) converter.

Although we model the A/D converter as a sampler followed by a quantizer and coder, in practice the A/D conversion is performed by a single device that takes $x_a(t)$ and produces a binary-coded number. The operations of sampling and quantization can be performed in either order but, in practice, sampling is always performed before quantization.

Sampling and quantization are treated in this section. In particular, we demonstrate that sampling does not result in a loss of information, nor does it introduce distortion in the signal if the signal bandwidth is finite. In principle, the analog signal can be reconstructed from the samples, provided that the sampling rate is sufficiently high to avoid the problem commonly called aliasing. On the other hand, quantization is a noninvertible or irreversible process that results in signal distortion. We shall show that the amount of distortion is dependent on the accuracy as measured by the number of bits, in the A/D conversion process. The factors affecting the choice of the desired accuracy of the A/D converters cost and sampling rate. In general, the cost increases with an increase in accuracy and/or sampling rate.

1.6.1 Sampling of Analog Signals

There are many ways to sample an analog signal. We limit our discussion to *periodic or uniform sampling*, which is the type of sampling used most often in practice. This is described by the relation

$$x(n) = x_a(nT), \quad -\infty < n < \infty \tag{1.32}$$

where x(n) is the discrete-time signal obtained by "taking samples" of the analog signal $x_a(t)$ every T seconds. This procedure is illustrated in Fig. 1.11.

The time interval T between successive samples is called the sampling period or sample interval and its reciprocal $1/T = F_s$ is called the sampling rate (samples per second) or the sampling frequency (hertz).

Periodic sampling establishes a relationship between the time variables't' and 'n' of continuous-time and discrete-time signals, respectively. Indeed, these variables are linearly related through the sampling period T or, equivalently, through the sampling rate $F_s = 1/T$, as

$$t = nT = \frac{n}{F_s} \tag{1.33}$$

As a consequence of (1.30), there exists a relationship between the frequency variable f (or ω) for analog signals and the frequency variable F (or Ω) for discrete-time signals. To establish this relationship, consider an analog sinusoidal signal of the form

$$x_a(t) = A \cos (2\pi ft + \theta) \tag{1.34}$$

which , when sampled periodically at a rate $F_s = 1/T$ samples per second, yields

$$\begin{aligned} x_a(nT) \equiv x(n) &= A \cos (2\pi f n T + \theta) \\ &= A \cos \left(\frac{2\pi nf}{F_s} + \theta\right) \end{aligned} \tag{1.35}$$

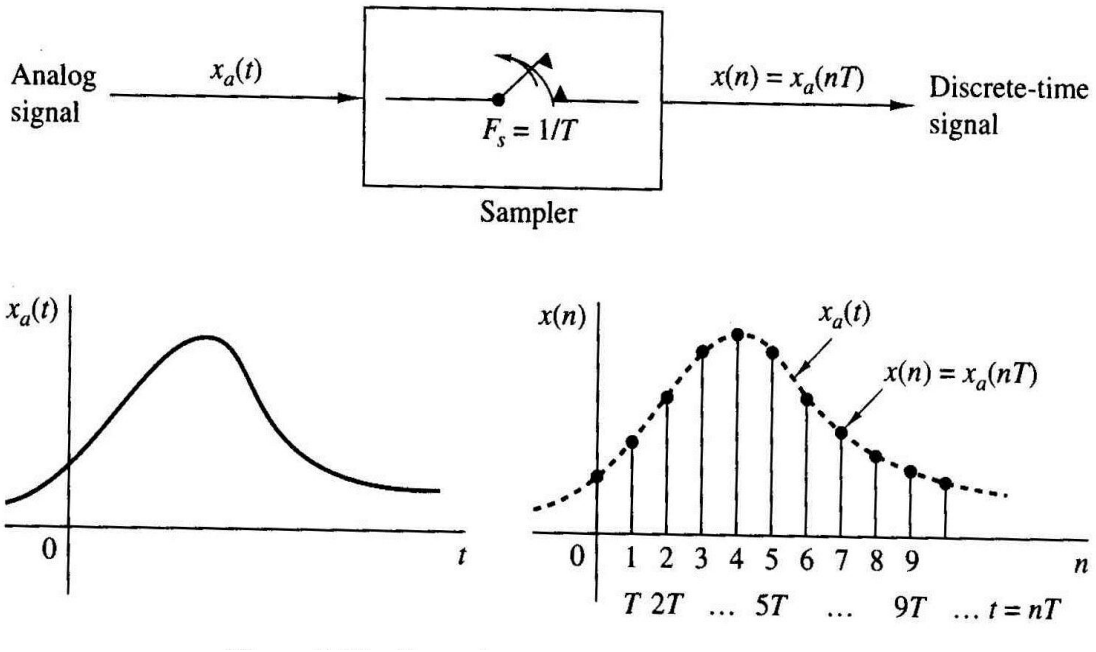


Fig. 1.10 Periodic sampling of an analog signal.

If we compare (1.32) with (1.14), we note that the frequency variables F and f are linearly related as

$$F = \frac{f}{F_s} \tag{1.36}$$

or, equivalently, as

$$\Omega = \omega T \tag{1.37}$$

The relation in (1.33) justifies the name relative or normalized frequency, which is sometimes used to describe the frequency variable “F”. As (1.33) implies, we can use “F” to determine the frequency f in hertz only if the sampling frequency F_s is known. We recall from section 1.3.1(continuous-time signal), that the range of the frequency variable f or ω for continuous-time sinusoids are

$$\begin{aligned} -\infty < f < \infty \\ -\infty < \omega < \infty \end{aligned} \tag{1.38}$$

However, the situation is different for discrete-time sinusoids. From section 1.3.2 (discrete- time signal), we recall that

$$-\frac{1}{2} < F < \frac{1}{2}$$
$$-\pi < \Omega < \pi \qquad (1.39)$$

By substituting from (1.33) and (1.34) into (1.36), we find that the frequency of the continuous-time sinusoid when sampled at a rate $F_s = 1/T$ must fall in the range

$$-\frac{1}{2T} = -\frac{F_s}{2} \le f \le \frac{F_s}{2} = \frac{1}{2T} \qquad (1.40)$$

or, equivalently,

$$-\frac{\pi}{T} = -\pi F_s \le \omega \le \pi F_s = \frac{\pi}{T}$$

These relations are summarized in Table 1.1.

Table1.1 Relations among Frequency Variables

Continuous-time signals		Discrete-time signals
$\omega = 2\pi f$		$\Omega = 2\pi F$
$\frac{radians}{\sec}$ Hz		$\frac{radians}{sample}$ $\frac{cycles}{sample}$
	$\Omega = \omega T, F = f / F_s$	$-\pi \le \Omega \le \pi$
		$-\frac{1}{2} \le F \le \frac{1}{2}$
	$\omega = \Omega / T, f = F..F_s$	
$-\infty < \omega < \infty$		$-\pi / T \le \omega \le \pi / T$
$-\infty < f < \infty$		$-F_2 / 2 \le f \le F_s / 2$

From these relations we observe that the fundamental difference between continuous-time and discrete-time signals is in their range of values of the frequency

variables f and F, or ω and Ω. Periodic sampling of a continuous-time signal implies a mapping of the infinite frequency range for the variable f (or ω) into a finite frequency range for the variable F (or Ω). Since the highest frequency in a discrete-time signal is $\Omega = \pi$ or $F = \frac{1}{2}$, it follows that, with a sampling rate F_s, the corresponding highest values of f and ω are

$$f_{max} = \frac{F_s}{2} = \frac{1}{2T} \tag{1.41}$$

$$\omega_{max} = \pi F_s = \frac{\pi}{T} \tag{1.42}$$

Therefore, sampling introduces an ambiguity, since the highest frequency in a continuous-time signal that can be uniquely distinguished when such a signal is sampled at a rate $F_s = 1/T$ is $f_{max} = F_s/2$, or $\omega_{max} = \pi F_s$. To see what happens to frequencies above $F_s/2$, let us consider the following examples.

1.6.2 Sampling Theorem

Given any analog signal, how should we select the sampling period "T" or, equivalently, the sampling rate F_s? To answer this question, we must have some information about the characteristics of the signal to be sampled. In particular, we must have some general information concerning the frequency content of the signal. Such information is generally available to us. For example, we know generally that the major frequency components of a speech signal fall below 3000 Hz. On the television signals, in general, contains important frequency components up to 5 MHz. The information content of such signals is contained in the amplitudes, frequencies, and phases of the various frequency components, but detailed knowledge of the characteristics of such signals is not available to us prior to obtaining the signals. In fact, the purpose of processing the signals is usually to extract this detailed information. However, if we know the maximum frequency content of the general class of signals (e.g., the class of speech signals, the class of video signals, etc.), we can specify the sampling rate necessary to convert the analog signals to digital signals.

Let us suppose that any analog signal can be represented as a sum of sinusoids of different amplitudes, frequencies, and phases, that is,

$$x_a(t) = \sum_{i=1}^{N} A_i \cos(2\pi f_i t + \theta_i) \quad (1.43)$$

where N denotes the number of frequency components. All signals, such as speech and video, lend themselves to such a representation over any short time segment. The amplitudes, frequencies, and phases usually change slowly with time from one time segment to another. However, suppose that the frequencies do not exceed some known frequency, say f_{max}. For example, f_{max} = 3000 Hz for the class of speech signals and f_{max} = 5 MHz for television signals. Since the maximum frequency may vary slightly from different realizations among signals of any given class (e.g., it may vary slightly from speaker to speaker), we may wish to ensure that f_{max} does not exceed some predetermined value by passing the analog signal through a filter that severely attenuates frequency components above f_{max}. Thus we are certain that no signal in the class contains frequency components (having significant amplitude or power) above f_{max}. In practice, such filtering is commonly used prior to sampling.

From our knowledge of f_{max}, we can select the appropriate sampling rate. We know that the highest frequency in an analog signal that can be unambiguously reconstructed when the signal is sampled at a rate $F_s = 1/T$ is $F_s/2$. Any frequency above $F_s/2$ or below $-F_s/2$ results in samples that are identical with a corresponding frequency in the rang $-F_s/2 \leq F \leq F_s/2$. To avoid the ambiguities resulting from aliasing, we must select the sampling rate to be sufficiently high. That is, we must select $F_s/2$ to be greater than $f_{max.}$ Thus to avoid the problem of aliasing, F_s is selected so that

$$F_s > 2f_{max} \quad (1.44)$$

Where f_{max} is the largest frequency component in the analog signal. With the sampling rate selected in this manner, any frequency component, say $|f_i| < f_{max}$, in the analog signal is mapped into a discrete-time sinusoid with a frequency

$$-\frac{1}{2} \le F_i = \frac{f_i}{F_s} \le \frac{1}{2} \tag{1.45}$$

or, equivalently,

$$-\pi \le \Omega_i = 2\pi F_i \le \pi \tag{1.46}$$

Since, $|F| = \frac{1}{2}$ or $|\Omega| = \pi$ is the highest (unique) frequency in a discrete-time signal, the choice of sampling rate according to (1.42) avoids the problem of aliasing. In other words, the condition $F_s > 2f_{max}$ ensures that all the sinusoidal components in the analog signal are mapped into corresponding discrete-time frequency components with frequencies in the fundamental interval. Thus all the frequency components of the analog signal are represented in sampled form without ambiguity, and hence the analog signal can be reconstructed without distortion from the sample values using an 'appropriate" interpolation (digital-to-analog conversion) method. The "appropriate" or ideal interpolation formula is specified by the sampling theorem.

Sampling theorem: If the highest frequency contained in an analog signal $x_a(t)$ is $f_{max} = B$ and the signal is sampled at a rate $F_s > 2f_{max} \equiv 2B$, then $x_a(t)$ can be exactly recovered from its sample values using the interpolation function

$$g(t) = \frac{\sin 2\pi Bt}{2\pi Bt} \tag{1.47}$$

Thus $x_a(t)$ may be expressed as

$$x_a(t) = \sum_{n=-\infty}^{\infty} X_a\left(\frac{n}{F_s}\right) g\left(t-\frac{n}{F_s}\right) \tag{1.48}$$

where $x_a(n/F_s) = x_a(nT) \equiv x(n)$ are the samples of $x_a(t)$.

When the sampling of $x_a(t)$ is performed at the minimum sampling rate $F_s = 2B$, the reconstruction formula in (1.51) becomes

$$x_a(t) = \sum_{n=-\infty}^{\infty} X_a\left(\frac{n}{2B}\right) \frac{\sin 2\pi B(t - n/2B)}{2\pi B(t - n/2B)} \tag{1.49}$$

The sampling rate $f_N = 2B = 2f_{max}$ is called the Nyquist rate. Figure 1.14 illustrates the ideal D/A conversion process using the interpolation function in (1.50).

As can be observed from either (1.51) or (1.52), the reconstruction of $x_a(t)$ from the sequence x(n) is a complicated process, involving a weighted sum of the interpolation function g(t) and its time-shifted versions g(t-nT) for $-\infty < n < \infty$ where the weighting factors are the samples x(n). Because of the complexity and the infinite number of samples required in (1.51) or (1.52), these reconstruction formulas are primarily of theoretical interest.

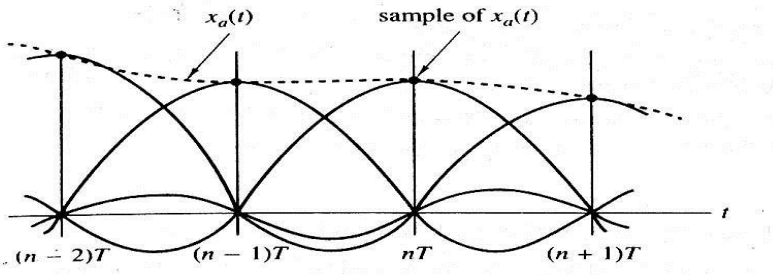


Fig. 1.11 Ideal D/A conversion (interpolation).

1.6.3 Quantization of continuous-amplitude signals

As we have seen, a digital signal is a sequence of numbers (samples) in which each number is represented by a finite number of digital (finite precision). The process of converting a discrete-time continuous-amplitude signal into a digital signal by expressing each sample value as a finite (instead of an infinite) number of digits, is called quantization. The error introduced in representing the continuous-valued signal by a finite set of discrete value levels is called quantization error or quantization noise. We denote the quantizer operation on the samples x(n) as Q[x(n)] and let $x_q(n)$ denote the sequence of quantized samples at the output of the quantizer. Hence

$$x_q(n) = Q\,[x(n)] \qquad (1.50)$$

Then the quantization error is a sequence $e_q(n)$ defined as the difference between the quantized value and the actual sample value. Thus

$$e_q(n) = x_q(n) - x(n) \qquad (1.51)$$

We illustrate the quantization process with an example. Let us consider the discrete-time signal

$$x(n) = \begin{cases} 0.9^n, n \geq 0 \\ 0, n < 0 \end{cases} \quad (1.52)$$

obtained by sampling the analog exponential signal $x_a(t) = 0.9^t, \quad t \geq 0$ with a sampling frequency $F_s = 1$ Hz (see Fig. 1.15(a)). Observation of table 1.2, which shows the values of the first 10 samples of x(n), reveals that the description of the sample value x(n) requires n significant digits. It is obvious that this signal can not be processed by using a calculator or a digital computer since only the first few samples can be stored and manipulated. For example, most calculators process numbers with only eight significant digits. However, let us assume that we want to use only one significant digit. To eliminate the excess digits, we can either simply discard them (truncation) or discard them by rounding the resulting number (rounding). The resulting quantized signals $x_q(n)$ are shown in Table (2) we discuss only quantization by rounding, although it is just as easy to treat truncation. The rounding process is graphically illustrated in Fig. 1.15b The values allowed in the digital signal are called the quantization levels, whereas the distance Δ between two successive quantization levels is called the quantization step size or resolution. The rounding quantizer assigns each sample of x(n) to the nearest quantization level. In contrast, a quantizer that performs truncation would have assigned each sample of x(n) to the quantization level below it. The quantization error $e_q(n)$ in rounding is limited to the range of $-\Delta/2$ to $\Delta/2$, that is,

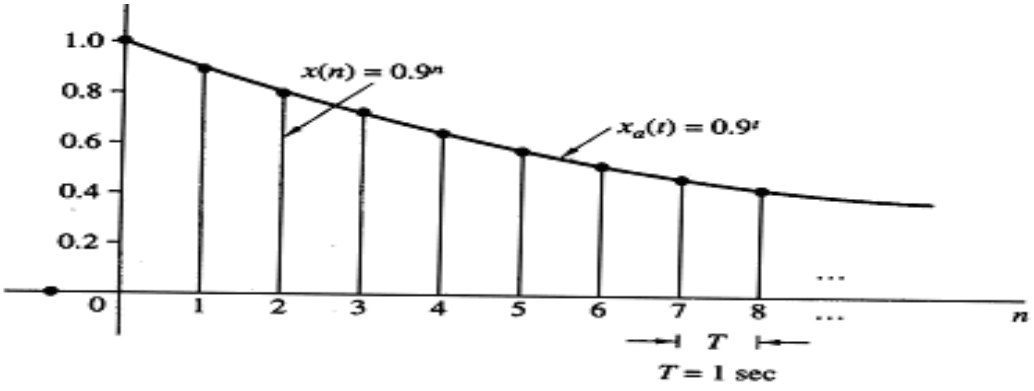
$$-\frac{\Delta}{2} \leq e_q(n) \leq \frac{\Delta}{2} \quad (1.53)$$

In other words, the instantaneous quantization error cannot exceed half of the quantization step (see Table (1.2)). If x_{min} and x_{max} represent the minimum and maximum value of x (n) and L is the number of quantization levels, then

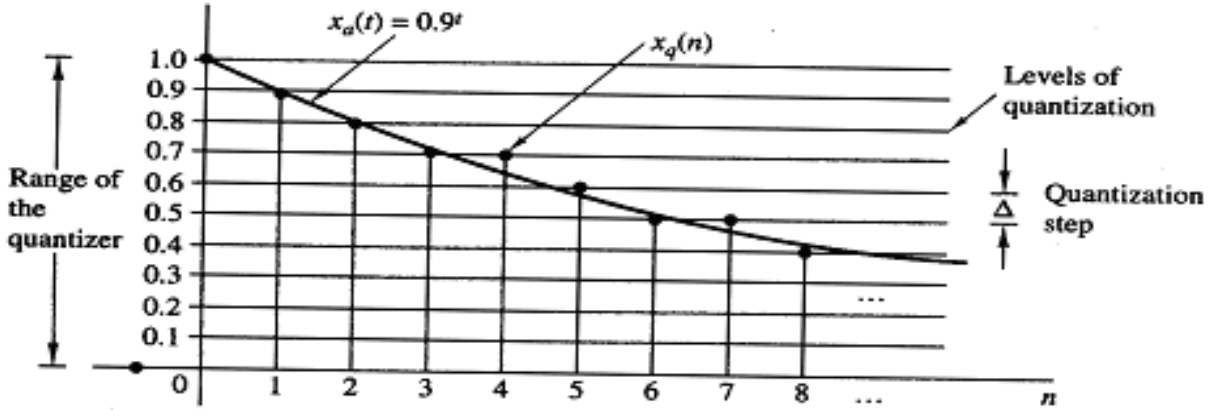
$$\Delta = \frac{x_{\max} - x_{\min}}{L-1} \quad (1.54)$$

We define the dynamic range of the signal as $x_{max} - x_{min}$. In our example we have x_{max} =1, x_{min} = 0 , and L = 11, which leads to Δ = 0.1. Note that if the dynamic range is fixed, increasing the number of quantization levels, L results in a decrease of the quantization step size. Thus the quantization error decreases and the accuracy of the quantizer increases. In practice we can reduce the quantization error to an insignificant amount by choosing a sufficient number of quantization levels.

Theoretically, quantization of analog signals always results in a loss of information. This is a result of the ambiguity introduced by quantization. Indeed, quantization is an irreversible or noninvertible process (i.e., a many-to-one mapping) since all samples in a distance $\Delta/2$ about a certain quantization level are assigned the same value. This ambiguity makes the exact quantitative analysis of quantization extremely difficult.



(a)



(b)

Fig. 1.12 Illustration of quantization

Table 1.2 Numerical Illustration of Quantization with One Significant Digit Using Truncation or Rounding

n	x(n) Discrete-time signal	$x_q(n)$ (Truncation)	$x_q(n)$ (Rounding)	$e_q(n) = x_q(n) - x(n)$ (Rounding)
0	1	1.0	1.0	0.0
1	0.9	0.9	0.9	0.0
2	0.81	0.8	0.8	-0.01
3	0.729	0.7	0.7	-0.029
4	0.6561	0.6	0.7	0.0439
5	0.59049	0.5	0.6	0.00951
6	0.531441	0.5	0.5	-0.031441
7	0.4782969	0.4	0.5	0.0217031
8	0.43046721	0.4	0.4	-0.03046721
9	0.387420489	0.3	0.4	0.012579511

1.6.4 Coding of Quantized Samples

The coding process in an A/D converter assigns a unique binary number to each quantization level. If we have “L” levels we need at least L different binary numbers. With a word length of b bits we can create 2^b different binary numbers.

Hence we have $2^b \geq L$, or equivalently, $b \geq \log_2 L$.

Thus the number of bits required in the coder is the smallest integer greater than or equal to $\log_2 L$. In our example it can easily be seen that we need a coder with b = 4 bits.

Commercially available A/D converters may be obtained with finite precision of b = 16 or less. Generally, the higher the sampling speed and the finer the quantization, the more expensive the device becomes.

1.7 DIGITAL TO ANALOG CONVERSION

To convert a digital signal into an analog signal we can use a D/A converter. As stated previously, the task of a D/A converter is to interpolate between samples.

In many cases of practical interest (e.g., speech processing) it is desirable to convert the processed digital signals into analog form. Obviously, we cannot listen to the sequence of samples representing a speech signal or see the numbers corresponding to a TV signal.) The process of converting a digital signal into an analog signal is known as digital-to-analog (D/A) conversion. All D/A converters "connect the dots" in a digital signal by performing some kind of interpolation, whose accuracy depends on the quality of the D/A conversion process. Figure 1.14 illustrates a simple form of D/A conversion, called a zero-order hold or a staircase approximation. Other approximations are possible, such as linearly connecting a pair of successive samples (linear interpolation), fitting a quadratic through three successive samples (quadratic interpolation), and so on. Is there an optimum (ideal) interpolator? For signals having a limited frequency content (finite bandwidth), the sampling theorem introduced in the following section specifies the optimum form of interpolation.

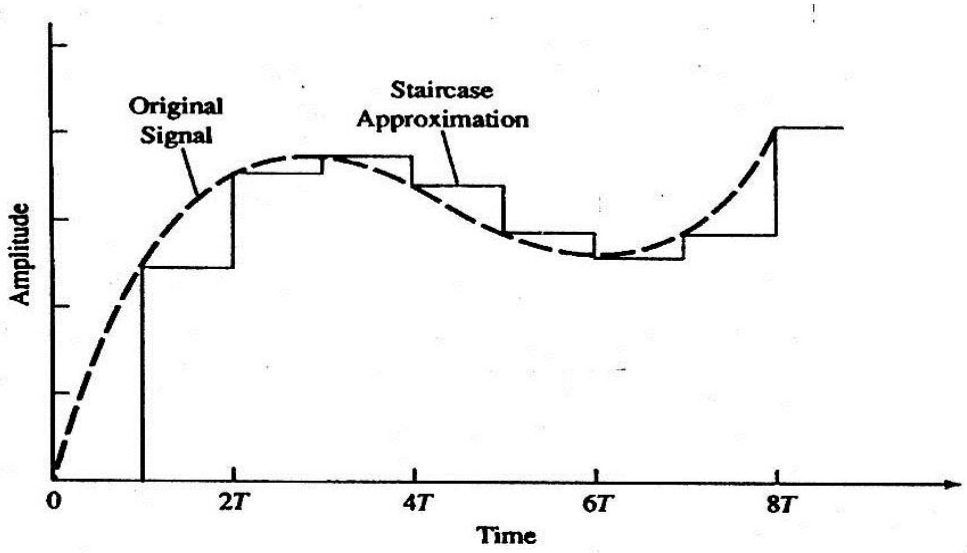


Fig. 1.13 Zero-order hold digital-to-analog (D/A) conversion.

The sampling theorem specifies the optimum interpolation for a band limited signal. However, this type of interpolation is too complicated and, hence impractical, as indicated previously. From a practical viewpoint, the simplest D/A converter is the zero-order hold shown in Fig. 1.14, which simply holds constant the value of one sample until the next one is received. Additional improvement can be obtained by using linear interpolation as shown in Fig. 1.15 to connect successive samples with straight-line segments. Better interpolation can be achieved by using more sophisticated higher-order interpolation techniques. In general, sub optimum interpolation techniques result in passing frequencies above the folding frequency. Such frequency components are undesirable and are usually removed by passing the output of the interpolator through a proper analog filter, which is called a post filter or smoothing filter. Thus D/A conversion usually involve a sub optimum interpolator followed by a post filter.

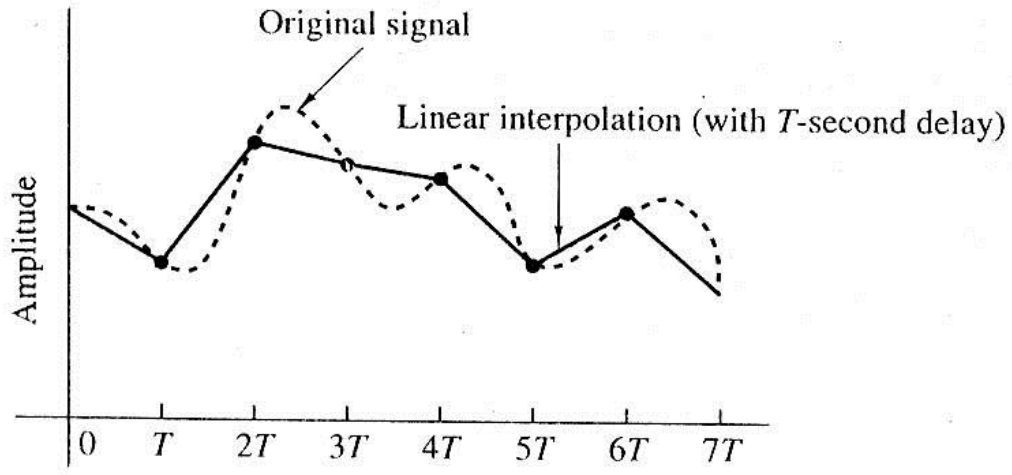


Fig. 1.14 Linear point connector (with T-second delay)

1.8 SOLVED EXAMPLES

Example 1.1

The using of these frequency relations can be fully appreciated by considering the two analog sinusoidal signals

$$x_1(t) = \cos 2\pi(10)t$$
$$x_2(t) = \cos 2\pi(50)t \qquad (1.55)$$

which are sampled at a rate F_s = 40 Hz. The corresponding discrete-time signals or sequences are

$$x_1(n) = \cos 2\pi\left(\frac{1}{40}\right)n = \cos\frac{\pi}{2}n$$
$$x_2(n) = \cos 2\pi\left(\frac{50}{40}\right)n = \cos\frac{5\pi}{2}n \qquad (1.56)$$

However, $\cos 5\pi n/2 = \cos(2\pi n + \pi n/2) = \cos \pi n/2$. Hence $x_2(n) = x_1(n)$.

Thus the sinusoidal signals are identical and, consequently, indistinguishable. If we are given the sampled values generated by $\cos(\pi/2)n$, there is some ambiguity as to whether these sampled values correspond to $x_1(t)$ or $x_2(t)$.

Since $x_2(t)$ yields exactly the same values as $x_1(t)$ when the two are sampled at f_s = 40 samples per second, we say that the frequency f_2 = 50 Hz is an alias of the frequency f_1 = 10 Hz at the sampling rate of 40 samples per second. It is important to note that f_2 is not the only alias of f_1. In fact at the sampling rate of 40 samples per second, the frequency f_3 = 90 Hz is also an alias of f_1, as is the frequency f_4 = 130 Hz, and so on.

All of the sinusoids $\cos 2\pi(f_1+40k)t$, k=1, 2, 3, 4, …. sampled at 40 samples per second, yield identical values. Consequently, they are all aliases of f_1 = 10 Hz.

In general, the sampling of a continuous-time sinusoidal signal:

$$x_a(t) = A \cos (2\pi f_0 t + \theta) \quad (1.57)$$

with a sampling rate $F_s = 1/T$ results in a discrete-time signal

$$x(n) A \cos (2\pi F_0 n + \theta) \quad (1.58)$$

where $F_0 = f_0/F_s$ is the relative frequency of the sinusoid. If we assume that $-F_s/2 \leq f \leq F_s/2$, the frequency F_0 of x(n) is in the range $-\frac{1}{2} \leq F_0 \leq \frac{1}{2}$, which is the frequency range for discrete-time signals. In this case, the relationship between f_0 and F_0 is one-to-one, and hence it is possible to identify (or reconstruct) the analog signal $x_a(t)$ from the sample x(n).

On the other hand, if the sinusoids

$$x_a(t) = A \cos (2\pi f_k t + \theta) \quad (1.59)$$

where

$$f_k = f_0 + kF_s, \qquad k = \pm 1, \pm 2, \ldots\ldots \quad (1.60)$$

are sampled at a rate "F_s", it is clear that the frequency "f_k" is outside the fundamental frequency rang $-F_s/2 \leq f \leq F_s/2$.

Consequently, the sampled signal is

$$x(n) \equiv x_a(nT) = A \cos \left(2\pi \frac{f_0 + kF_s}{F_s} n + \theta\right)$$

$$= A \cos (2\pi n f_0/F_s + \theta + 2\pi kn)$$

$$= A \cos (2\pi F_0 n + \theta)$$

which is identical to the discrete-time signal in (1.43) obtained by sampling (1.42). Thus an infinite number of continuous-time sinusoids is represented by sampling the same discrete-time signal (i.e., by the same set of samples).

Consequently, if we are given the sequence x(n), an ambiguity exists as to which continuous-time signal $x_a(t)$ these values represent. Equivalently, we can say that the frequencies $f_k = f_0 + kF_s$, $-\infty < k < \infty$ (k integer) are indistinguishable from the frequency f_0 after sampling and hence they are aliases of f_0.

The relationship between the frequency variables of the continuous-time and discrete-time signals is illustrated in Fig. 1.12

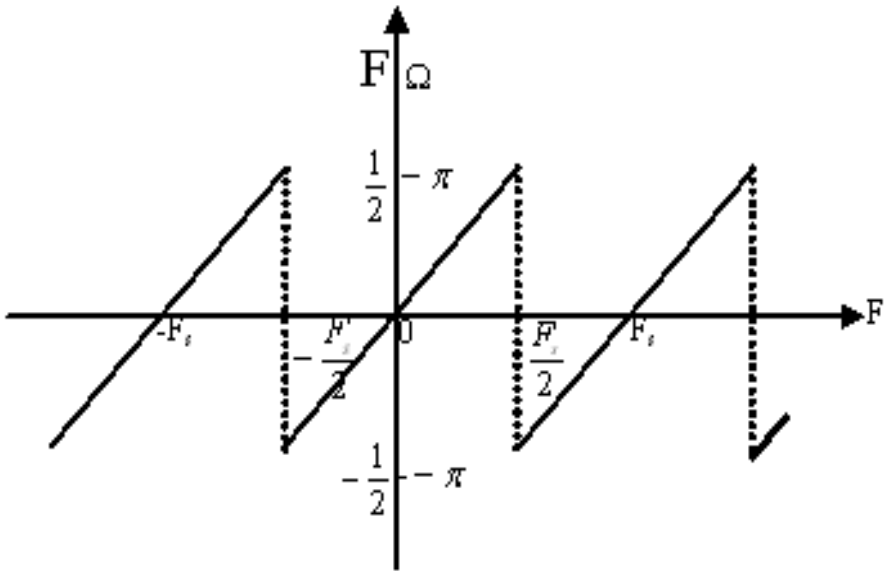


Fig. 1.15 Relationship between the continuous-time and discrete-time frequency variables in the case of periodic sampling

Example 1.2

Consider the analog signal $x_a(t) = 3 \cos 100\pi t$

(a) Determine the minimum sampling rate required to avoid aliasing.

(b) Suppose that the signal is sampled at the rate F_s = 200 Hz. What is the discrete-time signal obtained after sampling?

(c) Suppose that the signal is sampled at the rate F_s = 75Hz. What is the discrete-time signal obtained after sampling?

(d) What is the frequency $0 < f < F_s/2$ of sinusoid that yields samples identical to those obtained in part (c) ?

Solution

(a) The frequency of the analog signal is f = 50 Hz. Hence the minimum sampling rate required to avoid aliasing is F_s = 100 Hz.

(b) If the signal is sampled at F_s = 200 Hz, the discrete-time signal is

$$x(n) = 3 \cos \frac{100\pi}{200} n = 3 \cos \frac{\pi}{2} n$$

(c) If the signal is sampled at F_s = 75 Hz, the discrete-time signal is

$$x(n) = 3 \cos \frac{100\pi}{75} n = 3 \cos \frac{4\pi}{3} n = 3 \cos (2\pi - \frac{2\pi}{3})n = 3 \cos \frac{2\pi}{3} n$$

(d) For the sampling rate of F_s = 75 Hz, we have

$$f = F F_s = 75F$$

The frequency of the sinusoid in part (c) is $F = \frac{1}{3}$. Hence f = 25 Hz

Clearly, the sinusoidal signal

$$y_a(t) = 3 \cos 2\pi ft = 3 \cos 50\pi t$$

sampled at F_s = 75 samples/s yields identical samples.

Hence f = 50 Hz is an alias of f = 25 Hz for the sampling rate F_s = 75 Hz.

Example 1.3

Consider the analog signal

$$x_a(t) = 3 \cos 50\pi t + 10 \sin 300\pi t - \cos 100\pi t$$

what is the Nyquist rate for this signal?

Solution

The frequencies present in the signal above are

$$f_1 = 25 \text{ Hz.} \quad f_2 = 150 \text{ Hz,} \quad f_3 = 50\text{Hz}$$

Thus f_{max} = 150Hz and according to (1.47),

$$F_s > 2f_{max} = 300 \text{ Hz}$$

The Nyquist rate is

$$f_N = 2f_{max}.$$

Hence

$$f_N = 300 \text{ Hz}$$

Discussion:

It should be observed that the signal component 10 sin 300π t, sampled at the Nyquist rate f_N =300, results in the samples 10 sin π n, which are identically zero. In other words, we are sampling the analog sinusoid at its zero-crossing points, and hence we miss this signal component completely. This situation would not occur if the sinusoid is offset in phase by some amount θ. In such a case we have 10 sin (300π t+ θ) sampled at the Nyquist rate f_N = 300 samples per second, which yields the samples

$$10 \sin(\pi n + \theta) = 10(\sin \pi n \cos \theta + \cos \pi n \sin \theta)$$
$$= 10 \text{ Sin } \theta \cos \pi n$$
$$= (-1)^n 10 \sin \theta$$

Thus if $\theta \neq 0$ or π, the samples of the sinusoid taken at the Nyquist rate are not all zero. However, we still cannot obtain the correct amplitude from the samples when the phase θ is unknown. A simple remedy that avoids this potentially troublesome situation is to sample the analog signal at a rate higher than the Nyquist rate.

Example 1.4

Consider the analog signal

$$x_a(t) = 3 \cos 2000\pi t + 5 \sin 6000\pi t + 10 \cos 12.000\pi t$$

a- What is Nyquist rate for this signal?

b- Assume now that we sample this signal using a sampling rate F_s = 5000 samples/s. What is the discrete-time signal obtained after sampling?

c-What is the analog signal $y_a(t)$ we can reconstruct from the samples if we use ideal interpolation?

Solution

a- The frequencies existing in the analog signal are

$$f_1 = 1 \text{ kHz}, \quad f_2 = 3 \text{ kHz}, \quad f_3 = 6 \text{ kHz}$$

Thus $f_{max} = 6$ kHz, and according to the sampling theorem,

$$F_s > 2f_{max} = 12 \text{ kHz}$$

The Nyquist rate is: $f_N = 12$ kHz

b- Since we have chosen $F_s = 5$ kHz, the folding frequency is

$$\frac{F_s}{2} = 2.5 \text{ kHz}$$

and this is the maximum frequency that can be represented uniquely by the sampled signal.

By making use of (1.30) we obtain

$$x(n) = x_a(nT) = x_a\left(\frac{n}{F_s}\right)$$

$$= 3 \cos 2\pi\left(\frac{1}{5}\right)n + 5 \sin 2\pi\left(\frac{3}{5}\right)n + 10 \cos 2\pi\left(\frac{6}{5}\right)n$$

$$= 3 \cos 2\pi\left(\frac{1}{5}\right)n + 5 \sin 2\pi\left(1-\frac{2}{5}\right)n + 10 \cos 2\pi\left(1+\frac{1}{5}\right)n$$

$$= 3 \cos 2\pi\left(\frac{1}{5}\right)n + 5 \sin 2\pi\left(-\frac{2}{5}\right)n + 10 \cos 2\pi\left(\frac{1}{5}\right)n$$

Finally, we obtain

$$x(n) = 13 \cos 2\pi\left(\frac{1}{5}\right)n - 5 \sin 2\pi\left(\frac{2}{5}\right)n$$

The same result can be obtained using Fig. 1.12. Indeed, since F_s = 5 kHz, the folding frequency is $F_s/2$ = 2.5 kHz. This is the maximum frequency that can be represented uniquely by the sampled signal. From (1.45) we have $f_0 = f_k - kF_s$. Thus f_0 can be obtained by subtracting from f_k an integer multiple of F_s such that $-F_s/2 \leq f_0 \leq F_s/2$. The frequency f_1 is less than $F_s/2$ and thus it is not affected by aliasing. However, the other two frequencies are above the folding frequency and they will be changed by the aliasing effect. Indeed,

$$f^1_2 = f_2 - F_s = -2 \text{ kHz}$$

$$f^1_3 = f_3 - F_s = 1 \text{ kHz}$$

From (1.33) it follows that $F_1 = \frac{1}{5}$, $F_2 = -\frac{2}{5}$, and $F_3 = \frac{1}{5}$, which are in agreement with the result above.

c- Since only the frequency components at 1 kHz and 2 kHz are present in the sampled signal, the analog signal we can recover is

$$y_a(t) = 13 \cos 2000\pi t - 5 \sin 4000\pi t$$

which is obviously different from the original signal $x_a(t)$. This distortion of the original analog signal was caused by the aliasing effect, due to the low sampling rate used.

SHEET # 1

1.1 Define the following items: Signal, system, signal processing. (State one example for each).

1.2 Classify the different types of signal processing.

1.3 Classify the different signal types.

1.4 Explain the basic three operations performed in A/D converters.

1.5 State and explain the sampling theorem.

1.6 Determine which of the following sinusoid signals are periodic, and then compute their fundamental period.

(a) $x_{a(t)}$ = 3cos(5t +π/6) (b) $x_{a(t)}$ =3cos(100πt)

(c) x(n) = cos[π(30n/105)] (d) x(n) = 2exp.[j(n/6-π)]

1.7 Consider the following analog sinusoid signal: $x_{a(t)}$= 3sin (100πt)

(a) Sketch the signal for [0, 30] ms.

(b) The signal $x_{a(t)}$ is sampled with sampling rate F_S=300 samples per second.

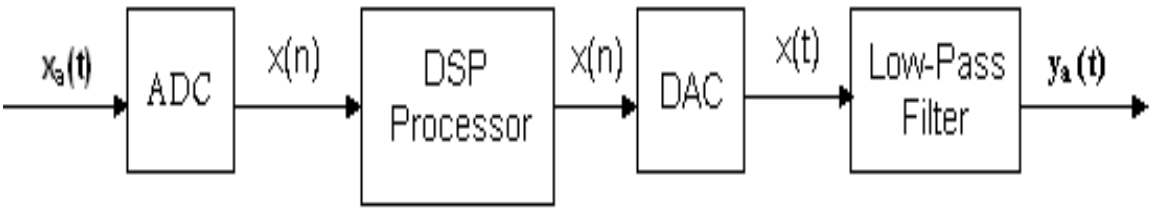
Determine the frequency of the discrete-time signal and show that it is periodic.

(c) Compute the sample values in one period of x (n). Sketch x (n) on the same diagram with $x_{a(t)}$. What is the period of this discrete-time signal, x (n).?

(d) Can you find a sampling rate F_S such that the signal X(n) reaches its peak value 3 ? What is the minimum F_S suitable for this task?

1.8 Consider a simple signal processing system as shown in the following figure. The sampling periods of the A/D and D/A converters are T_1=5 ms and T_2=1 ms respectively. Determine the output y_a (t) if the input is given by:

$x_{a(t)}$ = 3 cos (100πt) + 2 sin (250πt) (t in second)



Assignment # 1

A1.1 An analog signal contains up to 10 KHz.

a- What is the range of sampling frequencies allows exact reconstruction of this signal from its samples?

b- Suppose that we sample this signal with sampling frequency F_S = 8 KHz. Is it possible to reconstruct this signal without distortion?

c- For the same sampling frequency given in (b), examine what was happen if the frequency of this signal is 24 KHz. Cheek the periodicity of the obtained x (n) and if it is periodic find its fundamental period. Find the first four samples of x (n).

A1.2 Determine, which of the following sinusoid signals, is periodic, and then compute their fundamental period.

a- $x(n) = \cos(3n\pi)$ b- $x(n)= 3\cos(5n +\pi/6)$ c- $x(n)= \sin(3n)$

A1.3 Consider the analog signal $x_a(t) = 3 \cos 100\pi t$

a- Determine the minimum sampling rate required to avoid aliasing.

b- Assume that this signal is sampled at the rate $F_s = 200$ Hz. What is the discrete-time signal obtained after sampling, x(n)? Find the first four samples.

c- Plot $x_a(t)$ and x(n) obtained in (b) on the same graph.

CHAPTER 2

DISCRETE TRANSFORMS (DFT & FFT)

2.1 DISCRET FOURIER TRANSFORM (DFT)

In practice the Fourier component of data are obtained by digital computation rather than by analogue processing. Because the analogue waveform consists of an infinite number of contiguous points, the representation of all their values is a practical impossibility. Thus the analogue values have to be sampled at regular intervals and the sample values are then converted to a digital binary representation. This is achieved using a sample-and-hold circuit followed by an analogue-to-digital (AD) converter. Provided the number of samples recorded per second is high enough the waveform will be adequately represented. The theoretically necessary sampling rate is called the Nyquist rate and is $2f_{max}$ where f_{max} is the frequency of the highest frequency sinusoidal component in the signal of significant amplitude. In this chapter, therefore, it is assumed that the digital values of the data are available for transformation. Thus the data to be transformed is discrete and probably non-periodic. It is not possible to apply the Fourier transform because it is for continuous data. However, an analogous transform for use with discrete data, known as the discrete Fourier transform (DFT), is available.

Assume that a waveform has been sampled at regular time intervals T to produce the sample sequence {x(nT)} = x(0), x(T), ……., x[(N-1) T] of N sample values, where n is the sample number from n = 0 to n = N –1 . The data values x(nT) will be real only when representing the values of a time series such as a voltage waveform. The DFT of x(nT) is then defined as the sequence of complex values $\{X(k\Omega)\}$ = X(0), $X(\Omega)$, …., $X[(N-1)\Omega]$ in the frequency domain, where Ω is the first harmonic frequency given by $\Omega = 2\pi/NT$. Thus the $X(k\Omega)$ have real and imaginary components in general so that for the kth harmonic

$$X(k) = R(k) + jI(k) \tag{2.1}$$

and

$$|X(k)| = [R^2(k) + I^2(k)]^{1/2} \tag{2.2}$$

and X(k) has the associated phase angle

$$\phi(k) = \tan^{-1}[I(k)/R(k)] \tag{2.3}$$

where X(k) is understood to represent $X(k\Omega)$.

Note that N real data values (in the time domain) transform to N complex DFT values (in the frequency domain). The DFT values, X(k), are given by

$$X(k) = F_D[x(nT)] = \sum_{n=0}^{N-1} x(nT)e^{-jk\Omega nT},\ k = 0,1,\ldots,N-1 \tag{2.4}$$

where F_D denotes the discrete Fourier transformation. In this equation, k represents the harmonic number of the transform component.

The equation (2.4) can be seen to be analogous to the Fourier transform of Eq. F-1 when f(t) = 0 for t<0 and t>(N-1)T by putting

$$x(nT) = f(t),\ k\Omega = \omega, \text{ and } nT = t,$$

so the two transforms may be expected to have similar properties.

$$F(j\omega) = \int_{-\infty}^{\infty} f(t)e^{-j\omega t}dt \tag{F-1}$$

$F(j\omega)$ ….. is complex and is known as the ***Fourier integral*** or, ***Fourier transform***.
The transforms are not, however, equal. Thus, making theses substitutions in Eq. F-1, and putting dt = T, and replacing the integral by a summation gives

$$\sum_{n=0}^{N-1} x(nT)e^{-jk\Omega nT}T = F(j\omega) \qquad \text{for } 0 \le t \le (N-1)T. \tag{2.5}$$

Then a comparison of Equation 2.4 with 2.5 reveals that

$$F(j\omega) = T\,X(k) \tag{2.6}$$

showing that the Fourier transform components are related to the DFT components by the sampling interval, and may be obtained by multiplying the DFT components by the sampling interval.

Example 2.1

It is appropriate at this point to illustrate the use of Eq. 2.4 by evaluating a simple case. The DFT of the sequence {1,0,0,1} will be evaluated. It is worth noting here that if discontinuities in the actual data are present the average value of the data on each side of the discontinuity is taken for computational purposes to represent the value at the discontinuity. This will apply to the first and final data values as well as to other discontinuities. However, when obtaining the spectra of signals it is necessary to carry out a process known as windowing to avoid distortion of the spectra due to discontinuities at the beginning and end of the data. In this section it is assumed that the sequence {1,0,0,1} has already been preprocessed. Assume that this data represents four consecutive voltages x(0) = 1, x(T) = 0, x(2T) =0, x(3T) = 1, recorded at time intervals, T. Thus N =4.

It is required to find the complex values X(k) for k =0, k =1, k =2 and k =3 (since N-1=3).

With k=0, Eq. 2.4 becomes

$$X(0)= \sum_{n=0}^{3} x(nT)e^{-j0}$$

$$= \sum_{n=0}^{3} x(nT)$$

$$= x(0) + x(T) + x(2T) + x(3T)$$

$$= 1+0+0+1$$

$$= 2$$

so X(0) = 2 is entirely real, of magnitude 2 and phase angle $\phi(0)= 0$.

With k=1, Eq. 2.4 becomes

$$X(1) = \sum_{n=0}^{3} x(nT)e^{-j\Omega nT}$$

"T" had not been given, but may be eliminated using $\Omega = 2\pi/NT$, giving

$$X(1) = \sum_{n=0}^{3} x(nT)e^{-j\Omega n2\pi/N\Omega}$$

$$= \sum_{n=0}^{3} x(nT)e^{-j2\pi n/N}$$

$= 1 + 0 + 0 + 1\, e^{-j2\pi\, 3/4}$

$= 1 + e^{-j3\pi/2}$

$= 1 + \cos\left(\frac{3\pi}{2}\right) - j \sin\left(\frac{3\pi}{2}\right)$

$= 1 + j$

Thus

X(1) =1+j and is complex with magnitude $\sqrt{2}$ and has angle $\phi(\Omega) = \tan^{-1}(1) = 45^{o}$.

With k =2, Eq. 2.4 becomes

$$X(2) = \sum_{n=0}^{3} x(nT) e^{-j2\Omega nT}$$

$$= \sum_{n=0}^{3} x(nT) e^{-j2n2\pi/N}$$

$$= \sum_{n=0}^{3} x(nT) e^{-j4\pi n/N}$$

$= 1 + 0 + 1 + 1\, e^{-j4\pi n/N}$

$= 1 + 0 + 0 + e^{-j3\pi}$

$= 1 - 1 = 0$

Thus X(2) =0, of magnitude zero and phase angle $\phi(2)$=0.

Finally, for k = 3, Eq. 2.4 becomes

$$X(3) = \sum_{n=0}^{3} x(nT) e^{-j3n2\pi/N}$$

$= 1 + 0 + 0 + e^{-j9\pi/2}$

$= 1 - j$

Thus X(3) = 1– j, of magnitude $\sqrt{2}$ and phase angle $\phi(3) = -45^{o}$.

It has therefore been shown that the time series {1,0,0,1} has the DFT given by the complex sequence {2,1 +j,0,1-j}.

It is common practice to represent the DFT by the plots of $|X(k)|$ versus $k\Omega$ and of ϕ(k) versus $k\Omega$. This may be done in terms of harmonics of Ω, or in terms of frequency if Ω is known. To find Ω it is necessary to know the value of T, the sampling interval. If it is assumed that the above data sequence had been sampled at 8 kHz then T = 1/(8 x 10^3) = 125 μs. Then Ω = 2π/NT = 2π/(4x125x10^{-6}) = 1.57 kHz. Hence 2Ω = 25.14 kHz and 3Ω = 37.71 kHz Figure 2.1(a) is a plot of x(nT) versus t, Fig. 2.1(b) is a plot of $|X(k)|$ versus $k\Omega$, and Fig. 2.1(c) is a plot of ϕ(k) versus $k\Omega$. It is noteworthy that component, that is about harmonic number N/2, and that in Fig. 2.1(c) the phase angles are an odd function centered round this component. These results are more generally true.

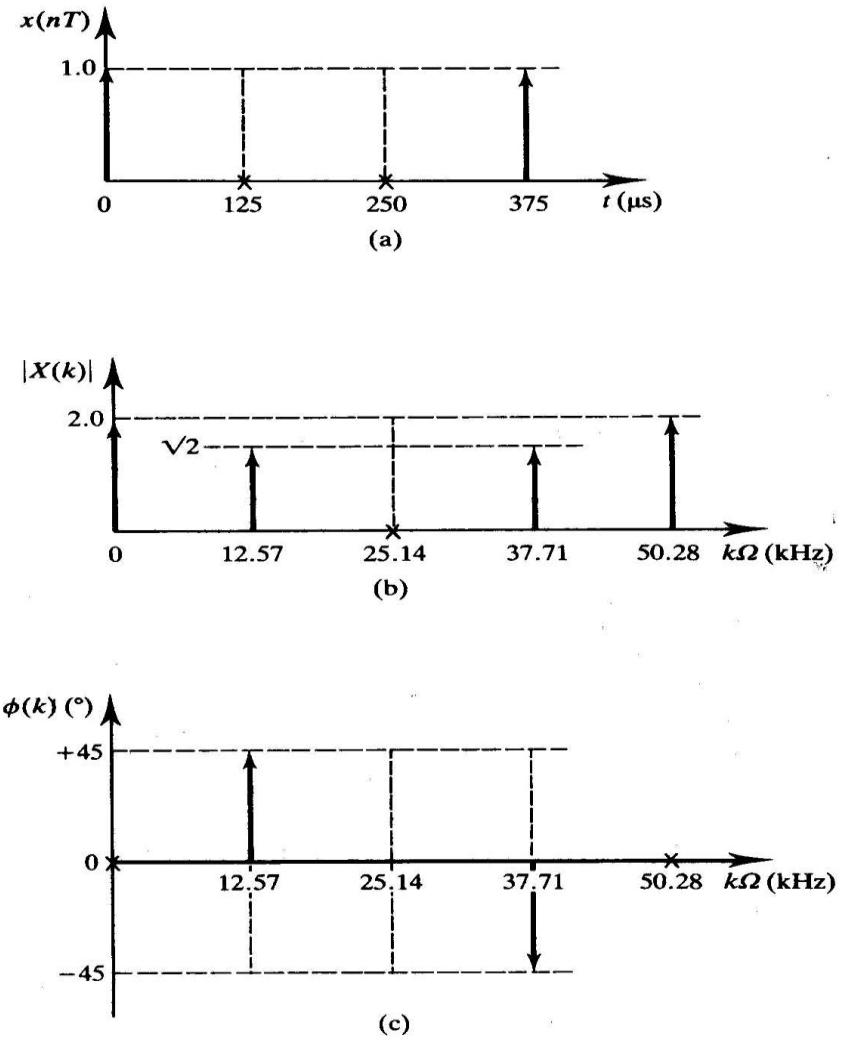


Fig. 2.1 (a) x(nT) versus t, (b) $|X(k)|$ versus k, and (c) ϕ(k) versus k,

An important property of the DFT may be deduced if the kth component of the DFT, X(k), is compared with the $(k+N)^{th}$ component, X(k+N). Thus

$$X(k) = \sum_{n=0}^{N-1} x(nT)e^{-jk\Omega nT}$$

$$= \sum_{n=0}^{N-1} x(nT)e^{-jk2\pi n/N}$$

and

$$X(k+N) = \sum_{n=0}^{N-1} x(nT)\, e^{-jk2\pi n/N} . e^{-jN2\pi n/N}$$

$$= \sum_{n=0}^{N-1} x(nT)\; e^{-jk2\pi n/N}\, e^{-j2\pi n}$$

$$= \sum_{n=0}^{N-1} x(nT)e^{-jk2\pi n/N} = X(k)$$

since n is integral so $e^{-j2\pi n} = 1$

The fact that X (k + N) = X(k) shows that the DFT is periodic with period N. This is the cyclical property of the DFT. The values of the DFT components are repetitive. If k =0, then k + N = N and X(0) = X(N). In the above example X(0) = 2 and therefore X(4) = 2 also. This is illustrated in Figure 2.3(b) where the fourth harmonic amplitude is drawn at 50.28 kHz. The symmetry of the amplitude distribution about the second harmonic is obvious. The general conclusion is that the amplitude spectrum of an N-point DFT is symmetrical about harmonic N/2. If $2f_{max}$ samples per second were taken of the signal for t seconds then $2f_{max}t$ = N, so $1/t = 2f_{max}/N$ is the first harmonic frequency. The symmetry at harmonic N/2 therefore occurs at the frequency $(N/2) / (2f_{max}/N) = f_{max}$, the maximum frequency present in the signal. Thus all the signal components are fully represented in an amplitude spectrum plotted up to f_{max} or harmonic component N/2, and it is unnecessary to plot further points. In this context, f_{max} is known as the folding frequency, since the spectrum between harmonics N/2 and N may be folded about the axis of symmetry at f_{max} to superimpose exactly the low frequency half of the spectrum. It is now seen that N real data values transform to N/2 complex DFT values of practical significance. The latter consist of N/2 real values and N/2 imaginary values giving a total of N values derived from the initial N real data values.

Finally the values of the Fourier transform components, F(jω), of the data {1,0,0,1} in Example 2.1 may be obtained by multiplying the DFT components by T = 125 μs. Therefore F(0) = 250 μV Hz $^{-1}$, F(12.57kHz) =(125+j125) μV Hz $^{-1}$, F(25.14kHz) = 0V Hz $^{-1}$, F(37.71kHz)=(125-j125) μ V Hz.

2.2 INVERSE DISCRET FOURIER TRANSFORM (IDFT)

As explained in the introduction to the chapter, it is also necessary to be able to carry out discrete transformation from the frequency to the time domain. This may be achieved using the inverse discrete Fourier transform (IDFT), defined by

x(nT)= F_D^{-1} [X(k)]

$$= \frac{1}{N} \sum_{k=0}^{N-1} X(k) e^{jk\Omega nT} \text{ , } n= 0, 1, \ldots\ldots, N-1 \qquad (2.7)$$

where F_D^{-1} denotes the inverse discrete Fourier transformation.

The analogy with Eq. F-2 for the inverse Fourier transform is obvious. This time it is quite simple to show that the inverse Fourier transform is obtainable from the IDFT by dividing the IDFT by T. The validity of Eq. 2.7 can be demonstrated by substituting for x(nT) in Eq. 2.4.

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(j\omega) e^{j\omega t} d\omega = \int_{-\infty}^{\infty} F(j\omega) e^{j\omega t} df \qquad (F\text{-}2)$$

Example 2.2

It is useful to illustrate the inverse discrete Fourier transform by using it to derive the time series {1,0,0,1} from its DFT components [2,1+j,0,1-j].

With n = 0,

x(nT) = x(0)

$= \frac{1}{N} \sum_{k=0}^{N-1} X(k)$

= ¼ [X(0) + X(1) + X(2) + X(3)]

= ¼ [2 + (1+j) + 0 + (1-j)]

= 1, as expected.

With n = 1,

x(nT)= x(T)

$= \frac{1}{N} \sum_{k=0}^{N-1} X(k)e^{jk\Omega T}$

$= \frac{1}{N} \sum_{k=0}^{N-1} X(k)e^{jk2\pi /N}$

$= ¼ \sum_{k=0}^{N-1} X(k)e^{jk\pi /2}$

$= ¼ [2 + (1 + j)e^{j\pi /2} + 0 + (1 – j)e^{j3\pi /2}]$

= ¼ [2 + (1 + j)j + (1 – j)(-j)]

= ¼ (2 + j – 1 – j – 1)

= 0 , as expected.

With n = 2,

x(nT) = x(2T)

$= \frac{1}{N} \sum_{k=0}^{N-1} X(k)e^{jk\Omega T.2}$, $\Omega = \frac{2\pi}{NT}$

$= \frac{1}{4} \sum_{k=0}^{3} X(k)e^{jk\frac{2\pi}{NT}.T.2}$

$= \frac{1}{4} \sum_{k=0}^{3} x(k)e^{jk\left(\frac{4\pi}{N}\right)}$, N = 4

$= \frac{1}{4} \sum_{k=0}^{3} X(k)e^{jk\pi}$

$= \frac{1}{4} [2 + (1 + j)\, e^{j\pi} + (1 - j)\, e^{j3\pi}]$

$= \frac{1}{4} [2 – (1 + j) – (1 – j)]$

$= \frac{1}{4}[2 - 1 - j - 1 + j]$

= 0, as expected.

With n = 3,

$$x(nT) = x(3T) = \frac{1}{4}\sum_{k=0}^{3} X(k)e^{jk\Omega 3T}$$

$$= \frac{1}{4}\sum_{k=0}^{3} X(k)e^{jk\frac{2\pi}{4NT}.T.3}$$

$$= \frac{1}{4}\sum_{k=0}^{3} X(k)e^{j\frac{3k\pi}{2}}$$

$$= \frac{1}{4}\sum_{k=0}^{3} X(k)e^{jk\left(\frac{3\pi}{2}\right)}$$

$$= \frac{1}{4}[2 + (1 + j)\, e^{j\frac{3\pi}{2}} + (1 - j)\, e^{j\frac{9\pi}{2}}$$

$$= \frac{1}{4}\,[2 + (1 + j)\,(-j) + (1 - j)j]$$

$$= \frac{1}{4}\,(2 - j + 1 + j + 1]$$

= 1

the correct final term of the series.

2.3 PROPERTIES OF THE DFT

The DFT has a number of mathematical properties which can be used to simplify problems or which lead to useful applications. Some of them are listed below. The data sequences x(nT) are written x(n).

(1) Symmetry:

$$\mathrm{Re}[X(N-k)] = \mathrm{Re}\, X(k) \qquad (2.8)$$

(where Re denotes the real part) states the symmetry of the amplitude spectrum, discussed previously, and

$$\mathrm{Im}\,[X(N-k)] = -\,\mathrm{Im}\,[X(k)] \qquad (2.9)$$

(where Im denotes the imaginary part)states the antisymmetrical property of the phase spectrum. This is a property to be aware of when considering component values.

(2) Even functions:

If x(n) is an even function $x_e(n)$, that is $x_e(n) = x_e(-n)$, then

$$F_D[x_e(n)] = X_e(k) = \sum_{n=0}^{N-1} x_e(n) \cos(k\Omega nT) \tag{2.10}$$

(3) Odd functions:

If x(n) is an odd function $x_0(n)$, that is $x_0(n) = -x_0(-n)$, then

$$F_D[x_0(n)] = X_0(k) = -j \sum_{n=0}^{N-1} x_0(n) \sin(k\Omega nT) \tag{2.11}$$

(4) **Parseval's Theorem:**

The normalized energy in the signal is given by either of the expressions

$$\sum_{n=0}^{N-1} x^2(n) = \frac{1}{N} \sum_{n=0}^{N-1} |X(k)|^2 \tag{2.12}$$

the right-hand side of Equation 2.12 is the mean square spectral amplitude, while the left-hand side is the sum of the squared magnitudes of the time series.

(5) Delta function:

$$F_D[\delta(nT)] = 1 \tag{2.13}$$

(6) The linear cross-correlation of two data sequences or series may be computed using DFT_s.

(7) DFTs may also be used in the computation of circular convolutions, and, by using augmenting zeros, in linear convolutions.

2.4 COMPUTATIONAL COMPLEXITY OF THE DFT

A large number of multiplications and additions are required for the calculation of the DFT. For an 8-point DFT the expansion for X(k) becomes (from Eq. 2.4).

$$X(k) = \sum_{n=0}^{7} x(n)e^{-jk2\pi n/8}, \qquad k=0,\ldots..,7 \tag{2.14}$$

and letting $k2\pi/8 = K$ this expands to

$$X(k) = x(0)e^{-jk0} + x(1)e^{-jk1} + x(2)e^{-jk2} + x(3)e^{-jk3} + x(4)e^{-jk4} + x(5)^{-jk5}$$
$$+x(6)e^{-jk6}+x(7)e^{-jk7},\ k=0,\ldots.,7 \tag{2.15}$$

Equation 2.15 contains eight terms on the right-hand side. Each term consists of a multiplication of an exponential term which is always complex by another term which is real or complex (for example, real for a voltage time series).

Each of the product terms is added together. There are therefore eight complex multiplications and seven complex additions to be calculated. For an N-point DFT, there will be N and N - 1 of them respectively. There are also eight harmonic components to be evaluated (k = 0, . . ., 7). This number becomes N for an N -point DFT. Therefore the calculation of the eight-point DFT requires $8^2 = 64$ complex multiplications and 8 x 7 = 56 complex additions. For an N-point DFT these become N^2 and N(N-1) respectively. If N = 1024, then approximately one million complex multiplications and one million complex additions are required. Clearly some means of reducing these numbers is desirable.

The amount of computation involved may be reduced if we note that there is a considerable amount of built-in redundancy in equations such as Eq.2.4.

For example,

if k =1 and n=2,

$$e^{-jk2\pi n/8} = e^{-j\pi/2}, \text{ and}$$

if k =2 and n =1,

$$e^{-jk2\pi n/8} = e^{-j\pi/2} \text{ also.}$$

2.4.1 The decimation-in-time fast Fourier transform algorithm

In this section it will be shown how the computational redundancy inherent in the DFT is used to reduce the number of different calculations necessary, thereby speeding up the computation. For a 1024-point DFT the number of calculations required can be reduced by a factor of 204.8. The algorithms which can achieve this are given the title 'fast Fourier transform', or FFT in short. When applied in the time domain the algorithm is referred to as a decimation-in-time (DIT) FFT. The first DIT algorithm was due to Cooley and Tukey (1965), after whom it is often named. Decimation then refers to the significant reduction in the number of calculations performed on time domain data. It is noteworthy that the computational savings will be seen to increase as $N^2 - (N/2)\log_2 N$.

First:

The notation will be simplified and some mathematical relationships will be established. Thus Eq. 2.4 will be re-written as

$$X_1(k) = \sum_{n=0}^{N-1} x_n e^{-j2\pi nk/N},\ k = 0, \ldots..,N-1 \qquad (2.16)$$

Also , the factor $e^{-j2\pi/N}$ will be written as W_N, thus

$$W_N = e^{-j2\pi/N} \qquad (2.17)$$

So that Equation 2.16 becomes

$$X_1(k) = \sum_{n=0}^{N-1} x_n W_N^{kn},\ k = 0 ,\ldots\ldots..,N-1 \qquad (2.18)$$

It is worthwhile at this point to note some relationships involving W_N.

First,

$$W_N^2 = (e^{-j2\pi/N})^2$$
$$= e^{-j2\pi 2/N}$$
$$= e^{-j2\pi/(N/2)}$$
$$= W_{N/2} \qquad (2.19)$$

Second,

$$W_N^{(k+N/2)} = W_N^k W_N^{N/2} = W_N^k\ e^{-j(2\pi/N)(N/2)}$$

$= W_N^k \; e^{-j\pi}$

$= - W_N^k$ (2.20)

Summarizing the useful results concerning W_N for convenience, we have

$$W_N = e^{-j2\pi/N} \quad (2.21a)$$

$$W_N^2 = W_{N/2} \quad (2.21b)$$

$$W_N^{(k+N/2)} = - W_N^k \quad (2.21c)$$

In exploiting the computational redundancy expressed by Eqs. 2.21, the data sequence is divided into two equal sequences, one of even-numbered data, and one of odd-numbered data. For the sequences to be of equal length, they must all contain an even number of data. If he original sequence consists of an odd number of data, then an augmenting zero should be added to render the number of data even. This allows the DFT, $X_1(k)$, to be written in terms of two DFTs, $X_{11}(k)$ and $X_{12}(k)$, which are the DFTs of the even-valued data and of the odd-valued data respectively (see Table 2.1). Thus the N-point DFT is converted into two DFTs each of N/2 points. This process is then repeated until $X_1(k)$ is decomposed into N/2 DFTs, each of two points, both of which are initial data. Thus, in practice, the initial data is re-ordered and the N/2 two-point DFTs are calculated by taking the data in pairs. These DFT outputs are suitably combined in fours to provide N/4 four-point DFTs which are computed and appropriately combined to produce N/8 eight-point DFTs which are computed, and appropriately so on until the final N -point DFT, ,$x_1(k)$, is obtained. At each stage common factors which are powers of W N are incorporated to reduce the number of complex calculations. This procedure is justified as follows.

The suffixes, n, in Eq. 2.18 extend from n = 0 -to- n=N - 1, corresponding to the data values $x_0, x_1, x_2, x_3, \ldots, x_{N-1}$. The even-numbered sequence is $x_0, x_2, x_4 \ldots, x_{N-2}$ and the odd-numbered sequence is $x_1, x_3, \ldots, x_{N-1}$. Both sequences contain N/2 points.

The terms in the even sequence may be designated X2n with n=0 -to- n=N/2 -1 while those in the odd sequence become x_{2n+1}. Then Eq. 2.18 may be re-written

$$X_1(k) = \underbrace{\sum_{n=0}^{N/2-1} x_{2n} W_N^{2nk}}_{evensequence} + \underbrace{\sum_{n=0}^{N/2-1} x_{2n+1} W_N^{(2n+1)k}}_{oddsequene}$$

$$= \sum_{n=0}^{N/2-1} x_{2n}\, W_N^{2nk} + W_N^k \sum_{n=0}^{N/2-1} x_{2n+1} W_N^{2nk}, \quad k = 0, \ldots\ldots, N-1 \qquad (2.22)$$

Using Eq. 2.21b gives $W_N^{2nk} = W_{N/2}^{nk}$ so Eq. 2.22 becomes

$$X(k) = \sum_{n=0}^{N/2-1} x_{2n}\, W_{N/2}^{nk} + W_N^k \sum_{n=0}^{n=N/2-1} x_{2n+1} W_{N/2}^{nk}, \quad k=0, \ldots\ldots, N-1 \qquad (2.23)$$

Equation 2.23 may be written as

$$X_1(k) = X_{11}(k) + W_N^k X_{12}(k), \quad k=0, \ldots\ldots, N-1 \qquad (2.24)$$

On comparison of Eq. 2.24 with Eq. 2.18, it is seen that $X_{11}(k)$ is indeed the DFT of the even sequence, while $X_{12}(k)$ is that of the odd sequence. Therefore, as previously stated, the DFT, $X_1(k)$, can be expressed in terms of two DFTs: $X_{11}(k)$ and $X_{12}(k)$. The factor $W_{N/2}^k$ occurs in both $X_{11}(k)$ and $X_{12}(k)$ and needs calculation once only.

Table 2.1 illustrates the process for an 8-point DFT. Line 1 gives the data while line 2 gives an expression for the DFT of the data in terms of the DFTs of the even and odd sequences, $X_{11}(k)$ and $X_{12}(k)$ respectively. Line 3 shows the re-ordered data from which $X_{11}(k)$ and $X_{12}(k)$ are derived. Line 4 gives the DFTs of the data sequences of line 3 in terms of the DFTs of their even and odd sequences, $X_{21}(k)$, $X_{22}(k)$, $X_{23}(k)$ and $X_{24}(k)$. These sequences are shown in line 5, and are seen to be the ultimate 2-point sequences, whose DFTs are $X_{21}(k)$, $X_{22}(k)$, $X_{23}(k)$ and $X_{24}(k)$ and are expressed in terms of the data in line 6. Thus the single 8-point DFT has been decomposed into four 2-point DFTs, each one of which produces two values, for example $X_{21}(0)$ and $X_{21}(1)$ in the case of $X_{21}(k)$. This process involved two decompositions, and the weights W_N^k were squared at each step.

Table 2.1 Structure of an 8- Point FFT

Line number	*Line content*								*k ranges*	*N ranges*
1	Data sequence A_0	A_0	$x_0\ x_1\ x_2\ x_3\ x_4\ x_5\ x_6\ x_7$							0, . . ., 7
2	8-point DFT of A_0	$X_1(k) = X_{11}(k) + W_N^k X_{12}(k)$							0, . . ., $N-1$ (0, . . ., 7)	0, . . ., 7
3	Re-ordered A_0: two sequences, A_1 and A_2	A_1	$x_0\ x_2\ x_4\ x_6$			A_2	$x_1\ x_3\ x_5\ x_7$			0, . . ., 3
4	4-point DFTs of A_1 and A_2	$X_{11}(k) = X_{21}(k) + W_{N/2}^k X_{22}(k)$				$X_{12}(k) = X_{23}(k) + W_{N/2}^k X_{24}(k)$			0, . . ., $N/2-1$ (0, . . ., 3)	0, . . ., 3
5	Re-ordered sequences A_1 and A_2: four sequences A_3, A_4, A_5, A_6	A_3	$x_0\ x_4$	A_4	$x_2\ x_6$	A_5	$x_1\ x_5$	A_6	$x_3\ x_7$	0, 1
6	2-point DFTs of A_3, A_4, A_5, A_6	$X_{21}(k) = x_0 + W_{N/4}^k x_4$		$X_{22}(k) = x_2 + W_{N/4}^k x_6$		$X_{23}(k) = x_1 + W_{N/4}^k x_5$		$X_{24}(k) = x_3 + W_{N/4}^k x_7$		0, 1

Using table 2.1, considering line 6, we can reach to the following decomposing equations (2.25)

$X_{11}(0) = X_{21}(0) + X_{22}(0) = X_{21}(0) + W_8^0 X_{22}(0)$ (2.25a)

$X_{11}(2) = X_{21}(0) - X_{22}(0) = X_{21}(0) - W_8^0 X_{22}(0)$ (2.25b)

$X_{11}(1) = X_{21}(1) - jX_{22}(1) = X_{21}(1) + W_8^2 X_{22}(1)$ (2.25c)

$X_{11}(3) = X_{21}(1) + jX_{22}(1) = X_{21}(1) - W_8^2 X_{22}(1)$ (2.25d)

Inspection of these equations shows how the DFTs $X_{11}(k)$ are related to the DFTs of their even-numbered and odd-numbered data, and that $X_{11}(0)$ and $X_{11}(2)$ are given by expressions with common terms which differ only in one sign. The same is true of $X_{11}(1)$ and $X_{11}(3)$. Equations such as these are known as recomposition equations because starting from the data pairs and forming $X_{21}(k)$, $X_{22}(k)$, $X_{23}(k)$ and $X_{24}(k)$ allows $X_{11}(k)$ and $X_{12}(k)$ to be found, and hence $X_1(k)$.

The number of complex additions and multiplications involved is reduced in this way because:

(i) the re-composition equations are expressed in terms of powers of the recurring factor W_N,

(ii) use is also made of relationships of the type $X_{21}(2) = X_{21}(0)$ and $X_{21}(3) = X_{21}(1)$, and

(iii) the presence of only sign differences in the pairs of expressions is exploited. This algorithm is known as the Cooley-Tukey algorithm.

2.4.2 The butterfly

Equations 2.25 may be represented diagrammatically by exploiting the symmetry centered on the sign differences and taking the equations in pairs. Thus from Eqs. 2.25a and 2.25b the outputs of the re-compositions are $X_{11}(0)$ and $X_{11}(2)$ formed from the inputs $X_{21}(0)$ and $X_{22}(0)$. This is illustrated in Fig. 2.2(a). The inputs are at the left-hand side of the cross, the outputs to the right. Figure 2.4(b) shows how the outputs $X_{11}(1)$ and

$X_{11}(3)$ are obtain diagrammatically. By overlapping Figs. 2.2(a) and 2.4(b), a composite diagram is obtained in which the output DFTs are arranged in order of increasing k. This is shown in Fig. 2.2(c). The structure of Fig. 2.2(a) or 2.2(b) is referred to as a 'butterfly', being reminiscent of the symbolic representation of the insect. The entire 8-point FFT may be depicted in this manner as Fig. 2.3.

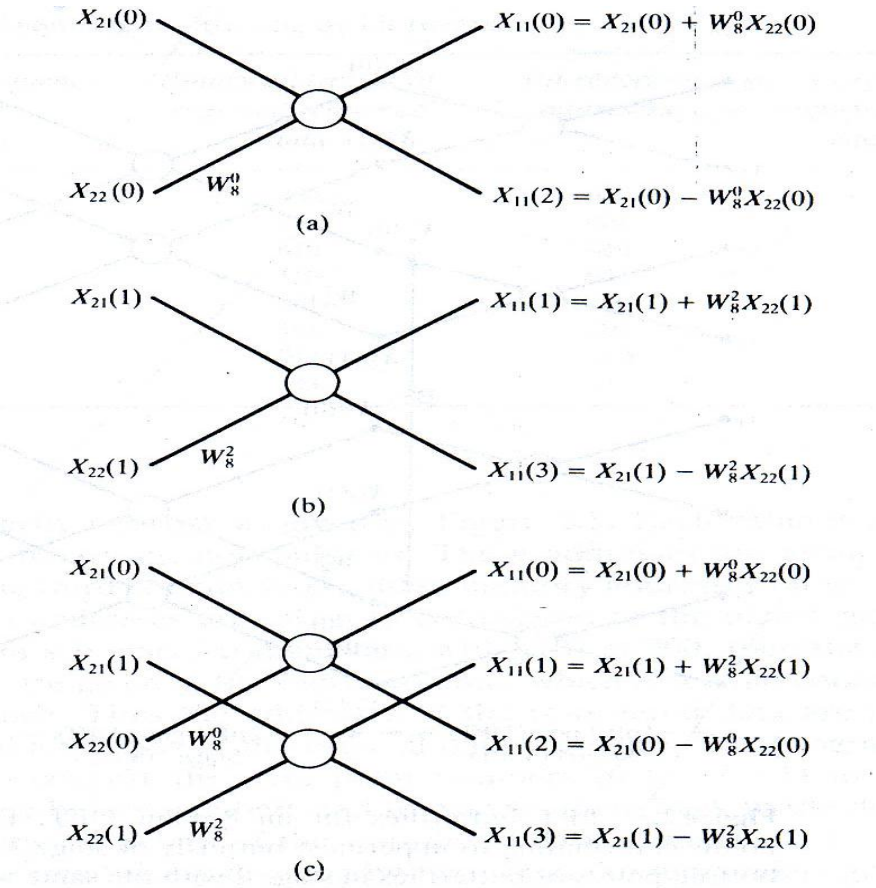


Fig. 2.2 FFT butterflies.

Example 2.3

It will now be instructive to obtain the DFT of the sequence {1, 0, 0, 1}, previously evaluated in Section 2.1, by means of the decimation-in-time FFT algorithm. This is a four-point DFT with $x_0 = 1$, $x_1 = 0$, $x_2 = 0$, $x_3 = 1$, and

$$X_1(k) = X_{11}(k) + W_N^k X_{12}(k), \qquad k = 0, 1,2,3.$$

The re-ordered sequence is x_0, x_2, x_1, x_3.

We can now utilize the top left-hand corner of Fig. 2.3 to work out the DFT. Points x_0, x_4, x_2, x_6 are replaced by x_0, x_2, x_1, x_3, and the required DFT values are $X_{11}(0)$, $X_{11}(1)$, $X_{11}(2)$ and $X_{11}(3)$.

Therefore ,

$$X_{21}(0) = x_0 + x_2 = 1 \qquad X_{21}(1) = x_0 - x_2 = 1$$

$$X_{22}(0) = x_1 + x_3 = 1 \qquad X_{22}(1) = x_1 - x_3 = -1$$

$$X_{11}(0) = X_{21}(0) + W_8^0 X_{22}(0) = 1 + 1 = 2$$

$$X_{11}(1) = X_{21}(1) + W_8^2 X_{22}(1) = 1 + e^{-j\pi/2}(-1) = 1 + j$$

$$X_{11}(2) = X_{21}(0) - W_8^0 X_{22}(0) = 1 - 1 = 0$$

$$X_{11}(3) = X_{21}(1) - W_8^2 X_{22}(1) = 1 - j$$

These values are the same as those obtained in Section 2.1, but were more readily obtained using the FFT algorithm. This conclusion is general and the computational savings increase as the number of data increases.

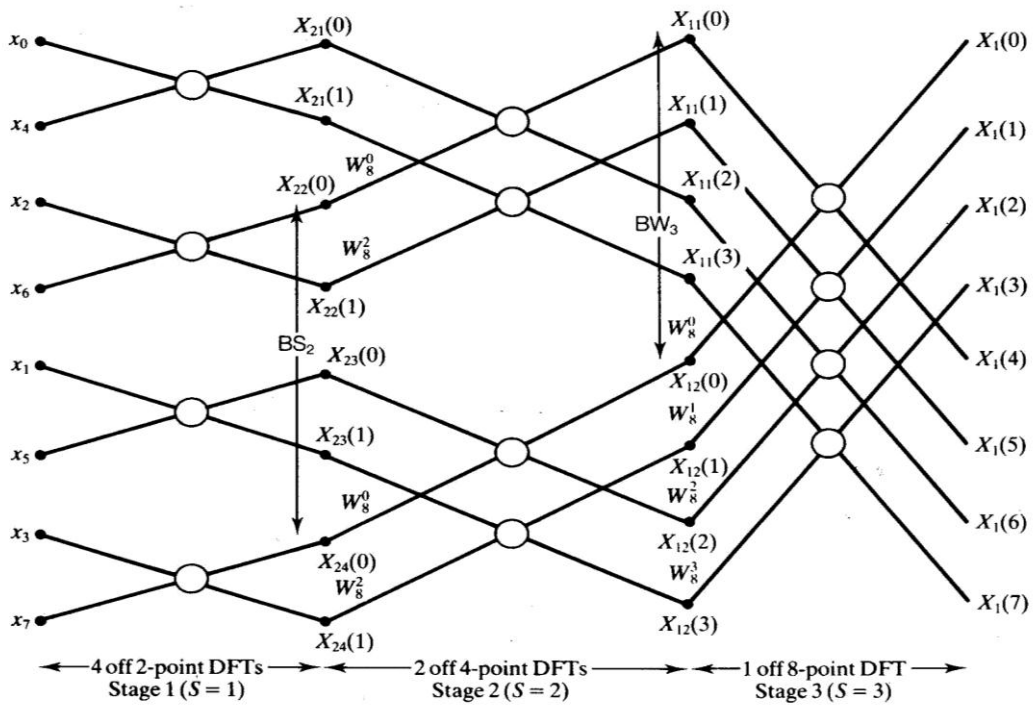


Fig. 2.3 DFT butterflies for an 8-point DFT.

2.4.3 Computational advantages of the FFT

The computational advantages of the FFT may be illustrated by considering first the FFT algorithm of Fig. 2.3. This figure shows that an N-point FFT contains N/2 butterflies per stage and $\log_2$ N stages, that is it contains a total of (N/2) $\log_2$ N butterflies. Figure 2.4(a) shows that each butterfly involves one complex multiplication of the form $W_N^R X_{ij}(k)$. Hence the FFT involves (N/2) $\log_2$ N complex multiplications compared with N^2 in the case of the DFT, as shown in Section 2.4. Thus the computational saving in complex multiplications is N^2 - (N/2) log2 N.

Each butterfly contains two complex additions so the FFT requires N $\log_2$ N complex additions compared with N (N - 1) for the DFT. Thus the saving in complex additions is N(N - 1) - N $\log_2$ N. These savings are illustrated in Table 2.2.

For a typical 1024-point DFT it is seen that the computation time will be reduced by two orders of magnitude if the FFT algorithm is employed.

2.5 INVERSE FFT ALGORITHM

An FFT algorithm for determining the inverse fast Fourier transform (IFFT) is readily obtainable from the FFT algorithm. Its use lies in transforming spectra into their corresponding waveforms and in checking that the FFT has been correctly computed by using basically the same algorithm to obtain the original data. To see how the IFFT is derived make the following substitutions in Eq. 2.4. Sum over the variable λ rather than n, let the variable k become μ, set $\Omega = 2\pi / NT$ so that the exponent of "e" becomes -jμ (2π/N)λ. Again using the notation x(λT) = x(λ) and so on. Equation 2.4 then becomes

$$X(\mu) = \sum_{\lambda=0}^{N-1} x(\lambda) e^{-j\mu(2\pi/N)\lambda} \qquad \mu = 0, 1, \ldots, N-1 \qquad (2.26)$$

Now make similar substitutions in Eq. 2.7, that is put k=λ, and n=μ so that Eq. 2.7 becomes

$$x(\mu) = \frac{1}{N} \sum_{\lambda=0}^{N-1} X(\lambda)\, e^{j\lambda(2\pi/N)\mu} \qquad \mu = 0, 1, \ldots, N-1 \qquad (2.27)$$

In the last two equations $X(\mu)$, $X(\lambda)$, $x(\lambda)$ and $x(\mu)$ are all elements of the equidimensional arrays X and x, and so it can be seen that the IFFT, $x(\mu)$, differs from the FFT, $X(\mu)$, only in the factor 1/N and the sign of the exponent. Thus with small modifications the FFT may be used to calculate the IFF/T.

Table 2.2 Saving in complex multiplications and additions when the FFT is used instead of the DFT.

	DFT		*FFT*		*Ratio of DFT multiplications to FFT multiplications*	*Ratio of DFT additions to FFT additions*
N	*Number of complex multiplications*	*Number of complex additions*	*Number of complex multiplications*	*Number of complex additions*		
2	4	2	1	2	4	1
4	16	12	4	8	4	1.5
8	64	56	12	24	5.3	2.3
16	256	240	32	64	8.0	3.75
32	1 024	992	80	160	12.8	6.2
64	4 096	4 032	192	384	21.3	10.5
128	16 384	16 256	448	896	36.6	18.1
256	65 536	65 280	1 024	2 048	64.0	31.9
512	262 144	261 632	2 304	4 608	113.8	56.8
1024	1 048 576	1 047 552	5 120	10 240	204.8	102.3
2048	4 194 304	4 192 256	11 264	22 528	372.4	186.1
4096	16 777 216	16 773 120	24 576	49 152	682.7	341.3
8192	67 108 864	67 100 672	53 248	106 496	1 260.3	630.0

Example 2.4

The first four sampled voltage values of a 10Hz bandwidth signal sampled at 125Hz were (0.5, 1, 1, 0.5). Demonstrate how the discrete Fourier transform of this sequence may be obtained using the fast Fourier transform, and hence obtain the Fourier transform of the data.

Solution

The flowgraph for the FFT is given in Fig. 2.4. We have

$$X(0) = G(0) + W_4^0 H(0) \qquad = G(0) + H(0)$$

$$G(0) = x(0) + W_2^0 x(2) \qquad = x(0) + x(2)$$

$$H(0) = x(1) + W_2^0 x(3) \qquad = x(1) + x(3)$$

Since $W^0 = 1$, substituting values gives

$$X(0) = x(0) + x(2) + x(1) + x(3)$$

$$= 0.5 + 1 + 1 + 0.5 \qquad = 3$$

$$X(1) = G(1) + W_4^1 H(1)$$

$$G(1) = x(0) - W_2^0 x(2) \qquad = x(0) - x(2)$$

$$H(1) = x(1) - W_2^0 x(3) \qquad = x(1) - x(3)$$

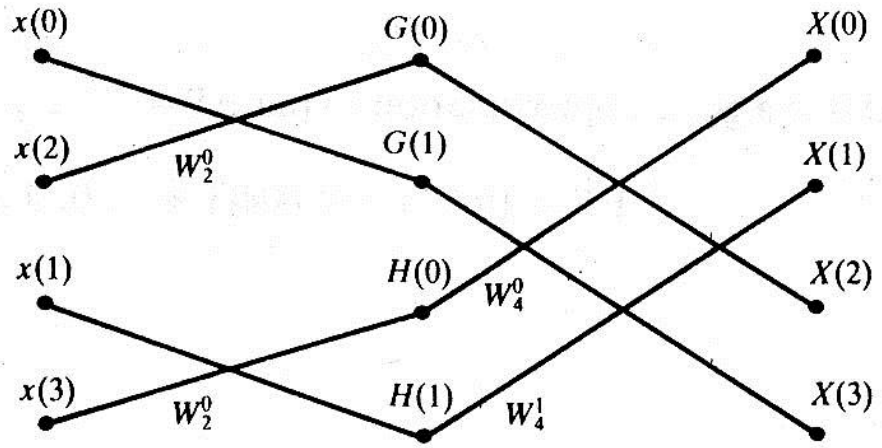


Fig. 2.4 Flowgraph of the FFT in Example 2.4.

Now, $W_N = e^{-j2\pi/N}$, and therefore

$$W_N^1 = e^{-j2\pi/4}$$

$$= e^{-j\pi/2}.$$

Substituting gives

$$X(1) = x(0) - x(2) + e^{-j\pi/2}[x(1) - x(3)]$$

$$= 0.5 - 1 + \left[\cos\left(\frac{\pi}{2}\right) - j\sin\left(\frac{\pi}{2}\right)\right](1 - 0.5)$$

$$= 0.5 - 1 + (0 - j)0.5 = -0.5 - j0.5 = -0.5(1 + j)$$

$$X(2) = G(0) - W_4^0 H(0) \qquad = G(0) - H(0)$$

$$= x(0) + x(2) - [x(1) + x(3)]$$

$$= 0.5 + 1 - (1 + 0.5) = 0$$

$$X(3) = G(1) - W_4^1 H(1)$$

$$= x(0) - x(2) - e^{-j\pi/2}[x(1) - x(3)]$$

$$= 0.5 - 1 - \left[\cos\left(\frac{\pi}{2}\right) - j\sin\left(\frac{\pi}{2}\right)\right](1 - 0.5)$$

$$= -0.5 - (-j)0.5 = -0.5 + j0.5 = 0.5(-1 + j)$$

Therefore $X(\Omega) = \{3, -0.5(1 + j), 0, 0.5(-1 + j)\}$

When T, the sampling interval, is small,

$$FT = T\,DFT$$

Where FT is the Fourier transform. Here, T = 1/125 s = 0.008 s.

The signal period is 1/10 s = 0.1 s; therefore

$$\frac{T}{period} = \frac{0.008}{0.1} = 0.08 << 1$$

and it is a good approximation to put FT = T DFT, so that

FT = { 0.024, - 0.004(1 + j), 0, 0.004(-1 +j)}

SHEET # 2

2.1 Prove that the DFT, X (k), of a sequence x (n) is periodic.

2.2 Calculate the DFT of the data sequence {0, 1, 1, 0} and check the validity of your answer by calculating its IDFT.

2.3 Use the results of prob. (2.2) to verify Parseval's theorem for the data {0,1,1,0}.

2.4 Calculate the DFT of the data sequence {0,1,1,0} using the decimation-in-time FFT algorithm. Check the answer with that of Prob. (2.2). Compare numbers of complex additions and multiplications in the two methods. Calculate the IFFT to verify your answer.

2.5 Calculate the FFT of the data sequence {0,0,1,1,1,1,0,0} and plot the amplitude and phase spectra. Check your answer by calculating the IFFT to obtain the original data sequence.

2.6 For the following signal, $X(n) = \cos(\frac{n\pi}{2})$, find the first 8-samples and then using the FFT algorithm, calculate the frequency components, X(k). Assume we use the DFT algorithm instead of the FFT algorithm, what is the computational saving.

Assignment # 2

A2.1 Calculate the FFT of the data sequence {1,0,1,0,1,0,1,0} and plot the amplitude and phase spectra. Check your answer by calculating the IFFT to obtain the original data sequence.

A2.2 Calculate the DFT of the data sequence {1,0,1,1} using the decimation-in-time FFT algorithm. What is the Computing saving?

CHAPTER 3

CORRELATION AND CONVOLUTION

3.1 CORRELATION DESCRIPTION

Consider how two data sequences, each consisting of simultaneously sampled values taken from the two corresponding waveforms, might be compared. If the two waveforms varied similarly point for point, then a measure of their correlation might be obtained by taking the sum of the products of the corresponding pairs of points. This proposal becomes more convincing when the case of two independent and random data sequences is considered. In this case the sum of the products will tend towards a vanishingly small random number as the number of pairs of points is increased. This is because all numbers, positive and negative, are equally likely to occur so that the product pairs tend to be self-cancelling on summation. By contrast, the existence of a finite sum will indicate a degree of correlation. A negative sum will indicate negative correlation, which is an increase in one variable is associated with a decrease in the other variable. The cross-correlation $r_{12}(n)$ between two data sequences $x_1(n)$ and $x_2(n)$ each containing N data might therefore be written as

$$r_{12} = \sum_{n=0}^{N-1} x_1(n)x_2(n) \tag{3.1}$$

This definition of cross-correlation, however, produces a result which depends on the number of sampling points taken. This is corrected for by normalizing the result to the number of points by dividing by N. Alternatively this may be regarded as averaging the sum of products. Thus, an improved definition is

$$r_{12} = \frac{1}{N} \sum_{n=0}^{N-1} x_1(n)x_2(n) \tag{3.2}$$

Example 3.1

The calculation of r_{12} is illustrated in the following example, in which the point numbers in the data sequences are the n, and the sequences are x_1 and x_2.

n	1	2	3	4	5	6	7	8	9
x_1	4	2	-1	3	-2	-6	-5	4	5
x_2	-4	1	3	7	4	-2	-8	-2	1

$$r_{12} = \frac{1}{9}(4\times -4 + 2\times 1 + -1\times 3 + 3\times 7 + -2\times 4 + -6\times -2 +$$

$$-5\times -8+4\times -2+5\times 1) = 5$$

However, this definition needs modification to be useful. In some cases it may indicate zero correlation although the two waveforms are 100% correlated. This may occur for example when the two waveforms are out of phase, which will often be the case. The situation is illustrated by the waveforms of Fig. 3.1. From this figure it is seen that each pair product in the correlation is zero, and hence the correlation is zero, because one of either x_1 or x_2 is always zero. However, the waveforms are clearly highly correlated, although they are out of phase. The phase difference could, for example, occur because x_1 is the reference signal while x_2 is the delayed output from a circuit. To overcome such phase differences it is necessary to shift, or lag, one of the waveforms with respect to the other. Typically x_2 is shifted to the left to align the waveforms prior to correlation. As illustrated in Fig. 3.2 this is equivalent to changing $x_2(n)$ to $x_2(n+j)$, where j represents the amount of lag which is the number of sampling points by which x_2 has been shifted to the left. An alternative, but equivalent, procedure is to shift x_1 to the right. The formula for the cross correlation thus becomes

$$r_{12}(j) = \frac{1}{N}\sum_{n=0}^{N-1} x_1(n)x_2(n+j)$$

$$= r_{21}(-j) = \frac{1}{N}\sum_{n=0}^{N-1} x_2(n)x_1(n-j) \qquad (3.3)$$

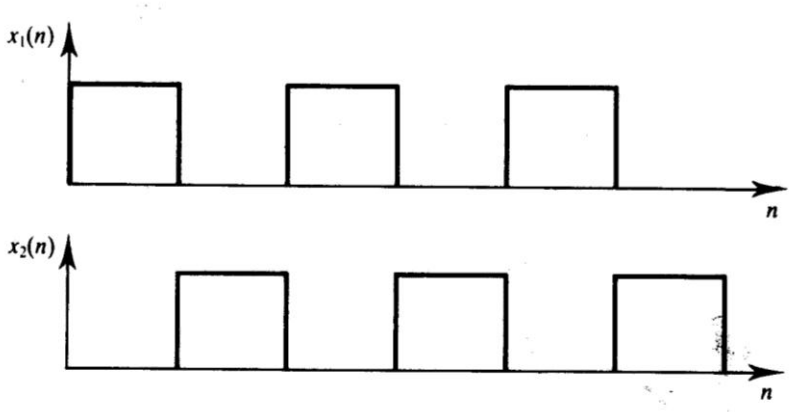


Fig. 3.1 Out-of-phase 100% correlated waveforms with zero correlation at lag zero.

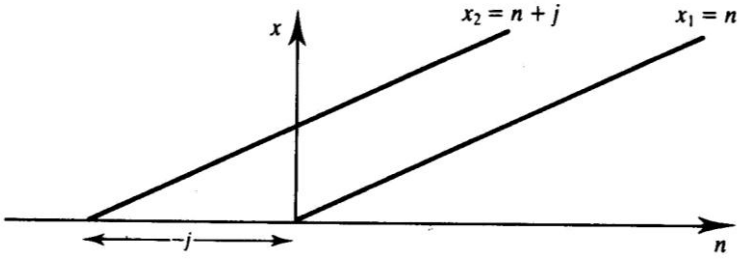


Fig. 3.2 Waveform $x_2 = x_1 + j$ shifted j lags to the left of waveform x_1.

In practice when two waveforms are correlated their phase relationship will probably not be known and so the correlation will be computed for a number of different lags in order to establish the largest value of the correlation which is then taken to be the correct value.

Example 3.2

Consider the cross-correlation of the above two sequences $x_1(n)$ and $x_2(n)$ at a lag of j = 3 , that is consider $r_{12}(3)$. The two sequences become

n	1	2	3	4	5	6	7	8	9
x_1	4	2	-1	3	-2	-6	-5	4	5
x_2	7	4	-2	-8	-2	1			

So

$$r_{12}(3) = \frac{1}{9}(4 \times 7 + 2 \times 4 + -1 \times -2 + 3 \times -8 + -2 \times -2 + -6 \times 1)$$

$$= 1.334$$

Correlation in continuous time domain:

Of course, it is also possible to consider correlation in the continuous time domain, and some analogue signal correlation is implemented this way. In the continuous domain $n \rightarrow t$ and $j \rightarrow \tau$ and

$$r_{12}(\tau) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} x_1(t) x_2(t+\tau) dt \qquad (3.4)$$

However, if $x_1(t)$ and $x_2(t)$ are periodic with period T_0 Eq. 3.4 simplifies to

$$r_{12}(\tau) = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} x_1(t) x_2(t+\tau) dt \qquad (3.5)$$

If the waveforms are of finite energy, for example non periodic pulse then, the average evaluated over the time T as $T \rightarrow \infty$ is not taken because when $1/T \rightarrow 0$ $r_{12}(\tau)$ is always vanishingly small. For this case Eq. 3.6 is used in principle:

$$r_{12}(\tau) = \int_{-\infty}^{\infty} x_1(t) x_2(t+\tau)\, dt \qquad (3.6)$$

In practice, a finite record length will be processed and so Eq. 3.7 or 3.3 will be applied:

$$r_{12}(\tau) = \frac{1}{T} \int_0^T x_1(t) x_2(t+\tau) dt \qquad (3.7)$$

There is another difficulty associated with cross-correlating finite lengths of data. This can be seen in the above example in which $r_{12}(3) = 1.334$ was determined. As x_2 is shifted to the left the waveforms no longer overlap and data at the ends of the sequences no longer form pair products. This is known as the end effect. In the example the number of pairs has dropped from nine to six for a lag of three. The result is a linear decrease in $r_{12}(j)$ as j increases, leading to debatable values of $r_{12}(j)$.

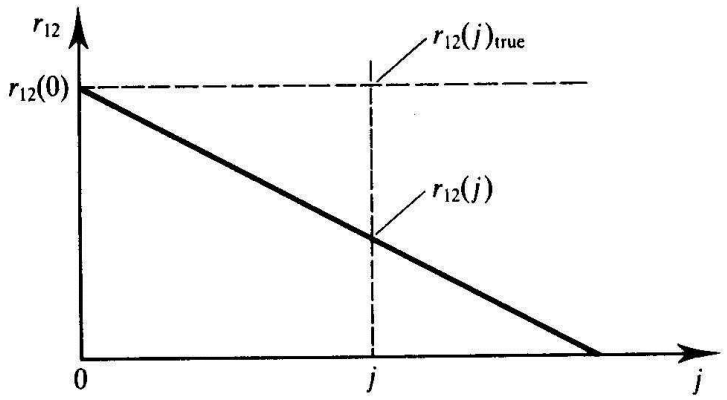


Fig. 3.3 The effect of the end-effect on the cross-correlation $r_{12}(j)$

One possible solution is to make one of the sequences twice as long as the required length for correlation. This could be achieved by recording more data, or, if one of the sequences were periodic, by repeating the sequence (taking care to match the two ends). Another possibility is to add a correction to all computed values. Figure 3.3 shows how $r_{12}(j)$ decreases with j purely as a result of the end effect, that is actual variations in $r_{12}(j)$ are not included. At j=0, $r_{12}(j) = r_{12}(0)$, which can be computed. At j = N, $r_{12}(N) = 0$, because the waveforms no longer overlap. In between, at some lag j, the true value of $r_{12}(j)$ is $r_{12}(j)_{\text{true}}$ while the actual value caused by the end effect is $r_{12}(j)$. Then, from the above figure:

$$\frac{r_{12}(j)_{\text{true}} - r_{12}(j)}{j} = \frac{r_{12}(0)}{N}$$

where,

$$r_{12}(j)_{\text{true}} = r_{12}(j) + \frac{j}{N} r_{12}(0) \qquad (3.8)$$

Computed values of the cross-correlation are therefore easily corrected for end effects by adding $jr_{12}(0)/N$ to the values of $r_{12}(j)$.

The cross-correlation values computed according to the above formulae depend on the absolute values of the data. It is often necessary to measure cross-correlations according to the fixed scale between –1 and +1. This can be achieved by normalizing the values by an amount depending on the energy content of the data. For example, consider the two pairs of waveforms $x_1(n)$, $x_2(n)$, and $x_3(n)$, $x_4(n)$. The data values are given in the table below:

n	0	1	2	3	4	5	6	7	8
$x_1(n)$	0	3	5	5	5	2	0.5	0.25	0
$x_2(n)$	1	1	1	1	1	0	0	0	1
$x_3(n)$	0	9	15	15	15	6	1.5	0.75	0
$x_4(n)$	2	2	2	2	2	0	0	0	2

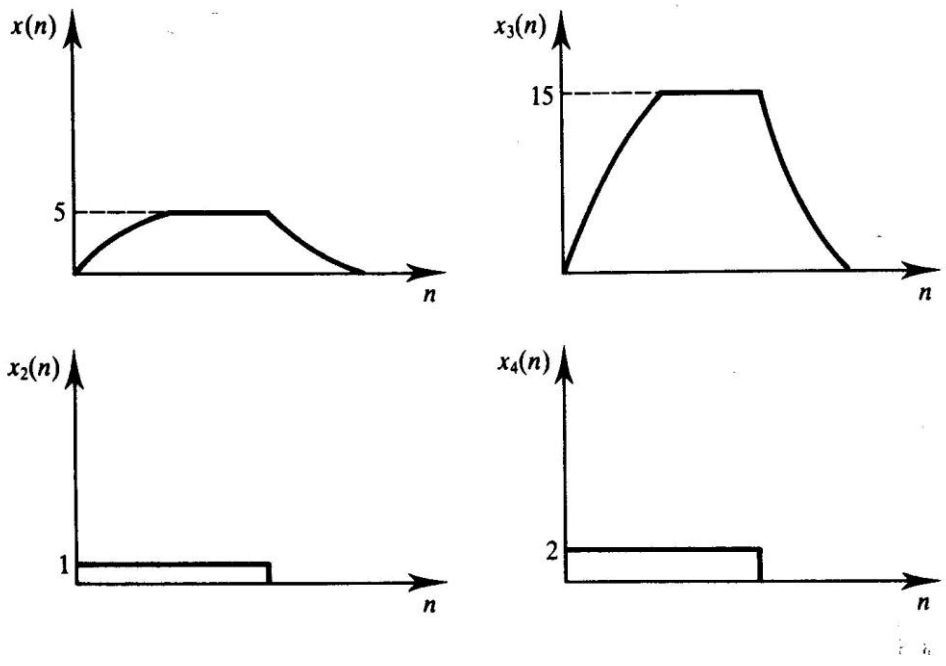


Fig. 3.4 Pairs of waveforms $\{x_1(n), x_2(n)\}$, $\{x_3(n), x_4(n)\}$ of different magnitudes but equal cross-correlations.

As may be seen from Fig. 3.4, waveforms $x_1(n)$ and $x_3(n)$ are alike, differing only in magnitude. The same is true of the pair $x_2(n)$ and $x_4(n)$. The correlation between $x_1(n)$ and $x_2(n)$ is therefore the same as that between $x_3(n)$ and $x_4(n)$. However, the cross-correlations $r_{12}(1)$ and $r_{34}(1)$ are 1.47 and 8.83 respectively. They are different because they depend on the absolute values of the data. This situation can be rectified by normalizing the cross-correlation $r_{12}(j)$ by the factor

$$\left[\frac{1}{N}\sum_{n=0}^{N-1} x_1^2(n) \times \frac{1}{N}\sum_{n=0}^{N-1} x_2^2(n)\right]^{1/2} = \frac{1}{N}\left[\sum_{n=0}^{N-1} x_1^2(n) \sum_{n=0}^{N-1} x_2^2(n)\right]^{1/2} \tag{3.9}$$

and similarly for $r_{34}(j)$. The normalized expression for $r_{12}(j)$ then becomes

$$\rho_{12}(j) = \frac{r_{12}(j)}{\frac{1}{N}\left[\sum_{n=0}^{N-1} x_1^2(n) \sum_{n=0}^{N-1} x_2^2(n)\right]^{1/2}} \tag{3.10}$$

$\rho_{12}(j)$ is known as the cross-correlation coefficient. Its value always lies between –1 and +1. +1 means 100% correlation in the same sense, -1 means 100% correlation in the opposing sense, for example signals in anti-phase. A value of 0 signifies zero correlation. This means the signals are completely independent. This would be the case, for example, if one of the waveforms were completely random. Small values of $\rho_{12}(j)$ indicate very low correlation.

The normalizing factor for $r_{12}(j)$ in the above illustration is

$$\frac{1}{N}\left[\sum_{n=0}^{N-1} x_1^2(n) \sum_{n=0}^{N-1} x_2^2(n)\right]^{1/2} = \frac{1}{9}(88.31 \times 6)^{1/2} = 2.56$$

and for $r_{34}(j)$ it is

$$\frac{1}{N}\left[\sum_{n=0}^{N-1} x_3^2(n) \sum_{n=0}^{N-1} x_4^2(n)\right]^{1/2} = \frac{1}{9}(794.8 \times 24)^{1/2} = 15.34$$

Therefore

$$\rho_{12}(1) = \frac{r_{12}(1)}{2.56} = \frac{1.47}{2.56} = 0.57$$

and

$$\rho_{34}(1) = \frac{r_{34}(1)}{15.34} = \frac{8.83}{15.34} = 0.57$$

Now $\rho_{12}(1) = \rho_{34}(1)$ which demonstrates that this normalization process indeed allows a comparison of cross-correlations independently of the absolute data values.

The Autocorrelation function:

A special case occurs when $x_1(n) = x_2(n)$. The waveform is then cross-correlated with itself. This process is known as autocorrelation. The auto-correlation of a waveform is given by

$$r_{11}(j) = \frac{1}{N}\sum_{n=0}^{N-1} x_1(n)x_1(n+j)$$

The autocorrelation function has one very useful property in that

$$r_{11}(0) = \frac{1}{N}\sum_{n=0}^{N-1} x_1^2(n) = S$$

where S is the normalized energy of the waveform. This provides a method for calculating the energy of a signal. If the waveform is completely random, for example corresponding to that of white, Gaussian noise in an electrical system then the autocorrelation will have its peak value at zero lag and will reduce to a random fluctuation of small magnitude about zero for lags greater than about unity (see Fig. 3.5.). This constitutes a test for random waveforms. It is also true that

$$r_{11}(0) \geq r_{11}(j)$$

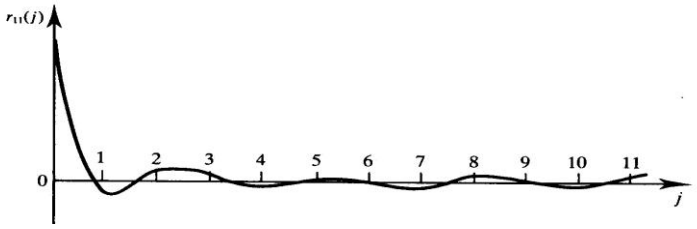


Fig 3.5 Autocorrelation function of a random waveform

3.2 CROSS CORRELATION

Care has to be exercised when cross-correlating two unequal length sequences when they are periodic. This is because the result of the correlation will be cyclic with the period of the shorter sequence. This result does not represent the full periodicity of the longer sequence and is, therefore, incorrect. This may be demonstrated by cross-correlating the sequences a = {4, 3, 1, 6} and b = {5, 2, 3} to obtain $r_{ab}(j)$. The sequence b is placed below sequence a, and b is shifted left by one lag on each of the subsequent rows, with the value of the cross-correlation appearing in the final column on the right.

Sequence				Lag	r_{ab} (j)
4	3	1	6		
3	5	2	3	0	47
5	2	3	5	1	59
2	3	5	2	2	34
3	5	2	3	3	47 $r_{ab}(j)$ repeats
5	2	3	5	4	59
etc					

The result shows that $r_{ab}(j)$ is cyclic, repeating every third lag, that is $r_{ab}(j)$ has the same period as that of the shorter sequence, b. This procedure is known as cyclic correlation. To obtain the correct value in which each value in a is multiplied by each value in b, all the elements in b have to be shifted in turn below each value in a as shown below:

```
4 3 1 6
      5 2 3
    5 2 3
  5 2 3
5 2 3
  5 2 3
    5 2 3
      5 2 3
```

This is seen to require 6 lags before the "b" sequence repeats. The sequence lengths are 4 and 3 and the number of lags necessary is 4 + 3 – 1 = 6. This reveals the general rule for obtaining the linear cross-correlation of two periodic sequences of lengths N_1 and N_2 : add augmenting zeros to each sequence to make the lengths of each sequence $N_1 + N_2 - 1$. This may be expressed as adding $N_2 - 1$ zeros to the sequence of length N_1 and adding $N_1 - 1$ zeros to the sequence of length N_2. This is now demonstrated for the given sequences "a" and "b":

Sequence						Lag	R_{ab} (j)
4	3	1	6	0	0		
5	2	3	0	0	0	0	29
2	3	0	0	0	5	1	17
3	0	0	0	5	2	2	12
0	0	0	5	2	3	3	30
0	0	5	2	3	0	4	17
0	5	2	3	0	0	5	35
5	2	3	0	0	0	6	29 r_{ab}(j) repeats
Etc							

Thus, the required linear cross-correlation of a and b is

$$r_{ab}(j) = \{29, 17, 12, 30, 17, 35\}$$

Analytical cross-correlation:

So far, the instances of cross-correlation taken have all assumed digitized data, but cross-correlation may also be performed analytically when analytical expressions can be written for the waveforms, including when this requires sectioning of the waveforms. In practice the analytical procedure has its equivalent in the use of analogue circuits to effect the cross-correlation. An example of analytical cross-correlation follows.

Example 3.3

Obtain the cross-correlation $r_{12}(-\tau)$ between the waveforms $\upsilon_1(t)$ and $\upsilon_2(t)$ of Fig. 3.6.

Solution

It is easy to express the waveforms analytically by dividing them into straight-line sections. It is only necessary to do this over one period, T, of the waveforms because $r_{12}(-\tau)$ will be periodic in τ with period T. Therefore, for $0 \le t \le T$, $\upsilon_1(t)$ = t/T, and for $0 \le t \le T/2$, $\upsilon_2(t)$ = 1.0, while for T/2 $\le$ t $\le$ T, $\upsilon_2(t)$ = -1.0.

- The requirement is to obtain an expression for $r_{12}(-\tau)$, that is $\upsilon_2(t)$, the rectangular waveform, is to be shifted right w.r.t. $\upsilon_1(t)$.
- **For** $0 \le \tau \le T/2$**,** the situation is described by Fig. 3.7 which shows that $\upsilon_1(t)$ has to be multiplied by three consecutive section of $\upsilon_2(t)$ in which $\upsilon_2(t)$ has the consecutive value –1, 1, -1.
- **For T/2** $\le \tau \le T$**,** Fig. 3.8 applies in which the consecutive values of the set of $\upsilon_2(t)$ have changed to 1, -1, +1. This means there are two parts to the solution which must match at τ = T/2.

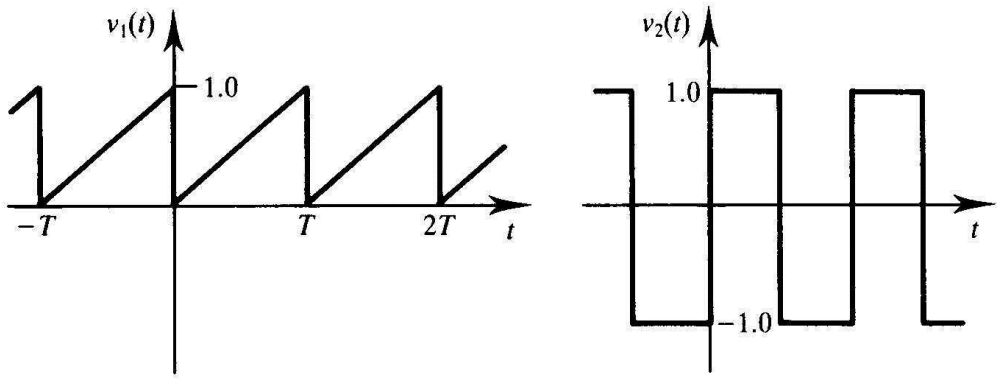


Fig. 3.6 The waveforms $\upsilon_1(t)$ and $\upsilon_2(t)$ for cross-correlation example.

Referring to Fig. 3.7, the cross-correlation is split into the three sections with boundaries at t = τ, t = τ+T/2, and t = T. Hence

$$r_{12}(-\tau) = \frac{1}{T}\int_0^T \upsilon_1(t)\upsilon_2(t-\tau)\,dt$$

$$= \frac{1}{T}\int_0^{\tau} \frac{t}{T}(-1)\,dt + \frac{1}{T}\int_{\tau}^{\tau+T/2} \frac{t}{T}(1)dt + \frac{1}{T}\int_{\tau+T/2}^{T} \frac{t}{T}(-1)dt$$

$$= \frac{-1}{T^2}\left[\frac{t^2}{2}\right]_0^{\tau} + \frac{1}{T^2}\left[\frac{t^2}{2}\right]_{\tau}^{\tau+T/2} - \frac{1}{T^2}\left[\frac{t^2}{2}\right]_{\tau+T/2}^{T}$$

$$r_{12}(-\tau) = -\frac{1}{4} + \frac{\tau}{T} \text{ for } 0 \le \tau \le \frac{T}{2} \qquad (3.11)$$

For $T/2 \le \tau \le T$, and referring to Fig. 3.8, it is seen that

$$r_{12}(-\tau) = \frac{1}{T}\int_0^{\tau-T/2} \frac{t}{T}(1)dt + \frac{1}{T}\int_{\tau-T/2}^{\tau} \frac{t}{T}(-1)dt + \frac{1}{T}\int_{\tau}^{T} \frac{t}{T}(1)dt$$

$$r_{12}(-\tau) = \frac{3}{4} - \frac{\tau}{T} \text{ for } \frac{T}{2} \le \tau \le T \qquad (3.12)$$

Substituting τ = T/2 into Equations 3.11 and 3.12 gives $r_{12}(-\tau)$ = 1/4 in both cases, confirming that the two functions match correctly. Figure 3.9 shows a plot of $r_{12}(-\tau)$ versus τ for $0 \le \tau \le T$.

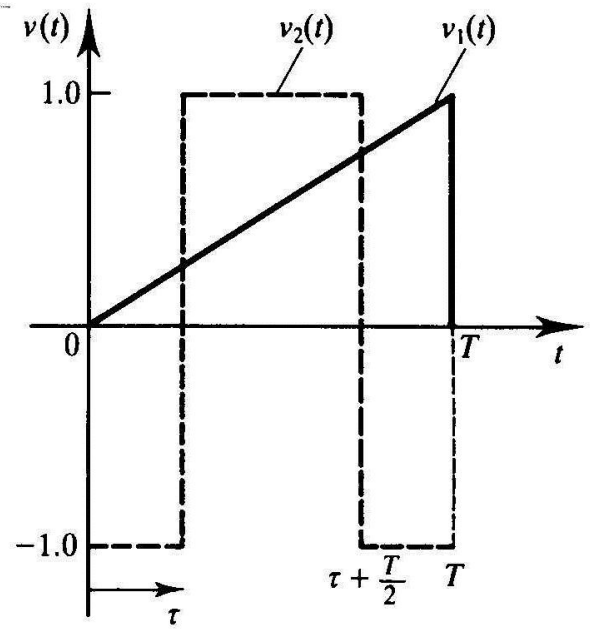


Fig. 3.7 Section of $\upsilon_2(t)$ for $0 \le \tau \le T$

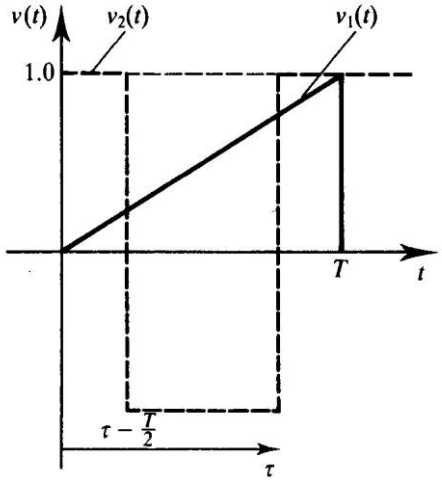


Fig. 3.8 Sections of $\upsilon_2(t)$ for $T/2 \le \tau \le T$.

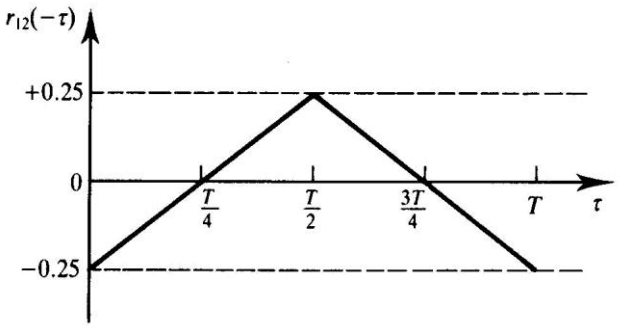


Fig. 3.9 $r_{12}(-\tau)$ as a function of τ.

3.3 FAST CORRELATION

The correlation computation may be speeded up by exploiting the correlation theorem, usually stated as

$$r_{12}(j) = F_D^{-1}[X_1^*(k)X_2(k)] \qquad (3.13)$$

but which is correctly written as

$$r_{12}(j) = \frac{1}{N} F_D^{-1}\left[X_1^*(k)X_2(k)\right] \qquad (3.14)$$

where F_D^{-1} denotes the inverse discrete Fourier transform. This approach requires computation of two discrete Fourier transforms (DFTs) and one inverse DFT, each of which is most easily executed using an FFT algorithm (see Chapter 2). If the number of terms in the sequences is sufficiently large, it is quicker to use this FFT method than to calculate the cross-correlation directly.

Example 3.4

Find the cross-correlation of the two sequences $x_1(n)$ and $x_2(n)$ below by applying the correlation theorem:

$$x_1(n) = \{1,0,0,1\}$$
$$x_2(n) = \{0.5,1,1,0.5\}$$

Solution

First use the correlation theorem, Eq. 3.14. $X_1(k)$ was found in Section 2.5 to be

$$X_1(k) = 2,1 + j, 0, 1 - j$$

and so

$$X_1^*(k) = 2, 1 - j. 0, 1 + j$$

$X_2(k)$ is most easily obtained using the FFT algorithm. Thus, with $x_0 = 0.5$, $x_2 = 1$, $x_1 = 1$, and $x_3 = 0.5$,

$$X_{21}(0) = x_0 + x_2 = 1.5$$
$$X_{21}(1) = x_0 - x_2 = - 0.5$$
$$X_{22}(0) = x_1 + x_3 = 1.5$$

$X_{22}(1) = x_1 – x_3 = 0.5$

$X_{11}(0) = X_{21}(0) + X_{22}(0) = 3$

$X_{11}(1) = X_{21}(1) + (-j)X_{22}(1) = -0.5 –j0.5$

$X_{11}(2) = X_{21}(0) – X_{22}(0) = 0$

$X_{11}(3) = X_{21}(1) – (-j)X_{22}(1) = -0.5 + j0.5$

Bringing the values of the FFTs together gives

$X_1^*(k) = 2, 1 – j, 0, 1+j$

$X_2(k) = 3, - 0.5 – j0.5, 0, -0.5 + j0.5$

So,

$X_1^*(1)X_2(1) = 2 \times 3 = 6$

$X_1^*(2)X_2(2) = (1 – j)(-0.5-j0.5) = -1$

$X_1^*(3)X_2(3) = 0 \times 0 = 0$

$X_1^*(4)X_2(4) = 0.5\ (1+j)(-1 + j) = -1$

Thus

$$[X_1^*(k)\ X_2(k)] = 6, -1, 0, -1$$

It is now necessary to take the inverse DFT (IDFT) of this. We know that, the IDFT is obtainable by changing the signs of the exponents (in the weighting factors W_N) in the above FFT algorithm and dividing the result by N. Hence, and without changing the notation in the algorithm,

$X_{21}(0) = x_0 + x_2 = 6$

$X_{21}(1) = x_0 – x_2 = 6$

$X_{22}(0) = x_1 + x_3 = -2$

$X_{22}(1) = x_1 – x_3 = 0$

$X_{11}(0) = X_{21}(0) + X_{22}(0) = 4$

$X_{11}(1) = X_{21}(1) + jX_{22}(1) = 6$

$X_{11}(2) = X_{21}(0) – X_{22}(0) = 8$

$X_{11}(3) = X_{21}(1) – jX_{22}(1) = 6$

The components of

$$F_D^{-1}\left[X_1^*(k)X_2(k)\right]$$

are obtained by dividing the values of $X_{11}(0)$, $X_{11}(1)$, $X_{11}(2)$, and $X_{11}(3)$ by N = 4.

Thus

$$F_D^{-1}\left[X_1^*(k)X_2(k)\right] = 1, 1.5, 2, 1.5$$

Now , from Equation 3.14,

$$r_{12}(j) = \frac{1}{4}F_D^{-1}\left[X_1^*(k)X_2(k)\right] = \{0.25, 0.375, 0.5, 0.375\} \qquad (3.15)$$

This correlation will be circular because all the data is periodic with period N.
The cross-correlation $r_{12}(j)$ may be worked out directly to be:

$r_{12}(0) = (1 \times 0.5 + 0 + 0 + 1 \times 0.5)/4 = 0.25$
$r_{12}(1) = (1 \times 1 + 0 + 0 + 1 \times 0.5)/4 = 0.375$
$r_{12}(2) = (1 \times 1 + 0 + 0 + 1 \times 1)/4 = 0.5$
$r_{12}(3) = (1 \times 0.5 + 0 + 0 + 1 \times 1)/4 = 0.375$

The next value, $r_{12}(4)$, is 0.25, the same as $r_{12}(0)$, and the sequence repeats periodically. This is circular correlation, as discussed in Section 3.2, and this result agrees with that derived above using the correlation theorem. The correlation theorem can be used to obtain the linear correlation by adding augmenting zeros to the two sequences as explained in Section 3.2. Thus, if the sequence lengths are N_1 for $x_1(n)$ and N_2 of $x_2(n)$, then $N_2 - 1$ zeros are added to $x_1(n)$ and $N_1 - 1$ zeros are added to $x_2(n)$. The cross-correlation is then computed using these two augmented sequences. This method of evaluating cross-correlations by using the correlation theorem and FFTs is known as fast correlation.

3.4 CONVOLUTION DESCRIPTION

The term convolution describes, amongst other things, how the input to a system interacts with the system to produce the output. Generally, the system output will be a delayed and attenuated or amplified version of the input. It is particularly useful to consider the output from the system owing to an impulse input. This is because any input may be represented as a sequence of impulses of different strengths. The output of the system owing to the impulse input will not be a corresponding impulse, but will vary with time, passing through a maximum, as shown in Fig. 3.10. This figure shows that at sampling instant m the output owing to the unit impulse applied at sampling instant 0 is h(m). The characteristic is known as the impulse response h(m) of the system.

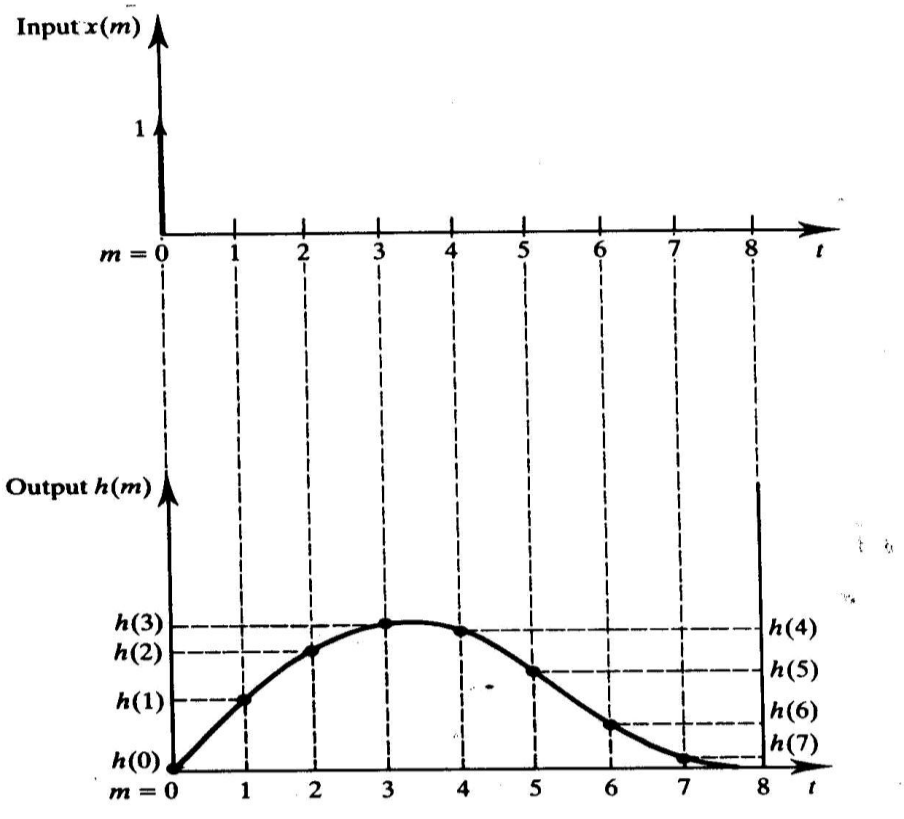


Fig. 3.10 Impulse input and corresponding impulse response of a system

Consider now the application of a sequence of impulses x(m) to the system, applied at the sampling instants m. Referring to Fig. 3.11, the output at instant m= 0 is y(0) given by:

$$y(0) = h(0)x(0)$$

At the sampling instant m = 1 the output will be given by h(0)x(1), the effect of the current input x(1), plus the delayed effect h(1)x(0) of the input applied at sampling instant m = 0. Thus

$$y(1) = h(1)x(0) + h(0)x(1)$$

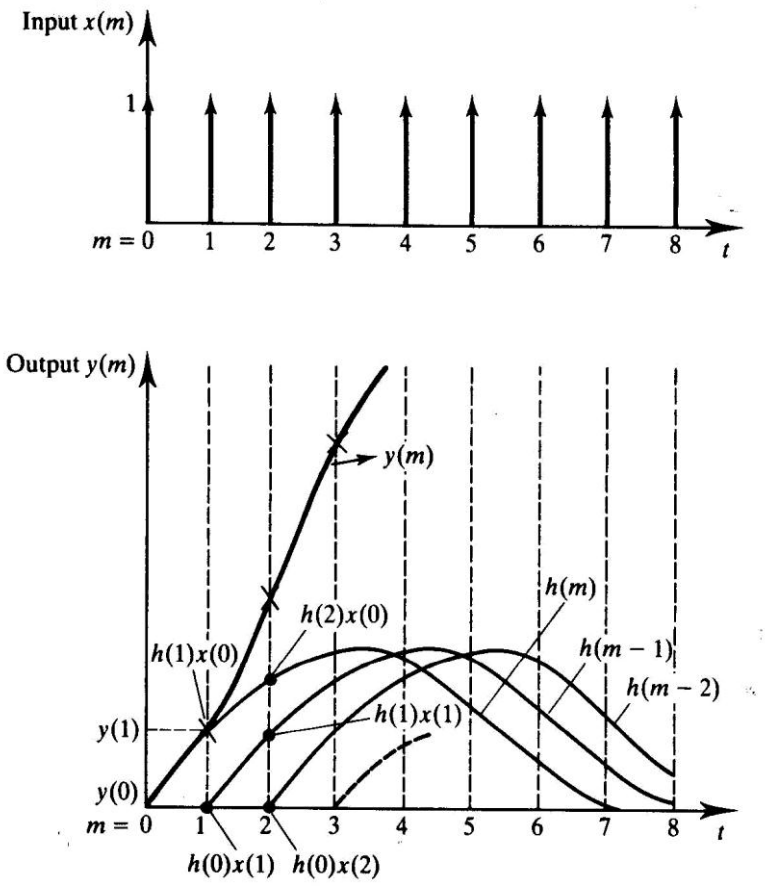


Fig. 3.11 Applied impulse sequence and system response derived from the individual impulse responses.

Similarly, subsequent outputs will be given by

$$y(2) = h(2)x(0) + h(1)x(1) + h(0)x(2)$$

$$y(3) = h(3)x(0) + h(2)x(1) + h(1)x(2) + h(0)x(3)$$

.

.

$$y(n) = h(n)x(0) + h(n–1)x(1) +\ldots+ h(0)x(n) \qquad (3.16)$$

The output may only be written in this manner as a linear sum of the effects of previous inputs if the system is a linear one. Equation 3.16 describes the output of a first-order linear system.

Inspection of the above expressions reveals that the output is obtained by multiplying the input sequence by the corresponding points of the time-reversed impulse response function. Alternatively, since Equation 3.16 may equally well be written as

$$y(n) = h(0)x(n) + h(1)x(n - 1) + \ldots + h(n)x(0) \tag{3.17}$$

the output may be regarded as the product of the corresponding pairs of points in the impulse response function by the time-reversed input sequence. Thus the convolution sum is equivalent to the cross-correlation of the one sequence by the time-reversed second sequence.

Equations 3.16 and 3.17 may be written compactly as

$$y(n) = \sum_{m=0}^{n} h(n - m)x(m) \tag{3.18}$$

and

$$y(n) = \sum_{m=0}^{n} h(m)x(n - m) \tag{3.19}$$

These latter two functions are referred to as the convolution sums of the inputs by the impulse response function, and the output is said to be given by the convolution of the input by the impulse response of the system.

Equations 3.18 and 3.19 may be extended to waveforms of infinite duration by writing them as

$$y(n) = \sum_{m=-\infty}^{\infty} x(m)h(n - m) = x(n) \otimes h(n) \tag{3.20}$$

and

$$y(n) = \sum_{m=-\infty}^{\infty} h(m)x(n - m) = h(n) \otimes x(n) \tag{3.21}$$

the generalized form of the convolution sum. In these equations the symbol $\otimes$ signifies the operation of convolution.

If the input consists of a continuous sequence of impulses the above summations may be replaced by integrals, so, for example, Eq. 3.20 becomes

$$y(t) = \int_{-\infty}^{\infty} x(\lambda)\, h(t - \lambda) d\lambda \qquad (3.22)$$

which is known as the convolution integral.

So far the term convolution has been taken to describe the result of convolving together the impulse response of a system and the input to the system. However, the idea can be extended to the convolution of any two sets of data and the term will henceforth be considered in this more general sense.

Example 3.5

Find the convolution of the two periodic time sequences (4,3,2,1{h(m)}) and (1,2,3,4{x(m)} will now be convolved shown in Fig. 3.12.

Solution:

Figure 3.12(a) shows the periodic sequence (4,3,2,1 {h(m)}) and Fig. 3.12(b) the time-reversed sequence h(-m) which has become (1, 2, 3, 4). (Recall that the convolution sum requires that one sequence has to be multiplied point by point by the time-reversed second sequence, which is it corresponds to the cross-correlation of the one sequence by the time reversal of the other.) The figure also shows a window of width equal to one period over which the convolution is evaluated. Clearly the result obtained will be periodic as for the corresponding case of cyclic correlation (Section 3.2) and so it is only necessary to evaluate the convolution over the windowed interval. Figure 3.12(f) shows the second sequence (1, 2, 3, 4 {x(m)}) for reference.

Now, when n = 0, Eq. 3.41 becomes

$$y(0) = \sum_{m=0}^{n} h(-m)x(m)$$

and is obtained by cross-correlating the windowed data in Figs. 3.12(b) and 3.12(f) :

$$y(0) = 4*1 + 1*2 + 2*3 + 3*4 = 24$$

When n = 1 Equation 3-41 becomes

$$y(1) = \sum_{m=0}^{n} h(1 - m)x(m)$$

and is obtained by cross-correlating the windowed data in Figures 3.12(c) and 3.12(f), giving

$$y(1) = 3 * 1 + 4*2 + 1 * 3 + 2 * 4 = 22$$

Similarly, it is found that

$$y(2) = 2*1 + 3*2 + 4*3 + 1*4 = 24$$

and

$$y(3) = 1*1 + 2* 2 + 3*3 + 4* 4 = 30$$

where after the output sequence repeats cyclically. This output sequence is shown in Fig. 3.12(g).

When the waveforms are well defined mathematically the convolution may be performed analytically. By considering a similar example, and also illustrating the steps involved graphically, it is possible to obtain a better understanding of the convolution process.

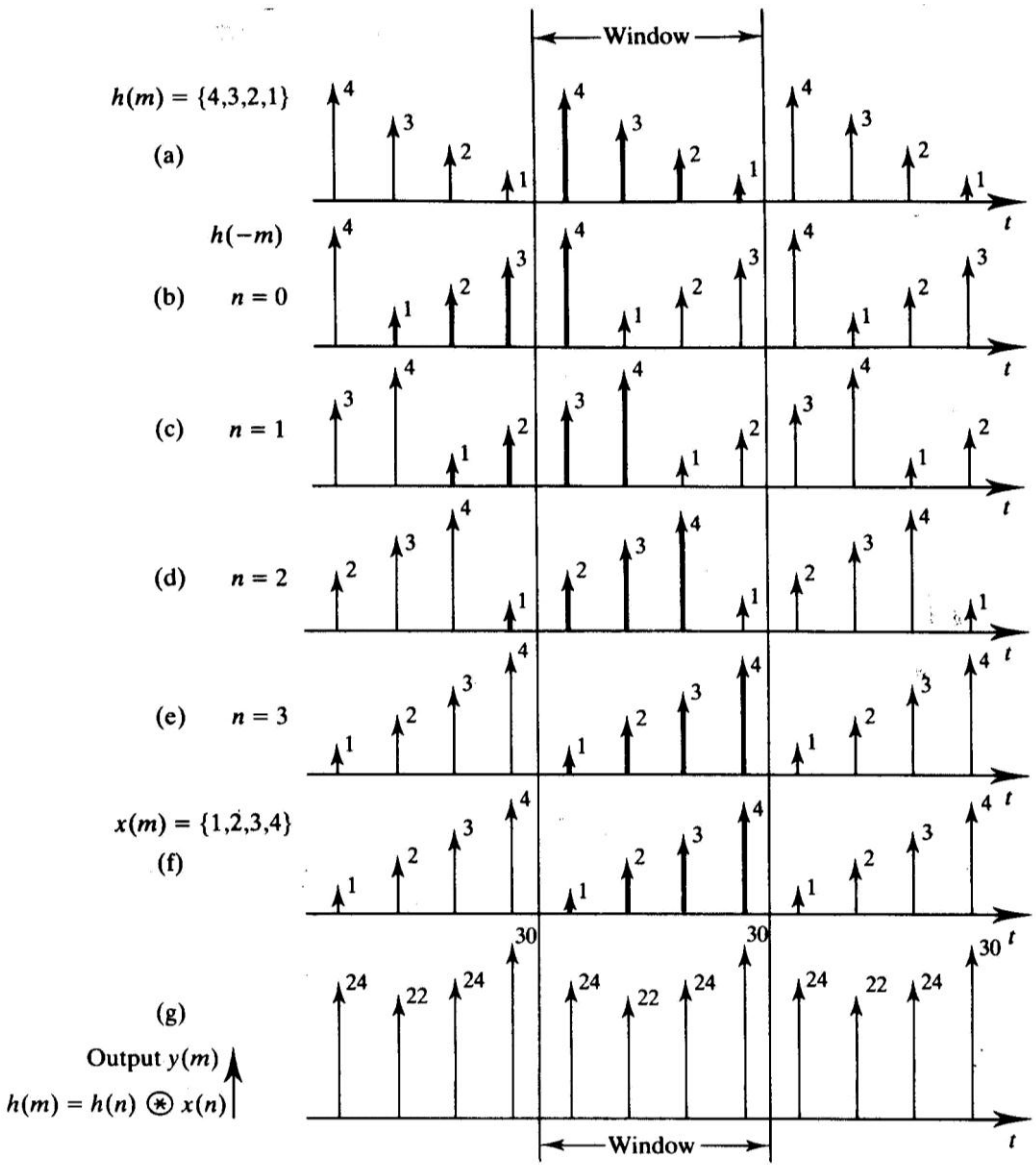


Fig. 3.12 The convolution y(m) of h(n) and x(n). (a) The periodic sequence h(m). (b) The time-reversed sequence h(-m). (c) to (e) Versions of h(-m) shifted right by incremental lags. (f) The sequence x(m). (g) The output sequence y(m) = h(n) $\otimes$ x(n).

Example 3.6

Convolve the waveforms x(t) and h(t) of Fig. 3.13(a) analytically.

Solution

Let the convolution integral be

$$y(t) = x(t) \otimes h(t) = \int_{-\infty}^{\infty} x(\tau)h(t - \tau)\, d\tau \qquad (3.23)$$

Equation 3.23 corresponds to Equation 3.22 in which the variable λ has been replaced by τ to indicate that time lags are being applied. The convolution integral depends on the variable τ and so Fig. 3.13(a) has to be replaced by Fig. 3.13(b).

It is now necessary to time reverse $h(\tau)$ as shown in Fig. 3.13 (c). $h(-\tau)$ in next shifted with respect to $x(\tau)$ in the direction of positive τ. The resulting waveform $h(t - \tau)$ then overlaps $x(\tau)$ in five separate geometrical stages as shown in Figs. 3.13(d), 3.13(e), 3.13(g), 3.13(h), and 3.13(i). For each of these stages there is a corresponding convolution integral. Thus $x(t) \otimes h(t)$ exists as five separate contiguous regions.

- Stage1:

$t < 0$ and $h(t - \tau)$ does not overlap $x(\tau)$ (Fig. 3.13(d)). As the functions do not overlap $x(\tau)\, h(t - \tau) = 0$ for all t and there is no contribution to the convolution integral.

- Stage2:

$0 < t \leq 2$ and partial overlap occurs between $h(t - \tau)$ and $x(\tau)$ (Fig. 3.13(e)). Over this range

$$y(t) = \int_{\tau=0}^{\tau=t} x(\tau)\, h(t-\tau)\, d\tau = \int_{\tau=0}^{\tau=t} (3) \times (2) d\tau$$

$$y(t) = 6\left[\tau\right]_0^t = 6t, \qquad 0 < t \leq 2 \qquad (3.24)$$

This geometrical stage terminates when $t = 2$, as shown in Fig. 3.13(f).

- Stage3:

$2 \le t \le 3$ and there is complete overlap of h(t - τ) and x(τ) (Fig. 3.13(g)). Over this range of t,

$$y(t) = \int_{\tau=t-2}^{t} (3) \times (2) d\tau = 6[\tau]_{t-2}^{t}$$

$$y(t) = 6(t - t + 2) = 12, \quad 2 \le t \le 3 \qquad (3.25)$$

- Stage4:

$3 \le t \le 5$. This is another type of overlap region shown in Fig. 3.13(h):

$$y(t) = \int_{\tau=t-2}^{\tau=3} (3) \times (2)\, d\tau = 6[\tau]_{t-2}^{3}$$

$$= 6 (5 - t) = 30 - 6t \qquad (3.26)$$

- Stage5:

t > 5. As seen in Fig. 3.13(i) this is a second region of no overlap, and so there is no contribution to the convolution integral.

Stages 2 to 4 thus each make a contribution to the convolution integral, with the convolution integral having a different expression for each of the three regions corresponding to the three stages, as summarized below:

$0 < t \le 2$	y(t) = 6t
$2 \le t \le 3$	y(t) = 12
$3 \le t \le 5$	y(t) = 30 – 6t

From these expressions y(t) may be plotted against t as shown in Fig. 3.14.

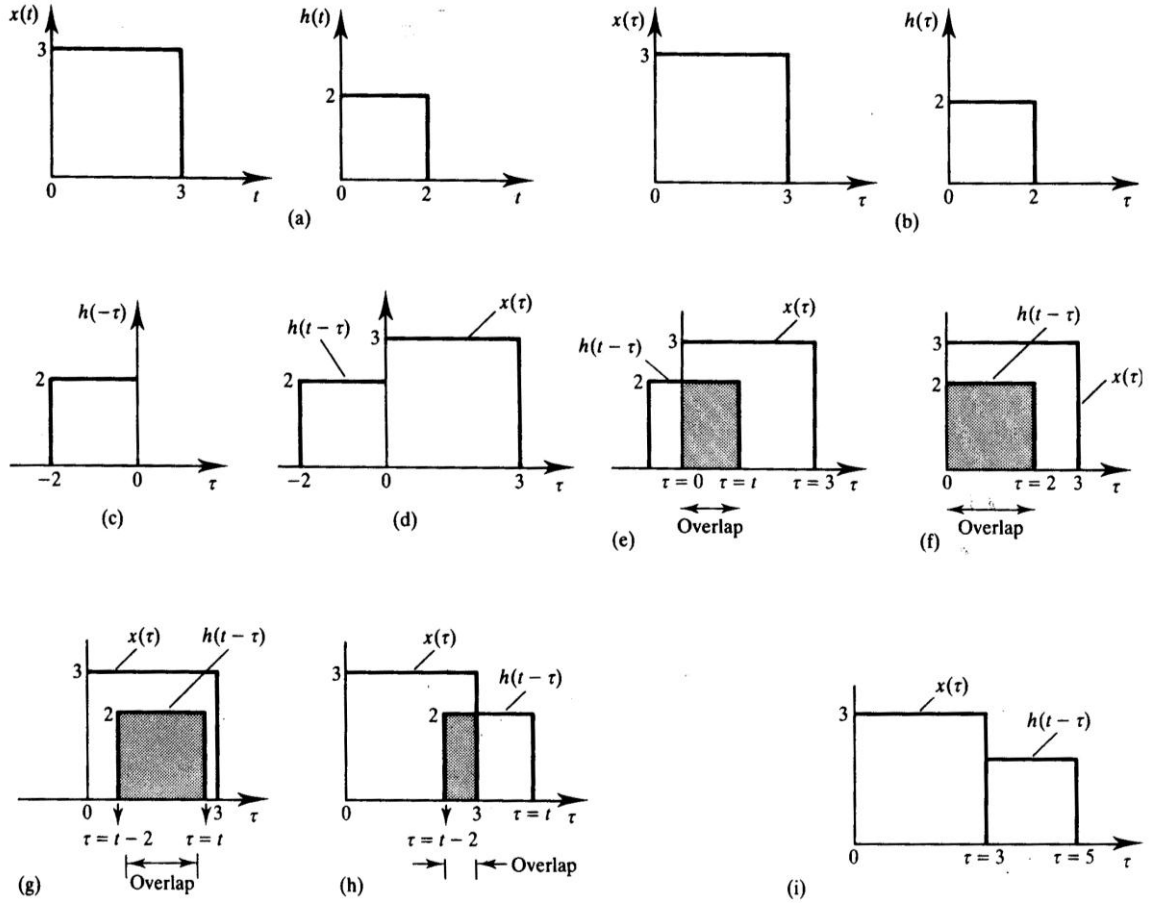


Fig. 3.13 (a) The waveforms x(t) and h(t) which are to be convolved analytically. (b) $x(\tau)$ and $h(\tau)$ vs τ. (c) $h(-\tau)$ vs τ. (d) $h(t-\tau)$ and $x(\tau)$ vs τ. $t < 0$, $h(t-\tau)$ does not overlap $x(\tau)$. (e) $h(t-\tau)$ and $x(\tau)$ vs τ. $0 < t \leq 2$. First partial overlap occurs between $h(t-\tau)$ and $x(\tau)$. (f) $h(t-\tau)$ and $x(\tau)$ vs (τ). $t = 2$. End of first partial overlap. (g) $h(t-\tau)$ and $x(\tau)$ vs(τ). $2 \leq t \leq 3$. Complete overlap between $h(t-\tau)$ and $x(\tau)$ occurs. (h) $h(t-\tau)$ and $x(\tau)$ vs (τ). $3 \leq t \leq 5$. Second partial overlap occurs between $h(t-\tau)$ and $x(\tau)$. (i) $h(t-\tau)$ and $x(\tau)$ vs τ. $t > 5$. $h(t-\tau)$ does not overlap $x(\tau)$.

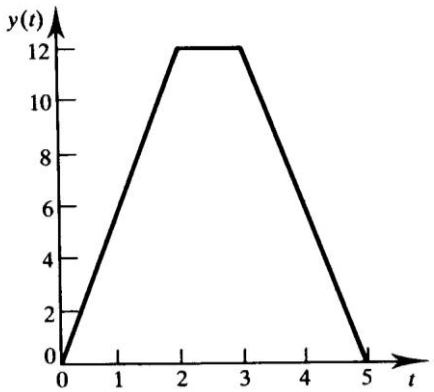


Fig. 3.14 The convolution $y(t) = x(t) \otimes h(t)$ vs t.

It is now convenient to repeat Eq.s. 3.27 and 3.28 here in order to discuss them.

$$y(n) = \sum_{m=-\infty}^{\infty} x(m)h(n - m) = x(n) \otimes h(n) \tag{3.27}$$

and

$$y(t) = \int_{-\infty}^{\infty} x(\lambda)\, h(t - \lambda)\, d\lambda \tag{3.28}$$

Inspection of these equations serves as a reminder that the convolution was carried out in time. This is known as convolution in the time domain. It is known that in the frequency domain the output component of a system at frequency, f, is Y(f), given by

$$Y(f) = H(f)\, X(f) \tag{3.29}$$

where H(f) is the frequency response function of the system at frequency, f, and X(f) is the Fourier transform of the input x(t). It can also be demonstrated that H(f) is the Fourier transform of h(t). The inverse Fourier transform of Equation 3.29 is

$$F^{-1}[Y(f)] = y(t) = F^{-1}[H(f)X(f)] \tag{3.30}$$

Bringing together Eqs. 3.28 and 3.30 shows that

$$y(t) = \int_{-\infty}^{\infty} x(\lambda)\, h(t - \lambda)\, d\lambda = x(t) \otimes h(t) = F^{-1}[H(f)X(f)] \tag{3.31}$$

Thus, it is seen that the convolution of two waveforms in the time domain is equivalent to the inverse Fourier transform of the product of the Fourier transforms of the two waveforms. This useful fact is often stated in the abbreviated form that convolution in the time domain is equivalent to multiplication in the frequency domain.

The dual of this relationship exists, that is convolution in the frequency domain is equivalent to multiplication in the time domain. Thus it can be shown (McGillem and Cooper, 1974) that

$$X(\omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega - u)\, H(u)\, du = X(f) \otimes H(f)$$

$$= F[y(t)] = F[x(t)h(t)] \tag{3.32}$$

Thus, the Fourier transform of the product of two time sequences corresponds to the convolution of the Fourier transforms of the two sequences.

3.4.1. Properties of Convolution

(1) Commutative law:

$$x_1(t) \otimes x_2(t) = x_2(t) \otimes x_1(t) \tag{3.33}$$

Note that this is identical to

$$\int_{-\infty}^{\infty} x_1(\tau)x_2(t-\tau)\,d\tau = \int_{-\infty}^{\infty} x_2(\tau)x_1(t-\tau)\,d\tau$$

(2) Distributive law:

$$x_1(t) \otimes [x_2(t) + x_3(t)] = x_1(t) \otimes x_2(t) + x_1(t) \otimes x_3(t) \tag{3.34}$$

(3) Associative law:

$$x_1(t) \otimes [x_2(t) \otimes x_3(t)] = [x_1(t) \otimes x_2(t)] \otimes x_3(t) \tag{3.35}$$

These properties may be proved either by manipulating the integrations involved or by considering the convolutions in terms of cross-correlating the one sequence by the time-reversed second sequence.

3.4.2. Circular convolution

Section (3.2) illustrated that the result of correlating two unequal length periodic sequences was a cyclic sequence of period equal to that of the shorter sequence, which was, therefore, an incorrect result. Because convolution is equivalent to the cross-correlation of one sequence by the reverse of a second sequence, the same will be true of convolution. Therefore, as with correlation, in convolution it is necessary that the two sequences be of the same length. Thus if the sequence lengths are N_1 and N_2, then $N_2 - 1$ augmenting zeros must be added to the sequence of length N_1, and $N_1 - 1$ augmenting zeros must be add to the sequence of length N_2. Both sequences will now be of identical length $N_1 + N_2 - 1$, and the correct linear convolution will be obtained, subject to taking other precautions as described for correlation.

3.4.3. Fast linear convolution

In Section 3.3 it was shown that correlation computations could be speeded up using the correlation theorem. A similar theorem, the convolution theorem, exists in the case of convolution. Thus, in discrete terminology and for the time domain

$$x_1(l) \otimes x_2(r) = F_D^{-1}[X_1(k)X_2(k)] \qquad (3.36)$$

Equation 3.36 is the convolution theorem, in which F_D^{-1} denotes the inverse discrete Fourier transform, $X_1(k)$ is the discrete Fourier transform of $x_1(\ell)$, and $X_2(k)$ is the discrete Fourier transform of $x_2(r)$. As in Section 3.3, $x_1(\ell)$ and $x_2(r)$ are periodic sequences of length, N.

3.4.4. Computational advantages of fast linear convolution

The method of fast linear convolution only offers the advantage of greater computational speed over the direct approach if the number of values to be convolved is sufficiently large. The number of multiplications required to perform the convolution by the direct and fast method is compared here as a measure of their relative computational efficiencies.

The necessary computations for the direct method were given in Eq. 3.16. It can be seen from these equations that to obtain the linear convolution of the two N-point sequences h(n – m) and x(m) it is necessary to multiply each value of h(n – m) by each value of x(m). Thus N values of h(n – m) are each to be multiplied by N values of x(m), making $N \times N = N^2$ multiplications in all.

Now consider linear convolution of the same two N-point sequences by the fast method according to Eq. 3.36. The addition of the necessary augmenting zeros means each sequence is of length 2N-1 points. Assume $2N - 1 \approx 2N$, for example $N \geq 8$, and that in order to use a radix-2 FFT N is given by an integer power of 2, that is $N = 2^d$ where d is an integer. The number of complex multiplications for an N-point FFT was shown to be $(N/2) \log_2 N$ (Section 2.5), so for the 2N-point FFT $(2N/2) \log_2 2N$ or $N \log_2 2N$

complex multiplications are necessary. Eq. 3.36 requires two DFTs and one inverse DFT to be computed. The inverse will be computed using the modified DFT (Sec. 2.3). The computation therefore requires computation of three 2N – point FFTs involving $3N \log_2 2N$ complex multiplications. Further, for each of the 2N values of Eq. 3.36 it is necessary to evaluate the complex multiplications $X_1(k)X_2(k)$, thus increasing the number of complex multiplications necessary to $3N \log_2 2N + 2N$. Now, each complex multiplication of the form (A + jB) (C +jD) requires four real multiplications: AC, AD, BC, and BD. Hence $12N \log_2 2N + 8N$ real multiplications are necessary.

It is therefore concluded that the direct method requires N^2 real multiplications while the method of fast convolution requires $12N \log_2 2N + 8N$ of them. Table 3.1 compares the numbers of real multiplications required for different values of N. The table demonstrates that fast convolution is faster than the direct method for sequences containing in excess of 128 data points, being approximately ten times faster for sequences containing 1024 points. The same conclusion also holds for direct and fast correlation.

Table 3.1 Numbers of real multiplications necessary for the convolution of two N-point sequences.

N	**Direct method**	Fast convolution	Number ratio, fast : direct
8	64	448	7
16	256	1088	4.25
32	1024	2560	2.5
64	4096	5888	1.4375
128	16384	13312	0.8125
256	65536	29696	0.4531
512	262144	65536	0.250
1024	1048576	143360	0.1367
2048	4194304	311296	0.0742

SHEET # 3

3.1 Describe the convolution and correlation operations.

3.2 Define the following:

(a)Cross-correlation between two sequences $x_1(n)$ and $x_2(n)$ having lengths N_1 and N_2 respectively.

(b) Correlation theorem.

(c) Correlation coefficients.

3.3 Use the correlation theorem to calculate the circular correlation of the data sequences {1,1,0,1} and {1,0,0,1}. Plot the correlation function against lag number, j.

3.4 Apply the correlation theorem to calculate the linear correlation of the data sequences {1,1,0,1} and {1,0,0,1}. Plot the correlation function against lag numbers and compare the results with the solution to prob.2.7, explain any differences.

3.5 Given the two sets of data {1.5, 2.0, 1.5, 2.0, 2.5} and {0, 0.33, 0.67, 1.0} calculate the following:

(a) Autocorrelation of the two sequences.

(b) Cross-correlation using direct method.

(c) Cross-correlation using correlation theorem.

3.6 Determine the output of an electrical system of impulse response function { 0, 0.899, 0.990, 0.991, 1} when the input{ 0, 2.5, 5.0, 0 } volts is applied

(a) By direct convolution.

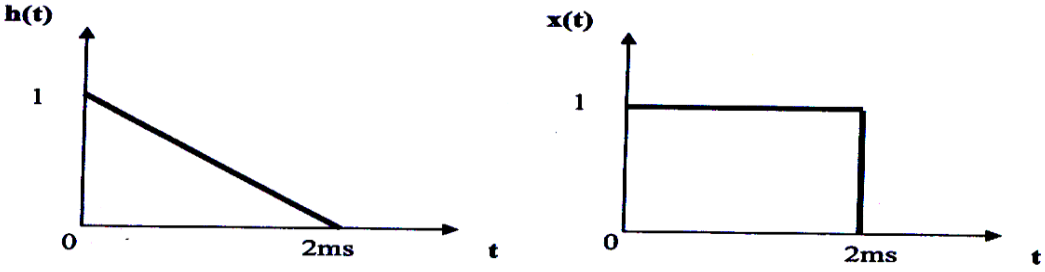
(b) By applying the convolution theorem.

3.7 A unit amplitude pulse of width 2ms (Fig.3.15b) is applied to a circuit which has an impulse response shown in Fig.3.15a. Determine the output waveform:

(a) Analytically.

(b) By sampling h(t) every 0.25ms and representing the pulse by an impulse function located at t = 0s.

(c) Compare your results in part (a) and (b).



(a) (b)

Fig. 3.15

3.8 For Prob. (3.7), find the cross-correlation between the given waveforms using direct method assume the sampling interval $T_S = 0.2\,msec$.

3.9 For the given waveform, Fig. 3.16, compute the convolution with itself y(t). Then assume the sampling interval $T_S = 0.2\ msec$ find y(n). Compare y(t) and y(n).

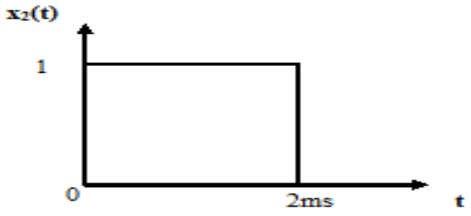


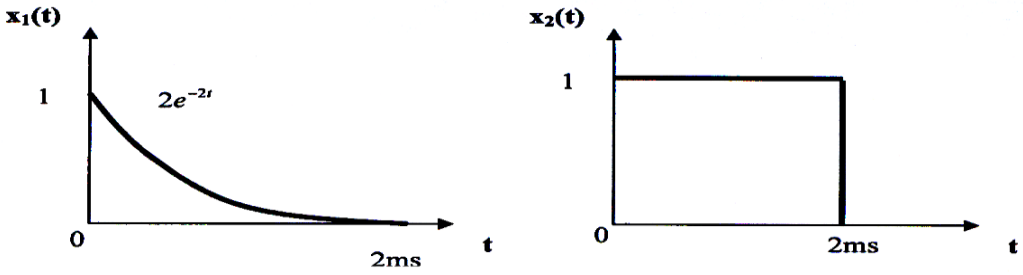
Fig. 3.16

3.10 Figure 3.17 show two functions x1(t) and x2(t). Evaluate the following:

(a) Their convolution, $x_3(t)$, numerically, taking sampled values at t = 0 , 0.5,1,1.5 and 2ms.

(b) x3(t) analytically.

(c) Sketch the function x3(t) and give reasons for any difference between them.



(a) (b)

Fig. 3.17

Assignment # 3

A rectangular pulse of height 5V and width 10μ sec shown in Fig. 3.18, is applied to the input of a system with the response function h(t) given by:

$$h(t) = \begin{cases} 0.1(1-e^{\frac{-t}{1.09\times10^{-6}}}) & 0 \le t \le 10\mu\text{sec} \\ 0 & 10\mu\text{sec} < t < \infty \end{cases}$$

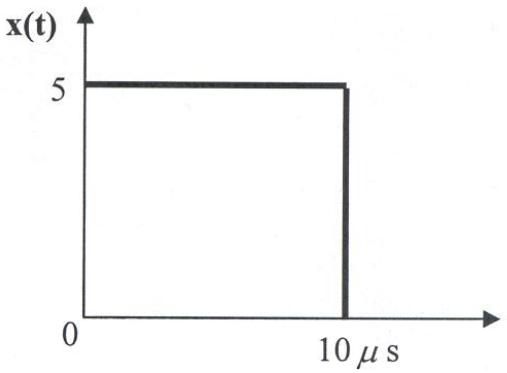


Fig. 3.18

Determine the system output:

(a) Analytically.

(b) By sampling h (t) every $1\mu\text{sec}$ and representing the pulse by impulse function located at t = 0sec.

(c) Compare your results in part (a) and (b).

CHAPTER 4

Z-TRANSFORM AND ITS APPLICATIONS IN DSP

4.1 INTRODUCTION

It is a convenient yet invaluable tool for representing, analog signal, and designing discrete time signals and system. It plays a similar role in discrete-time system to that which Laplace transform plays in continuous time system. A discrete time system is essentially a mathematical algorithms that takes an i/p seq., x(n), and produces an output sequence, y(n).

Examples of D-T systems are: Digital filters, digital controller ad digital spectrum analyzer. A discrete Time systems may be:

- Linear or non-linear systems.
- Time invariant or time varying system.
- Causal or a non-causal system.

- Linear time invariant systems (LTIS) form an important class of system used in DSP.

- <u>Linear systems</u>

A discrete-time system is said to be linear if it obeys the principles of superposition, i.e., the response of a linear system to two or more i/ps is equal to the sum of the responses of the system to all the other i/ps. e.g. if for input say $x_1(n)$ the system gives the output $y_1(n)$ and for input, say, $x_2(n)$ the system gives the output $y_2(n)$. The response of the system to both inputs will be:

$$a_1 x_1(n) + a_2 x_2(n) \longrightarrow a_1 y_1(n) + a_2 y_2(n)$$

where a_1 a_2 are arbitrary constants.

- Time Invariant systems

A discrete-time system is said to be time invariant (Some times referred to as shift invariant) if its output is independent of the time the input is applied. e.g., if the input is x(n), the system will give the output y(n) and for x(n-k) input, the system will gives the y(n-k) output,

$x(n) \longrightarrow y(n)$

$x(n-k) \longrightarrow y(n-k)$

i.e., delay in the input causes delay by the same amount in the output signal.

- Causal system

The input relationship of an LTIS is given by the convolution sum, given by:

$$y(n) = \sum_{k=-\infty}^{\infty} h(k)\, x(n-k)$$

Where h(k), is the impulse response of the system. The values of h(k) completely define the DTS in the time domain. The system is used to be causal system if it produces an o/p only where there is an input. Note that, all physical systems are causal.

In general, a causal D-T-System has zero impulse response before time zero. i.e.

$$h(k)=0 \quad \text{for all } k<0$$

Also the D-T-sequence, say, x(n) is causal if its value is zero before time zero. i.e.

$$x(n)=0 \quad \text{for all } n<0$$

For continuous signals or system, they are causal if

$x(t) = 0$ for all $t<0$ for signal.

$h(t) = 0$ for all $t<0$ for system.

- System stability

An LTIS is stable if its impulse response h(k), satisfied the following condition:

$$\sum_{k=-\infty}^{\infty} h(k) < \infty$$

This condition is satisfied if , h(k) is of finite duration. Or, if h(k) decays towards zero as k increase.

4.2 THE Z-TRANSFORM

The z-transform of a sequence, x(n), which is valid for all "n", is defined as

$$X(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n} \qquad (4.1)$$

Where z is complex variable.

In causal systems, x(n) may be nonzero only in the interval $0 < n < \infty$ and Eq. 4.1 reduces to the so – called one-sided z-transform:

$$X(z) = \sum_{n=0}^{\infty} x(n)z^{-n} \qquad (4.2)$$

Clearly, the z-transform is a power series with an infinite number of terms and so may not converge for all values of z. The region where the z-transform converges is known as the region of convergence (ROC), and in this region the values of X(z) are finite. Not surprisingly, the region of convergence is determined by the properties of x(n) or equivalently by those of X(z) as illustrated by the following set of examples.

Example 4.1

Find the z-transform and the region of convergence for each of the discrete-time sequence given in Fig. 4.1.

(1) The sequence of Fig. 4.1(a) is non-causal, since x(n) is not zero for n<0, but it is of a finite duration. The sequence has values x(-6) = 0, x(-5) = 1, x(-4) = 3, x(-3) = 5, x(-2) = 3 , x(-1) = 1 and x(0) = 0 . From Eq. 4.1, the z-transform is given by

$$X_1(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n} = z^5 + 3z^4 + 5z^3 + 3z^2 + z$$

It is readily verified that the value of X(z) becomes infinite when $z = \infty$ Thus the ROC is everywhere in the z-plane except at $z = \infty$.

(2) Again, the sequence in Fig. 4.1(b) is not causal. It is of a finite duration, and double sided. The values of the sequence are x(-3) =0, x(-2) = 1, x(-1) = 3 , x(0) = 5, x(1) = 3, x(2) = 1 and x(3) = 0. From Equation 3.1, the z-transform is given by

$$X_2(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n} = z^2 + 3z + 5 + 3z^{-1} + z^{-2}$$

It is evident that the value of X(z) is infinite if z =0 or if z = ∞. Therefore, the ROC is everywhere except at Z = 0 and Z = ∞.

(3) Figure 4.1(c) represents a causal, finite duration sequence with values x(0) = 0, x(1) = 1, x(2) =3, x(3) = 5, x(4) = 3, x(5) = 1 and x(6) =0. The z-transform is given by

$$X_3(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n} = z^{-1} + 3z^{-2} + 5z^{-3} + 3z^{-4} + z^{-5}$$

In this case, X(z) = ∞ for z = 0. Thus the region of convergence is every where except at z = 0.

(4) The discrete-time sequence in Fig. 4.1(d) may be defined mathematically as

$$x(n) = 1 \qquad 0 \le n \le \infty$$
$$= 0 \qquad n < 0$$

Clearly it is a causal sequence of infinite duration. From Eq. 4.1, the z-transform of the sequence is given by

$$X(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n}$$

$$= \sum_{n=0}^{\infty} z^{-n}$$

$$= 1 + z^{-1} + z^{-2} + \ldots\ldots$$

This is a geometric series with a common ratio of z^{-1}. The series converges if $|z^{-1}| < 1$ or equivalently if $|z| > 1$. Thus we may express X(z) in closed form provided that $|z| > 1$

$$X(z) = 1 + z^{-1} + z^{-2} + \ldots \qquad (4.3)$$
$$= 1/(1-z^{-1})$$
$$= z/(z-1)$$

In this case, the Z-transform is valid everywhere outside a circle of unit radius whose center is at the origin. The exterior of the circle is the region of convergence (Fig. 4.2). We can readily verify that when | z | > 1 X(z) converges whereas when | z | < 1 X(z) diverges. For example, if we let z = 2 (outside the unit circle) we find that the series on the RHS of Eq. 4.3 adds up to 2:

$$X(z) = 1 + 1/2 + (1/2)^2 + (1/2)^3 + \ldots. = 2/(2 - 1) = 2$$

as it is clearly a geometric series with a common ratio of 1/2 and a first term of 1, giving the sum to infinity of 2/(2 – 1) = 2. On the other hand if z = 1/2 (inside the unit circle) the series of Eq. 4.3 becomes

$$X(z) = 1 + 1/0.5 + (1/0.5)^2 + (1/0.5)^3 + (1/0.5)^4 + \ldots.$$
$$= 1 + 2 + 4 + 8 + \ldots\ldots$$

which is seen to be diverging. In Fig.4.2, the region of convergence (hatched) is seen to be bounded by the circle |z| =1, the radius of the pole of X(z). Values of z for which X(z) = ∞ are referred to as poles of X(z). Values of z for which X(z) = 0 are referred to as the zeros of X(z).

From the above examples, we may infer that for causal sequences of finite duration the z-transform converges everywhere except at z = 0. For causal infinite duration sequences the z-transform converges everywhere outside a circle bounded by the radius of the pole with the largest radius. For stable causal systems the ROC always encloses the circle of unit radius which is important for the systems to have a frequency response.

The z-transforms of common sequences are available, in closed form, and are given in the form of tables, such as Table 4.1. Such tables are useful in finding the inverse z-transform.

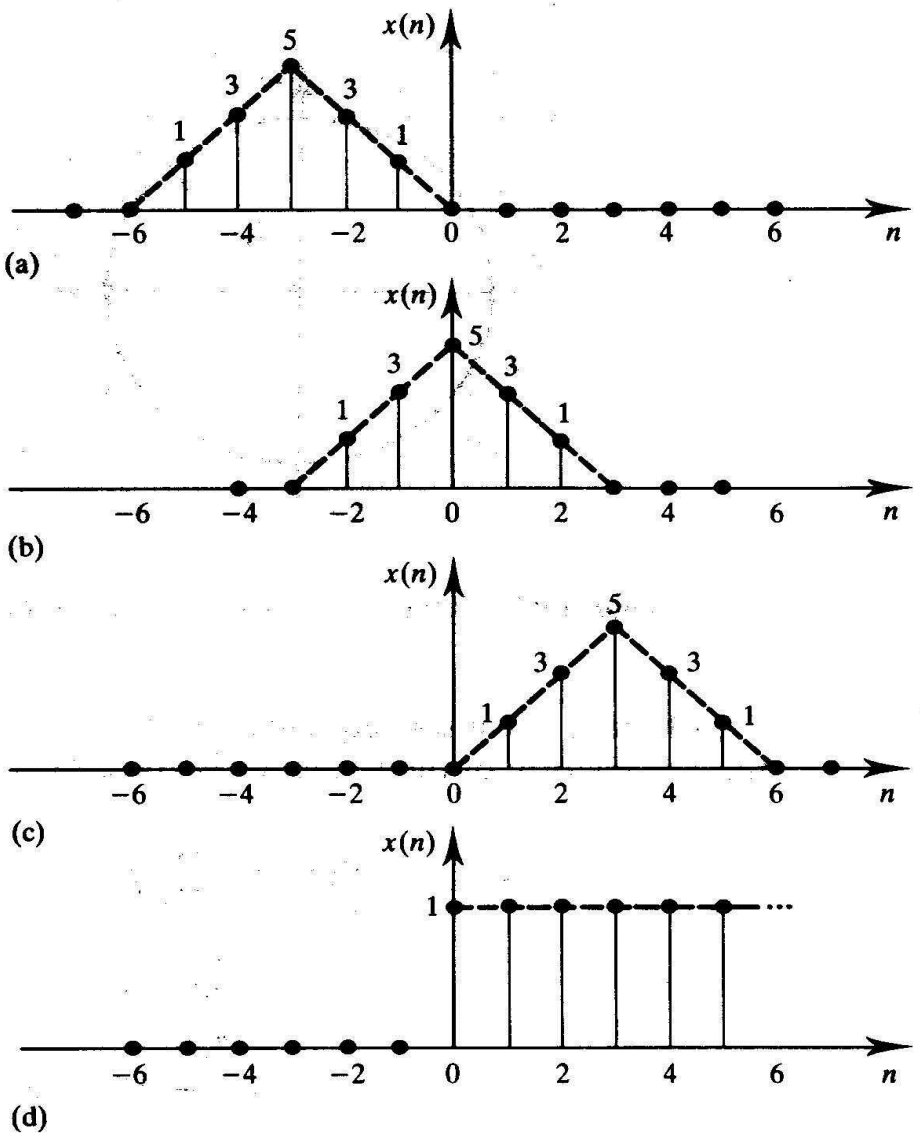


Fig. 4.1 Causal and non-causal discrete-time sequences.

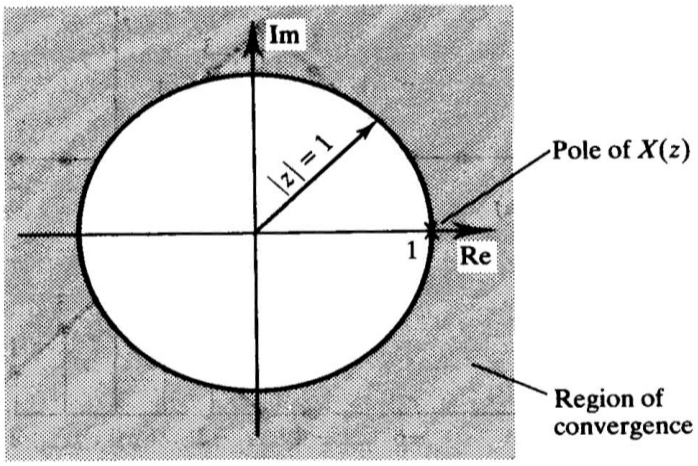


Fig. 4.2 Region of convergence for Example 4.1, part (4)

4.3 THE INVERSE Z-TRANSFORM

The inverse z-transform (IZT) allows us to recover the discrete-time sequence, x(n), given its z-transform. The IZT is particularly useful in DSP work, for example in finding the impulse response of digital filters. Symbolically, the inverse z-transform may be defined as

$$x(n) = Z^{-1}[X(z)] \tag{4.4}$$

where X(z) is the z-transform of x(n) and Z^{-1} is the symbol for the inverse z-transform.

Assuming a causal sequence, the z-transform, X(z), in Eq. 4.2 can be expanded into a power series as

$$\begin{aligned} X(z) &= \sum_{n=0}^{\infty} x(n)z^{-n} \\ &= x(0) + x(1)z^{-1} + x(2)z^{-2} + x(3)z^{-3} + \ldots \end{aligned} \tag{4.5}$$

It is seen that the values of x(n) are the coefficients of z^{-n} (n=0,1,….) and so can be obtained directly by inspection. In practice, X(z) is often expressed as a ratio of two polynomials in z^{-1} or equivalently in z :

$$X(z) = \frac{a_0 + a_1 z^{-1} + a_2 z^{-2} + \ldots + a_N z^{-N}}{b_0 + b_1 z^{-1} + b_2 z^{-2} + \ldots + b_M z^{-M}} \tag{4.6}$$

In this form, the IZT, x(n), may be obtained using one or several the following methods: power series expansion method, partial fraction expansion method and residue method.

Each method has its own merits and demerits. In terms of mathematical rigour, the residue methods are perhaps the most elegant. The power series method, however, lends itself most easily to computer implementation.

In the next few sections, we will describe each of the three methods in turn, using numerical examples to illustrate the principles involved.

Table 4.1 z-transforms of some common sequences.

Entry number	***Discrete-time sequence x(n), n ≥ 0***	***z-transform X(z)***	***Region of convergence of X(z)***
1	$k\delta(n)$	k	Everywhere
2	K	$\frac{kz}{z-1}$	$\|z\| > 1$
3	kn	$\frac{kz}{(z-1)^2}$	$\|z\| > 1$
4	kn^2	$\frac{kz(z+1)}{(z-1)^3}$	$\|z\| > 1$
5	$Ke^{-\alpha n}$	$\frac{kz}{z-e^{-\alpha}}$	$\|z\| > e^{-\alpha}$
6	$Kne^{-\alpha n}$	$\frac{kze^{-\alpha}}{(z-e^{-\alpha})^2}$	$\|z\| > e^{-\alpha}$
7	$1-e^{-\alpha n}$	$\frac{z(1-e^{-\alpha})}{z^2-z(1+e^{-\alpha})+e^{-\alpha}}$	$\|z\| > e^{-\alpha}$
8	$\cos(\alpha n)$	$\frac{z(z-\cos\alpha)}{z^2-2z\cos\alpha+1}$	$\|z\| > 1$
9	$\sin(\alpha n)$	$\frac{z\sin\alpha}{z^2-2z\cos\alpha+1}$	$\|z\| > 1$
10	$E^{-\alpha n}\sin(\alpha n)$	$\frac{ze^{-\alpha}\sin\alpha}{z^2-2e^{-\alpha}z\cos\alpha+e^{-2\alpha}}$	$\|z\| > e^{-\alpha}$
11	$e^{-\alpha n}\cos(\alpha n)$	$\frac{ze^{-\alpha}(ze^{\alpha}-\cos\alpha)}{z^2-2ze^{-\alpha}\cos\alpha+e^{-2\alpha}}$	$\|z\| > e^{-\alpha}$
12	$\cosh(\alpha n)$	$\frac{z^2-z\cosh\alpha}{z^2-2z\cosh\alpha+1}$	$\|z\| > \cosh\alpha$
13	$\sinh(\alpha n)$	$\frac{z\sinh\alpha}{z^2-2z\cosh\alpha+1}$	$\|z\| > \sinh\alpha$
14	$k\alpha^n$	$\frac{kz}{z-\alpha}$	$\|z\| > \alpha$
15	$kn\alpha^n$	$\frac{k\alpha z}{(z-\alpha)^2}$	$\|z\| > \alpha$
16	$2\|c\|\|p\|^n\cos(n\angle p+\angle c)$	$\frac{cz}{z-p}+\frac{c^*z}{z-p^*}$	

k and α are constants; c is a complex number.

4.3.1 Power series method

Given the z-transform, X(z), of a causal sequence as in Eq. 4.6, it can be expanded into an infinite series in z^{-1} or z by long division (sometimes called synthetic division) :

$$X(z) = \frac{a_0 + a_1 z^{-1} + a_2 z^{-2} + \ldots + a_N z^{-N}}{b_0 + b_1 z^{-1} + b_2 z^{-2} + \ldots + b_M z^{-M}}$$

$$= x(0) + x(1)z^{-1} + x(2)z^{-2} + x(3)z^{-3} + \ldots \qquad (4.7)$$

In this method, the numerator and denominator of X(z) are first expressed in either descending powers of z or ascending powers of z^{-1} and the quotient is then obtained by long division. We will illustrate the method by an example.

Example 4.2

Given the following z-transform of a causal LTI system, obtain its IZT by expanding it into a power series using long division:

$$X(z) = \frac{1 + 2z^{-1} + z^{-2}}{1 - z^{-1} + 0.3561z^{-2}}$$

Solution

First, we expand X(z) into a power series with the numerator and denominator polynomials in ascending power of z^{-1} and then perform the usual long division.

$$\frac{1 + 3z^{-1} + 3.6439z^{-2} + 2.5756z^{-3} + \ldots}{1 + 2z^{-1} + z^{-2}}$$

$$1 - z^{-1} + 0.3561z^{-2} \;\Big|$$

$$\frac{1 - z^{-1} + 0.3561z^{-2}}{3z^{-1} + 0.6439z^{-2}}$$

$$\frac{3z^{-1} - 3z^{-2} + 1.0683z^{-3}}{3.6439z^{-2} - 1.0683z^{-3}}$$

$$\frac{3.6439z^{-2} - 3.6439z^{-3} + 1.2975927z^{-4}}{2.5756z^{-3} - 1.2975927z^{-4}}$$

Alternatively, we may express the numerator and denominator in positive powers of z, in descending order, and then perform the long division:

$$\frac{z^2 + 2z + 1}{z^2 - z + 0.3561}$$

$$z^2 - z + 0.3561 \;\Big|\; \frac{1 + 3z^{-1} + 3.6439z^{-2} + 2.5756z^{-3} + \ldots}{z^2 + 2z + 1}$$

$$\underline{z^2 - z + 0.3561}$$

$$3z + 0.6439$$

$$\underline{3z - 3 + 1.0683z^{-1}}$$

$$3.6439 - 3.64391z^{-1} + 1.2975927z^{-2}$$

$$\underline{2.5756z^{-1} - 1.2975927z^{-2}}$$

Either way, the z-transform is now expanded into the familiar power series, that is

$$X(z) = 1+3z^{-1} + 3.6439z^{-2} + 2.576z^{-3} + \ldots$$

The inverse z-transform (x(n)) can now be written down directly :

$$x(0) = 1;\ x(1) = 3;\ x(2) = 3.6439;\ x(3) = 2.5756;\ \ldots\ldots$$

The long division approach can be reformulated so that the values of x(n) are obtained recursively as :

$$x(n) = \left[a_n - \sum_{i=1}^{n} x(n-i) b_i \right] / b_0,\ n = 1, 2, \ldots\ldots \qquad (4.8a)$$

where

$$x(0) = a_0 / b_0 \qquad (4.8b)$$

We will repeat the previous example to illustrate the recursive approach.

Example 4.3

Find the first four terms of the inverse z-transform, x(n), using the recursive approach. Assume that the z-transform, X(z), is the same as in Example 4.2, that is

$$X(z) = \frac{1 + 2z^{-1} + z^{-2}}{1 - z^{-1} + 0.3561z^{-2}}$$

Solution

Comparing the coefficients of X(z) above with those of the general transform in Equation 4.7 we have

$$a_0 = 1,\ a_1 = 2,\ a_2 = 1,\ b_0 = 1,\ b_1 = -1,\ b_2 = 0.3561;\ N = M = 2$$

From Equation 4.8 we have

$$x(0) = a_0/b_0 = 1$$

$$x(1) = [a_1 - x(0)b_1]/b_0 = [2 - 1 \times (-1)] = 3$$

$$x(2) = [a_2 - x(1)b_1 - x(0)b_2] = 1 - 3 \times (-1) - 1 \times 0.3561 = 3.6439$$

$$x(3) = [a_3 - x(2)b_1 - x(1)b_2 + x(0)b_3]$$

$$= 0 - x(2)b_1 - x(1)b_2 = 0 - 3.6439 \times (-1) - 3 \times 0.3561 = 2.5756$$

Thus the first four values of the inverse z-transform are

$$x(0) = 1,\ x(1) = 3,\ x(2) = 3.6439,\ x(3) = 2.5756$$

It is seen that both the recursive and direct, long division methods lead to identical solutions.

4.3.2 Partial fraction expansion method

In this method, the z-transform is first expanded into a sum of simple partial fractions. The inverse z-transform of each partial fraction is then obtained from tables, such as Table 4.1, and then summed to give the overall inverse z-transform. In many practical cases, the z-transform is given as a ratio of polynomials in z or z^{-1} and has the now familiar form

$$X(z) = \frac{a_0 + a_1 z^{-1} + a_2 z^{-2} + \ldots + a_N z^{-N}}{b_0 + b_1 z^{-1} + b_2 z^{-2} + \ldots + b_M z^{-M}} \qquad (4.9)$$

If the poles of X(z) are of first order and N=M, then X(z) can be expanded as

$$X(z) = B_0 + \frac{C_1}{1 - P_1 z^{-1}} + \frac{C_2}{1 - P_2 z^{-1}} + \ldots\ldots + \frac{C_M}{1 - P_M z^{-1}}$$

$$= B_0 + \frac{C_1 z}{z - P_1} + \frac{C_2 z}{z - P_2} + \ldots. + \frac{C_M z}{z - P_M}$$

$$= B_0 + \sum_{k=1}^{M} \frac{C_k z}{z - P_k} \qquad (4.10)$$

where P_k are the poles of X(z) (assumed distinct), C_k are the partial fraction coefficients and

$$B_0 = a_N / b_M \qquad (4.11)$$

The C_k are also known as the residues of X(z); see section 4.3.3.

If the order of the numerator is less than that of the denominator in Eq. 4.9, that is N < M, then B_0 will be zero. If N > M then X(z) must be reduced first, to make $N \leq M$, by long division with the numerator and denominator polynomials written in descending powers of z^{-1}. The remainder can then be expressed as in Eq. 4.10.

The coefficient, C_k, associated with the pole p_k may be obtained by multiplying both sides of Equation 4.10 by $(z - p_k)/z$ and then letting $z = p_k$:

$$C_k = \left. \frac{X(z)}{z} (z - p_k) \right|_{z=p_k} \qquad (4.12)$$

If X(z) contains one or more multiple-order poles (that is poles that are coincident) then extra terms are required in Equation 4-10 to take this into account. For example, if X(z) contains an mth-order pole at $z = P_k$ the partial fraction expansion must include terms of the form

$$\sum_{i=1}^{m} \frac{D_i}{(z - p_k)^i} \tag{4.13a}$$

The coefficients, D_i, may be obtained from the relationship

$$D_i = \frac{1}{(m-i)!} \frac{d^{m-i}}{dz^{m-i}} \left[(z - p_k)^m X(z)\right]_{z=p_k} \tag{4.13b}$$

Evaluation of inverse z-transforms by the partial fraction expansion method is best illustrated by examples.

Example 4.4 (X(z) contains simple, first-order poles)

Find the inverse z-transform of the following:

$$X(z) = \frac{z^{-1}}{1 - 0.25z^{-1} - 0.375z^{-2}}$$

Solution

For simplicity, we first express the z-transform in positive powers of z by multiplying the numerator and denominator by z^2 (the highest power of z):

$$X(z) = \frac{z}{z^2 - 0.25z - 0.375} = \frac{z}{(z - 0.75)(z + 0.5)}$$

X(z) Contains first-order poles at z = 0.75 and at z = - 0.5 (that is, only one pole occurs at each pole position). Since the order of the numerator is less than the order of the denominator (N < M), the partial fraction expansion has the form

$$X(z) = \frac{z}{(z - 0.75)(z + 0.5)} = \frac{C_1 z}{z - 0.75} + \frac{C_2 z}{z + 0.5} \tag{4.14}$$

To make it easier to find the values of the C_k we divide both sides by z:

$$\frac{X(z)}{z} = \frac{z}{z(z - 0.75)(z + 0.5)} = \frac{C_1}{z - 0.75} + \frac{C_2}{z + 0.5} \tag{4.15}$$

To obtain C_1, we simply multiply both sides of Equation 4-15 by (z-0.75) and let z = 0.75

$$\frac{(z-0.75)X(z)}{z} = \frac{(z-0.75)}{(z-0.75)(z+0.5)} = C_1 + \frac{C_2(z-0.75)}{z+0.5}$$

$$C_1 = \left.\frac{1}{z+0.5}\right|_{z=0.75} = \frac{1}{0.75+0.5} = \frac{4}{5}$$

Similarly, C_2 is obtained as

$$\frac{(z-0.5)X(z)}{z} = \frac{(z-0.5)}{z-0.75}C_1 + C_2\frac{(z-0.5)}{z+0.5}$$

$$C_2 = \left.\frac{(z+0.5)X(z)}{z}\right|_{z=-0.5}$$

$$= \left.\frac{(z+0.5)}{(z-0.75)(z+0.5)}\right|_{z=-0.5} = \frac{1}{-0.5-0.75} = -\frac{4}{5}$$

Using the values of C_1 and C_2 in Eq. 4.14 we have

$$X(z) = \frac{(4/5)z}{z-0.75} - \frac{(4/5)z}{z+0.5} \qquad (4.16)$$

From the z-transform table, entry 14 in Table 4.1, the inverse z-transform of each term on the right-hand side of Equation 4-16 is given as

$$Z^{-1}\left[\frac{(4/5)z}{z-0.75}\right] = \frac{4(0.75)^n}{5}$$

$$Z^{-1}\left[\frac{-(4/5)z}{z+0.5}\right] = \frac{-4(-0.5)^n}{5}$$

The desired inverse z-transform, x(n), is the sum of the two inverse z-transforms

$$X(n) = \frac{4}{5}[(0.75)^n - (-0.5)^n],\ n > 0$$

Example 4.5 (X(z) contains first-order, complex conjugate poles)

Find the discrete-time signal, x(n) , represented by the following z-transform using the partial fraction expansion method :

$$X(z) = \frac{1 + 2z^{-1} + z^{-2}}{1 - z^{-1} + 0.3561z^{-2}}$$

Solution

First, X(z) is expressed in positive powers of z:

$$X(z) = \frac{N(z)}{D(z)} = \frac{z^2 + 2z + 1}{z^2 - z + 0.3561}$$

The poles of X(z) are found by solving the quadratic $D(z) = z^2 - z + 0.3561 = 0$ using the formulae

$$p_1 = \frac{-b + (b^2 - 4ac)^{1/2}}{2a}$$

$$p_2 = \frac{-b - (b^2 - 4ac)^{1/2}}{2a} \qquad (4.17)$$

where a and b are the coefficients of z^2 and z, respectively, and c is the constant term. With a = 1, b = 1, c = 0.3561 the poles are

$$p_1 = \frac{-1 + (1 - 4 \times 0.3561)^{1/2}}{2} = 0.5 + 0.3257j = re^{j\theta}$$

$$p_2 = p_1^* = 0.5 - 0.3257j = re^{-j\theta}$$

where r = 0.5967 and $\theta = 33.08^o$. Thus we can express X(z) in terms of its poles:

$$X(z) = \frac{z^2 + 2z + 1}{(z - P_1)(z - p_1^*)}$$

Since the numerator and denominator of X(z) are of the same order, the partial fraction expansion has the form

$$\frac{X(z)}{z} = \frac{B_0}{z} + \frac{C_1}{z - P_1} + \frac{C_2}{z - p_1^*} \qquad (4.18)$$

From Eq. 4.11 B_0 = 1/0.3561 = 2.8082. To find C_1, we multiply both sides of Eq. 4.18 by z-p_1 and then let z = p_1:

$$\frac{(z-P_1)X(z)}{z} = \frac{B_0(z-P_1)}{z} + C_1 + \left.\frac{C_2(z-P_1)}{(z-P_2)}\right|_{z=P_1}$$

Thus

$$C_1 = \frac{(z-P_1)X(z)}{z} = \left.\frac{(z-P_1)(z^2+2z+1)}{z(z-P_1)(z-P_2)}\right|_{z=P_1=re^{j\theta}}$$

$$= \frac{(re^{j\theta})^2 + 2re^{j\theta} + 1}{re^{j\theta}(re^{j\theta} - re^{-j\theta})} \qquad (4.19)$$

where r = 0.5967, θ = 33.08°. After some manipulation and simplification have

$$C_1 = \frac{2.1439 + 0.97719j}{-0.2122 + 0.3257j}$$

$$= -0.904099\ 9 - 5.992847j = 6.06066\angle -98.58^\circ$$

since p_1 and p_2 are complex conjugate pairs then

$$C_2 = C_1^* = -0.904\ 099\ 9 + 5.992847j = 6.06066\angle -98.58^\circ$$

Thus the z-transform can be expressed as (from Eq. 4.18)

$$X(z) = 2.8082 + \frac{C_1 z}{z-p_1} + \frac{C_2 z}{z-p_1^*} \qquad (4.20)$$

Where

p_1 = 0.5 + 0.3257j p_2 = 0.5 – 0.3257j

C_1= -0.9041 – 5.59928j C_2 = - 0.9041 + 5.59928j

From the z-transform table, entries 1 and 16 in Table 4.1, the inverse z-transform of the terms on the right-hand side of Eq. 4.20 is

$$Z^{-1}(2.8082) = 2.8082\ u(n)$$

$$Z^{-1}\left[\frac{C_1 z}{z-p_1}+\frac{C_2^* z}{z-p_1^*}\right] = 2 \times 6.060\,66(0.5967)^n \cos(33.08n-98.58^o)$$

$$= 12.1213(0.5967)^n \cos(33.08n - 98.58^o)$$

Thus the discrete-time signal becomes

$$x(n) = 2.8082u(n)+ 12.1213(0.5967)^n \cos(33.08n–98.58^o),\ n \geq 0$$

A useful check for partial faction results is to compute the values of x(n) for n = 0, 1, 2 (say) and then to compare these with values obtained by the power series method. For example, from the expression for x(n) we find that

x(0) = 2.8082 – 1.808 38 = 1;

x(1) = 2.999 59 = 3;

x(2) = 3.6436

which checks with the results obtained in Example (4.3) using the power series method.

Example 4.6 (X(z) contains a second-order pole)

Find the discrete-time sequence, x(n), with the following z-transform :

$$X(z) = \frac{z^2}{(z-0.5)(z-1)^2}$$

Solution

X(z) has a first-order pole at z = 0.5 and a second-order pole at z = 1. In this case, the partial fraction expansion has the form

$$X(z) = \frac{CZ}{z-0.5}+\frac{D_1 Z}{z-1}+\frac{D_2 Z}{(z-1)^2} \qquad (4.21)$$

To obtain C, we proceed as before and multiply both sides of Eq. 4.21 by z – 0.5, set z = 0.5 and evaluate the expression:

$$C = \left.\frac{(z-0.5)z^2}{z(z-0.5)(z-1)^2}\right|_{z=0.5} = 0.5/(0.5\text{-}1)^2 = 2$$

To obtain D_1 we use Equation 4-13b, with i = 1 and m = 2. Thus

$$D_1 = \frac{d}{dz}\left[\frac{(z-1)^2 X(z)}{z}\right]_{z=1} = \frac{d}{dz}\left[\frac{(z-1)^2 z^2}{z(z-0.5)(z-1)^2}\right]_{z=1}$$

$$= \frac{d}{dz}\left[\frac{z}{(z-0.5)}\right]_{z=1} = \frac{z-0.5-z}{(z-0.5)^2}\Bigg|_{z=1}$$

$$= \left(\frac{-0.5}{z-0.5}\right)\Bigg|_{z=1} = \frac{-0.5}{(0.5)^2} = \frac{-0.5}{0.5\times 0.5} = -2$$

Similarly, D_2 is obtained from Eq. 4.13b by letting i = m = 2

$$D_2 = \frac{1}{(m-i)}\frac{d^{m-i}}{dz^{m-i}}\left[(z-P_k)^m X(z)\right]_{z=P_k}$$

$$= \left(\frac{(z-1)^2 X(z)}{z}\right)\Bigg|_{z=1} = \left(\frac{(z-1)^2 z^2}{z(z-0.5)(z-1)^2}\right)\Bigg|_{z=1}$$

$$= \left[\frac{z}{z-5}\right]_{z=1} = \frac{1}{1-0.5} = 2$$

Combining the results, X(z) becomes

$$X(z) = \frac{2z}{z-0.5} - \frac{2z}{z-1} + \frac{2z}{(z-1)^2}$$

The inverse z-transform of each term on the right-hand side is obtained from Table 4.1 and then summed to give x(n) :

$$x(n) = 2(0.5)^n - 2 + 2n = 2[(n-1)+(0.5)^n],\ n \geq 0 \qquad (4.22)$$

The reader can verify that the result is correct by comparing the first few values of x(n) with values calculated with the power series method.

4.3.3 Residue method

In this method the IZT is obtained by evaluating the contour integral

$$x(n) = \frac{1}{2\pi j}\oint_C z^{n-1} X(z)\, dz \qquad (4.23)$$

where C is the path of integration enclosing all the poles of X(z). For rational polynomials, the contour integral in Eq. 4.23 is evaluated using a fundamental result in complex variable theory known as Cauchy's residue theorem (Mathews, 1982):

$$x(n) = \frac{1}{2\pi j}\oint_C z^{n-1} X(z)\, dz \qquad (4.24)$$

= sum of the residues of $z^{n-1}X(z)$ at all the poles inside C.

In the last section, it was stated that the partial fraction coefficients, the C_k, are also referred to as residues of X(z) and a way of obtaining their values given. The key point to remember is that every residue, C_k, is associated with a pole, p_k. In the present method, the residue of $z^{n-1}X(z)$ at the pole p_k (not the residue of X(z)) is given by

$$\text{Res}\,[F(z), p_k] = \frac{1}{(m-1)!}\frac{d^{m-1}}{dz^{m-1}}[(z-p_k)F(z)]_{z=Pk} \qquad (4.25)$$

where $F(z) = z^{n-1}X(z)$, m is the order of the pole at p_k and Res $[F(z), p_k]$ is the residue of F(z) at $z = p_k$. For a simple (distinct) pole, Equation 4.25 reduces to

$$\text{Res}[F(z), p_k] = (z - p_k)\,F(z) \qquad (4.26)$$

$$= (z - p_k)z^{n-1}\ X(z)\big|_{z=P_k}$$

Example 4.7

Find, using the residue method, the discrete-time signal corresponding to the following z-transform:

$$X(z) = \frac{z}{(z-0.75)(z+0.5)}$$

Assume C is the circle |z| = 1.

Solution

This problem is the same as Example (4.4). In factored form, X(z) is given by

$$X(z) = \frac{z}{(z-0.75)(z+0.5)}$$

If we let $F(z) = z^{n-1} X(z)$ then

$$F(z) = \frac{z^{n-1} z}{(z-0.75)(z+0.5)}$$

$$= \frac{z^n}{(z-0.75)(z+0.5)}$$

F(z) has poles at z = 0.75 and z=-0.5. A sketch of the contour with the positions of the poles indicated by crosses is given in Fig. 4.3. Both poles lie inside the contour (unit circle). From Eq. 4.24, the inverse z-transform is given by

$$x(n) = \text{Res}\,[F(z), 0.75] + \text{Res}\,[F(z), -0.5]$$

Since the poles are first order, Equation 4-26 will be used. Thus

$$\text{Res}[F(z), 0.75] = (z-0.75)\,F(z)\Big|_{z=0.75}$$

$$= \frac{(z-0.75)z^n}{(z-0.75)(z+0.5)}\Bigg|_{z=0.75} = \frac{(0.75)^n}{0.75+0.5}$$

$$= \frac{4}{5}(0.75)^n$$

$$\text{Res}[F(z), -0.5] = (z+0.5)\,F(z)\Big|_{z=-0.5}$$

$$= \frac{(z+0.5)z^n}{(z-0.75)(z+0.5)}\Bigg|_{z=-0.5} = -\frac{4}{5}(-0.5)^n$$

The inverse z-transform is the sum of the residues at z = 0.75 and at z = -0.5:

$$x(n) = (4/5)[(0.75)^n - (-0.5)^n]$$

which is identical to the result obtained by the partial fraction expansion.

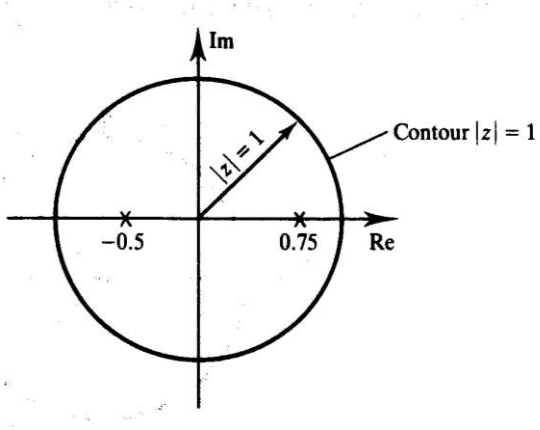


Fig. 4.3 A sketch of the contour of integration showing the poles of X(z).

Example 4.8 (The poles of X(z) are complex conjugate poles)

Find the inverse z-transform, using the residue method, given the following z-transform:

$$X(z) = \frac{z^2 + 2z + 1}{z^2 - z + 0.3561}$$

Solution

In factored form X(z) is given as:

$$X(z) = \frac{z^2 + 2z + 1}{(z - p_1)(z - p_2)}$$

where $p_1 = 0.5 + 0.3557j$ and $p_2 = 0.5 - 0.3557j$, that is $p_2 = p_1^*$. To find the inverse z-transform we evaluate the residues of F(z), where in this case

$$F(z) = z^{n-1} X(z) \quad = \frac{z^{n-1}(z^2 + 2z + 1)}{z^2 - z + 0.3561} \quad = \frac{z^n (z^2 + 2z + 1)}{z(z^2 - z + 0.3561)}$$

F(z) has the same poles as X(z), that is at $z = p_1$ and $z = p_2$, plus a pole at z = 0 when n = 0. Figure 4-4 shows a sketch of the contour with the positions of the poles indicated. All the poles lie inside the contour. The pole at z = 0 does not exist for n > 0 and so we need to consider the two cases separately.

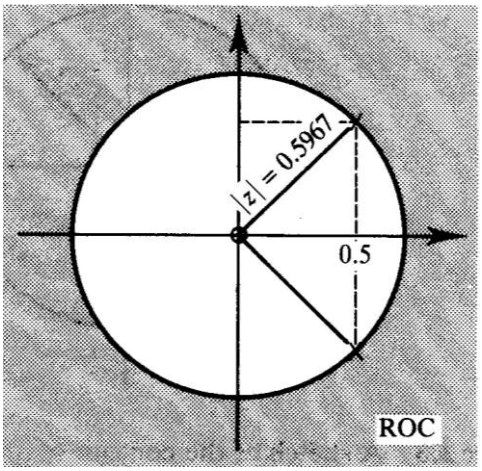


Figure 4.4 Contour for Example 4.8 showing the ROC.

When n = 0, F(z) reduces to $F(z) = \dfrac{z^2 + 2z + 1}{z(z^2 - z + 0.3561)}$

and x(0) = Res [F(z), 0] + Res [F(z), p_1] + Res [F(z), p_2]

Therefore,

$$\text{Res } [F(z), 0] = zF(z)\big|_{z=0} = \frac{z(z^2 + 2z + 1}{z(z^2 - z + 0.3561)}\bigg|_{z=0} = 1/0.3561 = 2.8082$$

$$\text{Res}[F(z), p_1] = (z-p_1)\,F(z)\big|_{z=P1} = \frac{(z - p_1)(z^2 + 2z + 1)}{z(z - p_1)(z - p_2)}$$

$$= \frac{(re^{j\theta})^2 + 2re^{j\theta} + 1}{re^{j\theta}(re^{j\theta} - re^{-j\theta})}$$

where r = 0.5967 and θ = 33.08^o . Noting that this expression is identical to that of Equation 4-19, we can write

Res [F(z), p_1] = - 0.9041 – 5.9928j

Since p_1 and p_2 are complex conjugate pairs then

Res [F(z), p_2] = - 0.9041 + 5.9928j

Thus

x(0) = Res [F(z), 0] + Res [F(z), p_1] + Res [F(z), p_2]

= 2.8082 – 0.9041 – 5.9928j – 0.9041 + 5.9928j = 1

when n > 0 , the pole at z = 0 vanishes and we have

$$F(z) = \frac{z^n(z^2 + 2z + 1)}{z(z^2 - z + 0.3561)}$$

$$x(n) = \text{Res}\,[F(z), p_1] + \text{Res}\,[F(z), p_2]$$

$$\text{Res}\,[F(z),p_1] = (z-p_1)\ F(z)\Big|_{z=P1}$$

$$= \frac{(z - p_1)z^n(z^2 + 2z + 1)}{z(z - p_1)(z - p_2)}\Bigg|_{z=P_1} \qquad (4.27)$$

$$= \frac{(re^{j\theta})^n[(re^{j\theta})^2 + 2re^{j\theta} + 1]}{re^{j\theta}(re^{j\theta} - re^{-j\theta})}$$

where r = 0.5967 and θ = 33.08°. Noting that this expression is similar to Equation 4-19, we can write

$$\text{Res}\,[F(z), p_1] = (0.5967\, e^{j33.08})^n\ (6.060\ 66\ e^{-j98.58})$$

$$= 6.060\ 66\ (0.5967)^n\ [\cos(33.08\ n - 98.58) + j \sin(33.08n - 98.58]$$

Since p_2 and p_1 are complex conjugate pole pairs we can write:

$$\text{Res}\,[F(z), p_2] = 6.060\ 66\ (0.5967)^n\ [\cos(3308\ n - 98.58) - j \sin(33.08\ n - 98.58)]$$

Thus

$$x(n) = \text{Res}[F(z), p_1] + \text{Res}[F(z), p_2]$$

$$= 12.1213\ (0.5967)^n\ \cos(33.08\ n - 98.58^o), \qquad n > 0$$

which checks with the results for the partial fraction expansion.

Example 4.9 (X(z) contains a second-order pole)

Find the discrete-time sequence, x(n), with the following z-transform :

$$X(z) = \frac{z^2}{(z-0.5)(z-1)^2}$$

Solution

This example is the same as Example (4.6) under partial fraction expansion. According to the residue method the discrete-time sequence is given by

$$x(n) = \sum_{k=1}^{M} \text{Res}\,[F(z), p_k]$$

where, $F(z) = z^{n-1} X(z) = \frac{z^{n+1}}{(z-0.5)(z-1)^2}$

F(z) has a simple pole at z = 0.5 and a second-order pole at z = 1; thus x(n) is given by

$$x(n) = \text{Res}\,[F(z), p_1] + \text{Res}\,[F(z), p_2]$$

$$\text{Res}\,[F(z), 0.5] = \frac{(z-0.5)z^{n+1}}{(z-0.5)(z-1)^2} = \frac{z^{n+1}}{(z-1)^2}\bigg|_{z=0.5}$$

$$= 0.5(0.5)^n/(0.5)^2 \qquad = 2(0.5)^2 \qquad = (0.5)^{n-1}$$

$$\text{Res}\,[F(z), 1] = \frac{d}{dz}\left[\frac{(z-1)^2 z^{n+1}}{(z-0.5)(z-1)^2}\right] \qquad = \frac{(z-0.5)(n+1)z^n - z^{n+1}}{(z-0.5)^2}\bigg|_{z=1}$$

$$= [(0.5)(n+1) - 1] / (0.5)^2 \qquad = 2(n-1)$$

Combining the results, we have

$$x(n) = 2[(n-1) + (0.5)^n]$$

which is the same result as for the partial fraction expansion method.

The reader may have noticed that the partial fraction and residue methods are related. Both methods required the evaluation of residues albeit performed in different ways. The partial fraction method requires the evaluation of the residues of X(z), that is the C_k, while the residue method requires the evaluation of the residues of $z^{n-1}X(z)$. When X(z) has first-order poles we have

$$\text{Res}[z^{n-1}X(z), p_k] = z^n \text{Res}[X(z), p_k] = z^n C_k \qquad (4.28)$$

4.3.4 Comparison of the inverse z-transform methods

We have discussed in some detail three methods of obtaining the inverse z-transform: the power series, partial fraction expansion and the residue method. A limitation of the power series method is that it does not lead to a closed form solution (although this can be deduced in simple cases), but it is simple and lends itself to computer implementation. However, because of its recursive nature care should be taken to minimize possible build-up of numerical errors when the number of data points in the inverse z-transform is large, for example by using double precision.

Both the partial fraction expansion and the residue methods lead to closed form solutions. The main disadvantage with both methods is the need to factorize the denominator polynomial, that is finding the poles of X(z). If the order of X(z) is high finding the poles of X(z), if X(z) is not in factored from, is quite a difficult task. Both methods may also involve high-order differentiation if X(z) contains multiple-order poles. Clearly, if closed form solution is required then the partial fraction or residue method is the most appropriate. The partial fraction method is particularly useful in generating the coefficients of parallel structures for digital filters. The residue method is widely used in the analysis of quantization errors in discrete-time systems.

4.4 PROPERTIES OF THE Z-TRANSFORM

Some useful properties of the z-transform which have found practical use in DSP are described briefly below. The proofs for some of these properties are given as problems at the end of the chapter.

(1) **Linearity:** If the sequences $x_1(n)$ and $x_2(n)$ have z-transforms $X_1(z)$ and $X_2(z)$, then the z-transform of their linear combination is

$$ax_1(n) + bx_2(n) \rightarrow aX_1(z) + bX_2(z) \qquad (4.29)$$

(2) **Delays or shifts:** If the z-transform of a sequence, x(n), is X(z) then the z-transform of the sequence delayed by m samples is $z^{-m}X(z)$. This property is widely used in converting the z transfer function of discrete-time systems into time domain difference equations and vice versa.

$$x(n) \rightarrow X(z)$$

$$x(n - m) \rightarrow z^{-m}X(z)$$

(3) **Convolution:** Given a discrete-time LTI system with input, x(n), and impulse response, h(k), the output of the system is given by

$$y(n) = \sum_{k=-\infty}^{\infty} h(k)\, x(n - k) \qquad (4.30a)$$

In terms of the z-transform, the input and output are related as

$$Y(z) = H(z)\, X(z) \qquad (4.30b)$$

where X(z), H(z) and Y(z) are, respectively, the z-transform of x(n), h(k) and y(n). Given X(z) and H(z), the output y(n) can be obtained by inverse z-transform Y(z).

It is seen that the convolution operation in Equation 4-30a has become a multiplicative process in the z-domain. H(z) is often referred to as the system transfer function.

(4) **Differentiation:** If X(z) is the z-transform of x(n), then the z-transform of n x(n) can be obtained by differentiating X(z):

$$x(n) \rightarrow X(z)$$

$$n\, x(n) \rightarrow -z \frac{dX(z)}{dz} \qquad (4.31)$$

This property is useful in obtaining the inverse z-transform when X(z) contains multiple order poles.

(5) **Relationship with the Laplace transform.**

Continuous-time systems or signals are normally described using the Laplace transform. If we let $z = e^{sT}$, where s the complex Laplace variable given by

$$s = d + j\omega$$

then

$$z = e^{ST}\ e^{(d+j\omega)T} = e^{dT}\, e^{j\omega T} \qquad (4.32)$$

Thus

$$|z| = e^{dT} \text{ and } \angle z = \omega T = 2\pi f/F_s = 2\pi\, \omega/\omega_s$$

where ω_s (rads^{-1}) is the sampling frequency.

As ω varies from $-\infty$ to ∞ the s-plane is mapped to the z-plan as shown in Fig.4.5. The entire $j\omega$ axis in the s-plane is mapped onto the unit circle. The left-hand s-plane is mapped to the inside of the unit circle and the right-hand s-plane maps to the outside of the unit circle.

In terms of frequency response, the $j\omega$ axis is the most important in the s-plane. In this case, d = 0, and frequency points in the s-plane are related to points on the z-plane unit circle by

$$Z = e^{j\omega T} \qquad (4.33)$$

Table 4.2 shows how some specific frequencies are mapped from the s-plane to the z-plane. It is clear that the mapping is not unique since for example the two frequencies $\omega = \omega_s$ and $\omega = 2\omega_s$ in the s-plane map to the same point on the unit circle.

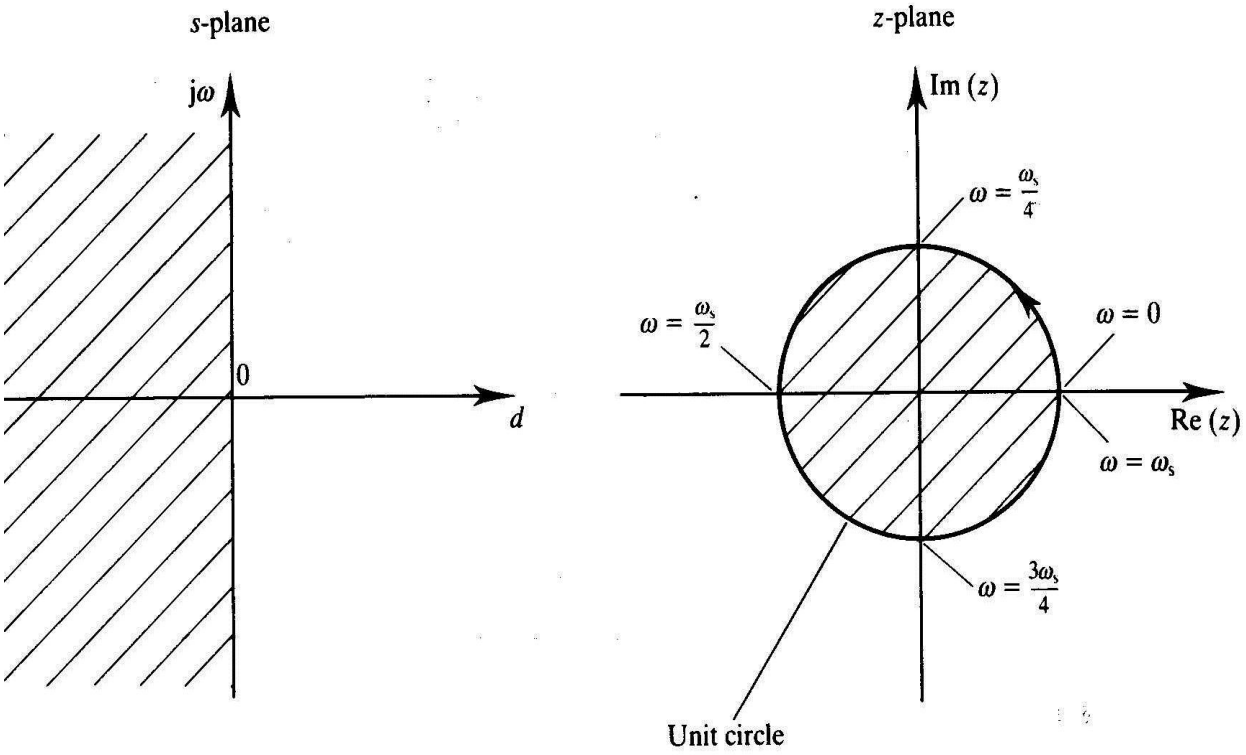


Fig. 4.5 Mapping of s-plane to z-plane. The left-hand side of the s-plane maps to the interior of the z-plane, the right-hand side maps to the exterior and the j ω-axis maps onto the unit circle.

Table 4.2 Mapping of frequencies from the s-plane to the z-plane.

s-plane : ω (rads^{-1})	z-plane : ωT (rad)
0	0
$\omega_s/4$	$\pi/2$
$\omega_s/2$	π
$3\omega_s/4$	1.25π
ω_s	2π
$1.25\omega_s$	$\pi/2$
$1.5\omega_s$	π
$1.75\omega_s$	1.25π
$2\omega_s$	2π

4.5 SOME APPLICATIONS OF THE Z-TRANSFORM IN DSP

4.5.1 Pole-zero description of discrete-time systems

In most practical discrete-time systems the z-transform, that is the system transfer function H(z), can be expressed in terms of its poles and zeros. Consider for example, the following z-transform representing a general, N^{th}-order discrete-time filter (where N = M):

$$H(z) = \frac{N(z)}{D(z)} \tag{4.34}$$

Where :

$$N(z) = a_0z^N + a_1z^{N-1} + a_2z^{N-2} + \ldots\ldots + a_N$$

$$D(z) = b_0z^N + b_1z^{N-1} + b_2z^{N-2} + \ldots\ldots + b_N$$

The a_k and b_k are the coefficients of the filter.

If H(z) has poles at $z = p_1, p_2, \ldots\ldots\ldots, p_N$ and zeros at $z = z_1, z_2, \ldots\ldots, z_N$,
then H (z) can be factored and represented as

$$H(z) = \frac{K(z-z_1)(z-z_2).....(z-z_N)}{(z-p_1)(z-p_2).....(z-z_N)} \tag{4.35}$$

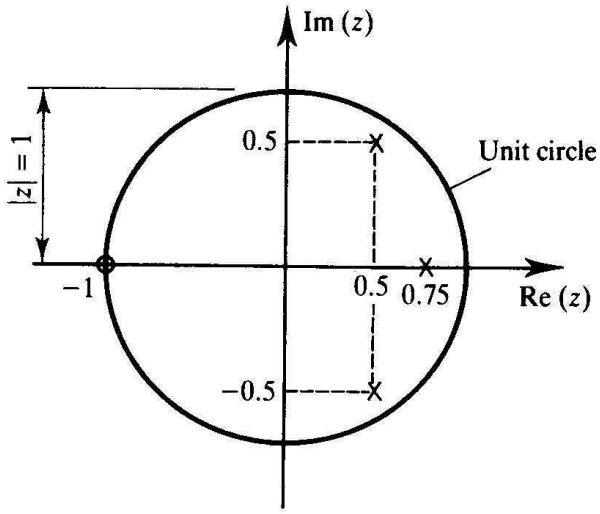


Fig. 4.6 Description of a z-transform in the form of a pole-zero diagram: X, pole; o, zero.

Where z_i is the ith zero, p_i is the ith pole and K is the gain factor. You will recall that the poles of a z-transform such as H(z) are the values of z for which H(z) becomes infinity. The values of z for which H(z) becomes zero are referred to as zeros. The poles and zeros of H(z) may be real or complex. When they are complex, they occur in conjugate pairs, to ensure that the coefficients, a_k and b_k, are real. It should be clear from Eq. 4.35 that if the locations of the poles and zeros of H(z) are known, then H(z) itself can be readily reconstructed to within a constant.

The information contained in the z-transform can be conveniently displayed as a pole-zero diagram; see for example Fig. 4.6. In the diagram, X marks the position of a pole a o denotes the position of a zero. For this example, the poles are located at z = 0.5 $\pm$ 0.5j, and at z = 0.75. A single zero is at z = -1. An important feature of the pole-zero diagram is the unit circle, that is the circle defined by |z| = 1; see Fig. 4.6. As will become clear, the unit circle plays an important role in the analysis and design of discrete-time systems.

The pole-zero diagram provides an insight into the properties of a given discrete-time system. For example, from the locations of the poles and zeros we can infer the frequency response of the system as well as its degree of stability. For a stable system, all the poles must lie inside the unit circle (or be coincident with zeros on the unit circle).

Often, the z-transform is not available in factored form but as a ratio of polynomials such as Eq. 4.34. In these cases, describing the z-transform, H(z), in terms of its poles and zeros will require finding the roots of the denominator polynomial, D(z), and those of the numerator polynomial, N(z).

For a second-order polynomial which has the form:

$$ax^2+bx+c,$$

the roots are given by

$$\frac{-b \pm (b^2 - 4ac)^{1/2}}{2a} \qquad (4.36)$$

For higher-order polynomials, finding the roots of N(z) or D(z) is a difficult task. In practice, this is often achieved using numerical methods involving, for example, Newton's and/or Baistow's algorithms (see for example, Atkinson and Harley (1983)). The need to find the poles and zeros often arises in connection with the design of discrete filters and in stability analysis. Fortunately, in the case of discrete-time filter design the poles and zeros are automatically generated by the filter design software, obviating the need to find the roots directly.

Example 4.10

(1) Express the following transfer function in terms of its poles and zeros and sketch the pole-zero diagram:

$$H(z) = \frac{1 - z^{-1} - 2z^{-2}}{1 - 1.75z^{-1} + 1.25z^{-2} - 0.375z^{-3}}$$

(2) Determine the transfer function, H(z), of a discrete-time filter with the pole-zero diagram shown in Fig. 4.7.

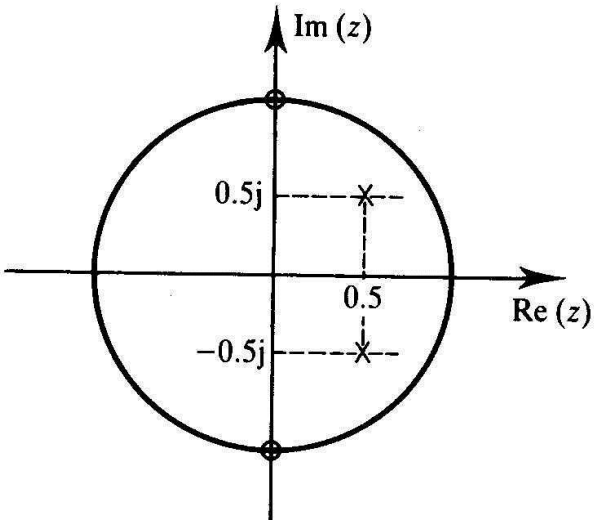


Fig. 4.7 Pole-zero diagram for Example (4.10), part (2)

Solution

(1) First, we express H(z) positive powers of z and then factorize it so that its poles and zeros can be determined. If we multiply the denominator and numerator by z^3, the highest power of z, we obtain.

$$H(z) = \frac{z(z^2 - z - 2)}{z^3 - 1.75z^2 + 1.25z - 0.375}$$

Factoring, we have

$$H(z) = \frac{z(z-2)(z+1)}{(z-0.5+j0.5)(z-0.5-j0.5)(z-0.75)}$$

Thus, the pole locations are at z = 0.5 $\pm$ 0.5j and at z = 0.75. The zeros are at z = 0, z=2 and z= -1. The pole-zero diagram is depicted in Fig. 4.8.

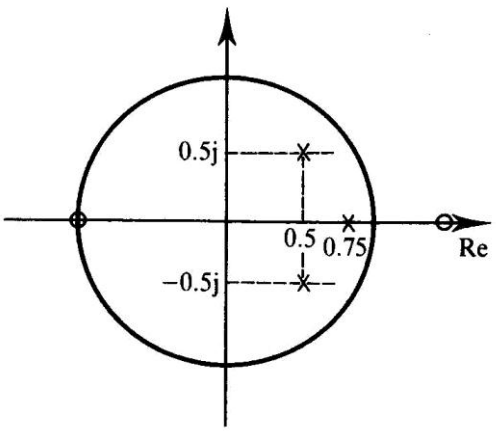


Fig. 4.8 Pole-zero diagram for Example (4-10), part (1).

(2) From the pole-zero diagram of Fig. 4-7, the zeros of the transfer function are at z = $\pm$j and the poles are at z = 0.5 $\pm$ 0.5j. The transfer function can be written down directly:

$$H(z) = \frac{K(z - j)(z + j)}{(z - 0.5 - 0.5j)(z - 0.5 + 0.5j)}$$

$$= \frac{K(z^2 + 1)}{z^2 - z + 0.5}$$

$$= \frac{K(1 + z^{-2})}{1 - z^{-2} + 0.5z^{-2}}$$

4.5.2 Stability Considerations of discrete-time systems

Stability analysis is often carried out as part of the design of discrete-time systems. A useful stability criterion for LTI systems is that all bounded inputs produce bounded outputs. This is the so-called BIBO (bounded input, bounded output) condition. An LTI system is said to be BIBO stable if and only if it satisfies the criterion.

$$\sum_{k=0}^{\infty} |h(k)| < \infty \qquad (4.37)$$

where h(k) is the impulse response of the system. It is obvious that if the impulse response is of finite length the condition above is satisfied since the sum of the impulse response coefficients will be finite. Thus, stability considerations apply only to systems with impulse response of infinite duration.

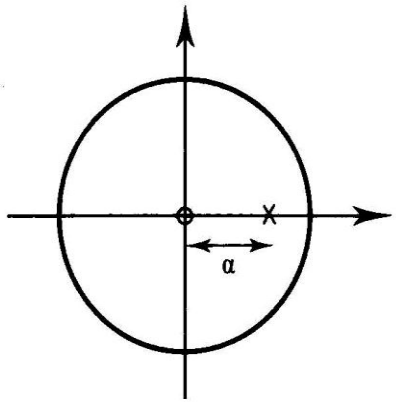
For the output to be bounded all the poles must lie inside the unit circle. When a pole lies outside the unit circle the system is unstable. In practice, a system with a pole on the unit circle is also regarded as unstable or potentially unstable, since a minor disturbance or error will invariably push the system into instability. An exception is when a pole is coincident with a zero on the unit circle so that its effects are nullified. For an unstable system, the impulse response will increase indefinitely with time.

In principle, testing for stability is simple: find the pole positions of the z-transform. If any pole is on or outside the unit circle (unless it is coincident with a zero on the unit circle) then the system is unstable. In practice, finding the positions of the poles may not be an easy task.

A simple test that may be used when the system z-transform, H(z), is not available in factored form is to obtain and plot a sufficient number of values of the impulse response, by finding the inverse z-transform. If the impulse response increases indefinitely with time or fails to decay fast enough then the system is either unstable or marginally stable. Figure 4.9 shows examples of the behavior of the impulse response of a simple discrete-time system for various degrees of stability.

	h(n)			
n	α=0.5	α=0.99	α=1	α=1.5
0	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00
1	1.00000e+01	1.00000e+01	1.00000e+01	1.00000e+01
2	5.00000e+00	9.90000e+00	1.00000e+01	1.50000e+01
3	2.50000e+00	9.80100e+00	1.00000e+01	2.25000e+01
4	1.25000e+00	9.70299e+00	1.00000e+01	3.37500e+01
5	6.25000e−01	9.60596e+00	1.00000e+01	5.06250e+01
6	3.12500e−01	9.50990e+00	1.00000e+01	7.59375e+01
7	1.56250e−01	9.41480e+00	1.00000e+01	1.13906e+02
8	7.81250e−02	9.32065e+00	1.00000e+01	1.70859e+02
9	3.90625e−02	9.22745e+00	1.00000e+01	2.56289e+02

(a)



(b)

Fig. 4.9 An illustration of the behavior of the impulse response of a system for various degrees of stability: (a) impulse response; (b)z-plane pole-zero diagram.
The system z-transform is $10z^{-1}/(1-\alpha z^{-1})$.
(i) For $\alpha = 0.5$ system is stable,
(ii) for $\alpha = 0.99$ it is marginally stable,
(iii) for $\alpha = 1$ it is potentially unstable, and
(iv) for $\alpha = 1.5$ it is unstable.
Notice, for example, that for $\alpha = 0.5$ the impulse response values decay rapidly as n increases, while, for $\alpha = 1.5$, the impulse response values increase rapidly.

4.5.3 Difference equations of discrete-time systems

The difference equation specifies the actual operations that must be performed by the discrete-time system on the input data, in the time domain, in order to generate the desired output. The difference equation, for most practical cases of interest, may be written as

$$y(n) = \sum_{k=0}^{N} a_k x(n-k) - \sum_{k=1}^{M} b_k y(n-k) \qquad (4.38)$$

where x(n) is the input sample, y(n) is the output sample, y(n–k) is the previous output and a_k, b_k are system coefficients. Equation 4-38 indicates that the current output, y(n), is obtained from present and past input samples and previous outputs, y(n-k).

The difference equations for discrete-time systems are readily obtained from their transfer functions and vice versa using the delay property of the z-transform:

$$a_k x(n) \leftrightarrow a_k X(z)$$

$$a_k x(n-k) \leftrightarrow a_k z^{-k} X(z)$$

Thus, Eq. 4.38 may be written as

$$Y(z) = \sum_{k=0}^{N} a_k z^{-k} X(z) - \sum_{k=1}^{M} b_k z^{-k} Y(z) \qquad (4.39)$$

Simplifying, we obtain the z-domain transfer function of the discrete system, H(z) :

$$H(z) = \frac{Y(z)}{X(z)} = \sum_{k=0}^{N} a_k z^{-k} \; / \; (1 + \sum_{k=1}^{M} b_k z^{-k}) \qquad (4.40)$$

If all the denominator coefficients, b_k, are zero. Eqs. 4.38 and 4.39 reduce to

$$y(n) = \sum_{k=0}^{N} a_k x(n-k) \qquad (4.41a)$$

$$H(z) = \frac{Y(z)}{X(z)} = \sum_{k=0}^{N} a_k z^{-k} \qquad (4.41b)$$

The output of the system, y(n), now depends only on the present and past input samples but not on previous outputs as in Eq. 4.38. The coefficients, the a_k, in this case represent the impulse response of the system and are normally denoted by the symbol h(k). This class of LTI system is referred to as finite impulse response (FIR) systems, since the length of h(k) is clearly finite.

Systems characterized by Eqs. 4.38 and 4.40 where at least one of the denominator coefficients is nonzero are referred to as infinite impulse response (IIR) systems. In IIR systems at least one of the poles will be non-zero, but FIR systems normally contain no poles.

4.5.4 Impulse response estimation of discrete-time systems

In the design of discrete-time systems the need often arises to obtain values of impulse responses. For example, in FIR system design the impulse response is required to implement the system, and in IIR system design the values are required for stability

analysis. The impulse response may also be used to evaluate the frequency response of the system.

The impulse response of a discrete-time system may be defined as the inverse z-transform of the system's transfer function, H(z):

$$h(k) = Z^{-1}[H(z)],\ k = 0, 1, \ldots\ldots$$

If the z-transform, H(z), is available as a power series, that is

$$H(z) = \sum_{n=0}^{\infty} h(n)z^{-n}$$

$$= h(0) + h(1)z^{-1} + h(2)z^{-2} + h(3)z^{-3} + \ldots\ldots \qquad (4.42)$$

the coefficients of the z-transform give directly the impulse response. For IIR systems, H(z) is often expressed as a ratio of polynomials such as Eq. 4.9. In this case, the IZT methods described in Section 4.2 can be used to obtain the impulse response of the system.

$$H\left(e^{j\omega T}\right) = \left.\frac{a_0 + a_1 z^{-1} \ldots + a_N z^{-N}}{b_0 + b_1 z^{-1} + \ldots + b_M z^{-M}}\right|_{z=e^{j\omega T}} \qquad (4.9)$$

The impulse response may also be viewed as the response of a discrete-time system to a unit impulse, u(n), which has a value of 1 at n=0 and a value of 0 at all other values of n. This view arises from the fact that if we make the input to the system equal to the unit impulse, x(n) = u(n), the output of the system is in fact equal to h(n), the system's impulse response :

$$y(n) = \sum_{n=0}^{\infty} h(k)x(n-k) = \sum_{n=0}^{\infty} h(k)u(n-k)$$

$$= h(0)u(n)+h(1)u(n-1) + h(2)u(n-2) + \ldots\ldots$$

$$= h(n),\ n = 0, 1, \ldots\ldots\ldots \qquad (4.43)$$

This provides a simple alternative method of computing h(n) (indeed it provides another method of obtaining the inverse z-transform) as illustrated by the following example.

Example 4.11

Find the impulse response of the discrete-time filter characterized by the following z-transfer function:

(1) by using the power series method and

(2) by applying a unit impulse to the system

$$H(z) = \frac{1 - z^{-1}}{1 + 0.5z^{-1}}$$

Solution

(1) Using the power series method, the values of the impulse response are obtained as:

$$1 - 1.5z^{-1} + 0.75z^{-2} - 0.325z^{-3} .....$$

$$1 + 0.5z^{-1} \overline{)\, 1 - z^{-1}}$$

$$1 + 0.5z^{-1}$$

$$-1.5z^{-1}$$

$$-1.5z^{-1} - 0.75z^{-2}$$

$$0.75\, z^{-2}$$

$$0.75z^{-2} + 0.375z^{-3}$$

$$-\, 0.3725z^{-3}$$

From the quotients the impulse response values are

$$h(0) = 1,\ h(1) = -\,1.5,\ h(2) = 0.75,\ h(3) = -0.325$$

(2) First, we need to obtain the difference equation of the filter from the transfer function:

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1 - z^{-1}}{1 + 0.5z^{-1}}$$

Cross-multiplying and using the delay property of the z-transform we obtain the difference equation:

$$Y(z) + 0.5Y(z)z^{-1} = X(z) - X(z)z^{-1}$$

$$y(n) + 0.5y(n - 1) = x(n) - x(n - 1)$$

Simplifying we have

$$y(n) = x(n) – x(n – 1) – 0.5y\ (n – 1)$$

The impulse response of the filter can now be obtained by letting x(n) = u(n) where

$$u(n) = 1, \qquad n = 0$$
$$= 0, \qquad n \neq 0$$

and assuming an initial condition of y(– 1) = 0 :

$$y(0) = 1$$
$$y(1) = x(1) – x(0) – 0.5y(0) = 0 – 1 – 0.5 = - 1.05$$
$$y(2) = x(2) – x(1) – 0.5y(1) = -0.5 \times -1.05 = 0.75$$
$$y(3) = x(3) – x(2) – 0.5y(2) = -0.5 \times 0.75 = -0.325$$

From this the impulse response values are

$$h(0) = 1,\ h(1) = - 1.5,\ h(2) = 0.75,\ h(3) = - 0.375$$

It is seen that both method lead to identical results.

4.5.5 Realization structures for digital filters

Discrete-time filters are often represented in the form of block or signal flow diagrams. The diagrams are a convenient way of representing the difference equations or equivalently the transfer functions. Consider for example, a simple discrete filter with the following difference equation:

$$y(n) = x(n-1) – b_1y(n-1) + b_2y(n-2) + b_3y(n-3) \qquad (4.44)$$

The block diagram representation of this equation is shown in Fig. 4.10(a). In the figure, the symbol z^{-1} represents a delay of 1 unit of time as may be deduced from the signals at various nodes; the arrows represent multipliers and the constants next to them the multiplication factors. The relationship between the difference equation and the block diagram should be evident. A signal flow diagram representation of the same difference

equation is shown in Fig. 410(b). It is common to refer to the block or flow diagram as a realization diagram.

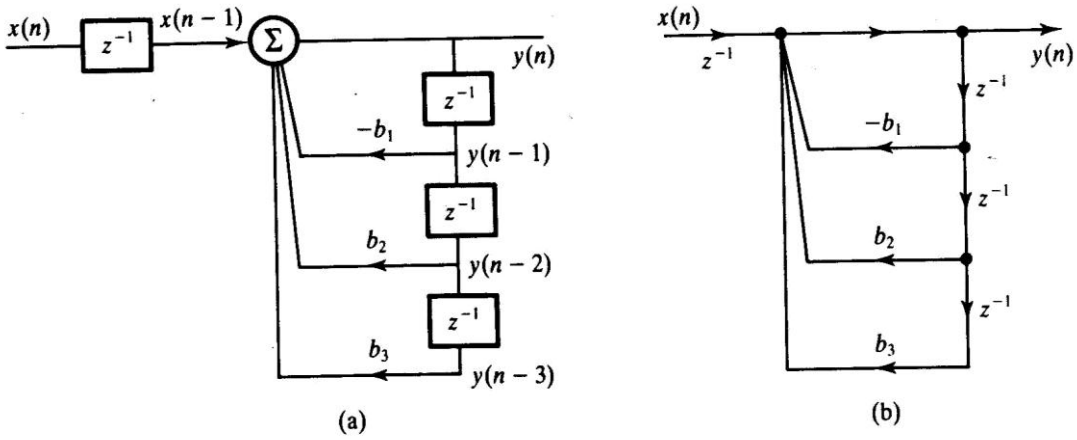


Fig. 4.10 Realization diagrams for a difference equation: (a) in block diagram form; (b) signal flow diagram.

When the order of H(z) is high, the discrete-time filter is rarely realized directly as Fig. 4.10 implies because large errors will result if a small number of bits is used to represent the coefficients and to execute the difference equations. The common practice is to decompose the transfer function into a cascade or parallel combination of second- and/or first-order z-transforms. For cascade realization, the transfer function, H(z), is factored as

$$H(z) = H_1(z)\, H_2(z) \ldots H_k(z) = \prod_{i=1}^{k} H_i(z) \qquad (4.45)$$

where $H_i(z)$ is either a second- or a first-order section:

$$H_i(z) = \frac{a_0 + a_{1i} z^{-1} + a_{2i} z^{-2}}{1 + b_{1i} z^{-1} + b_{2i} z^{-2}} \qquad \text{second order}$$

$$H_i(z) = \frac{a_0 + a_{1i} z^{-1}}{1 + b_{1i} z^{-1}} \qquad \text{first order}$$

and K is the integer part of (M + 1)/2. The overall z-transform is the product of the individual z-transforms: see Fig. 4.11.

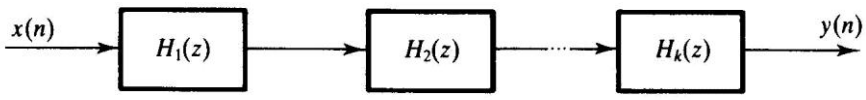


Fig. 4.11 General structure for cascade realization.

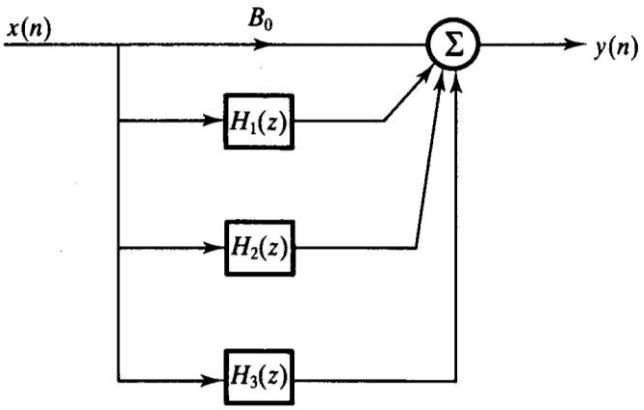


Fig. 4.12 General structure for parallel realization.

For parallel realization, the transfer function is decomposed, using partial fractions, to give

$$H(z) = B_0 + \sum_{i=1}^{k} H_i(z) \qquad (4.46)$$

where, as above, $H_i(z)$ is either a second-order or a first-order section but of the form

$$H_i(z) = \frac{a_{0i} + a_{1i}z^{-1}}{1 + b_{1i}z^{-1} + b_{2i}z^{-2}} \qquad \text{second order}$$

$$H_i(z) = \frac{a_0}{1 + b_{1i}z^{-1}} \qquad \text{first order}$$

where K is the integer part of (M+1)/2 and

$$B_0 = a_N / b_M$$

Figure 4.12 shows the general structure for the parallel realization.

In the design of digital filters, software packages are normally used to obtain the coefficients, a_{ki} and b_{ki}, above. Unfortunately, most software packages produce coefficients only for the cascade structure. The coefficients for the parallel structure can be obtained from those of the cascade structure by using a partial fraction expansion.

SHEET # 4

4.1 Define the following items: Time invariant discrete – time system, Linear discrete – time system, and causal signal.

4.2 Find and plot the z-transform of the following discrete-time sequences:

(a) $x(n)=\{0,1,3,5,3,1,0\}$

(b) $x(-3)=0$, $x(-2)=1$, $x(-1)=3$, $x(0)=5$, $x(1)=3$, $x(2)=1$ & $x(3)=0$.

4.3 Find and plot the z-transform of each of the following discrete-time sequences:

(a) $x(n) = \sin(nT)$, $n=0,1,\ldots$

(b) $x(n) = a^n$, $n > 0$

$= 0$ $n < 0$

4.4 An exponential sequence is defined as

$$x(n) = e^{-kn}, \quad n \geq 0$$

Find and plot its z-transform, including the constraint on the z-transform to converge, for: (a) k is real. (b) k is complex.

4.5 Find, using the power series method the first five values of causal discrete-time sequence corresponding to each of the following z-transform:

(a) $X(z) = (z - 1)/(z - 0.7071)^2$

(b) $X(z) = 1/(z - 0.5)(z + 0.9)^3$

4.6 Repeat problem (5) using the partial fraction expansion method.

4.7 Repeat problem (5) using the residue method.

4.8 Given the difference equation

$$y(n) + B_1 y(n-1) + B_2 y(n-2) = A$$

where A, B_1 and B_2 are arbitrary constants, find an expression for z-transform, Y(z). Use an appropriate inverse z-transform technique to obtain a closed form for y (n).

4.9 For the discrete-time system shown in Fig.4.13, obtain the difference equation, and then derive the system transfer function H (z).

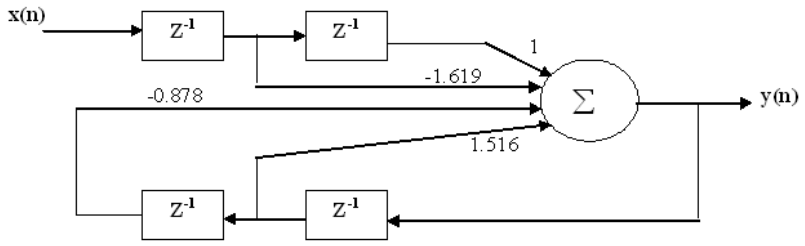


Fig.4.13

4.10 The transfer function of a discrete-time system has poles at z=0.5, z=0.1±j0.02 and a third-order zero at z=-1 and z=1.

(a) Sketch the pole-zero diagram for the system.

(b) Derive the system transfer function, H (z), from the pole-zero diagram.

(c) Develop the difference equation.

(d) Draw the realization diagram in signal flowgraph form.

4.11 The signal flow graph of a discrete-time system is shown in Fig. 4.14 Find the two-step difference equation relating the output, y (n), and input, x (n). Derive the transfer function, H (z), from the difference equation.

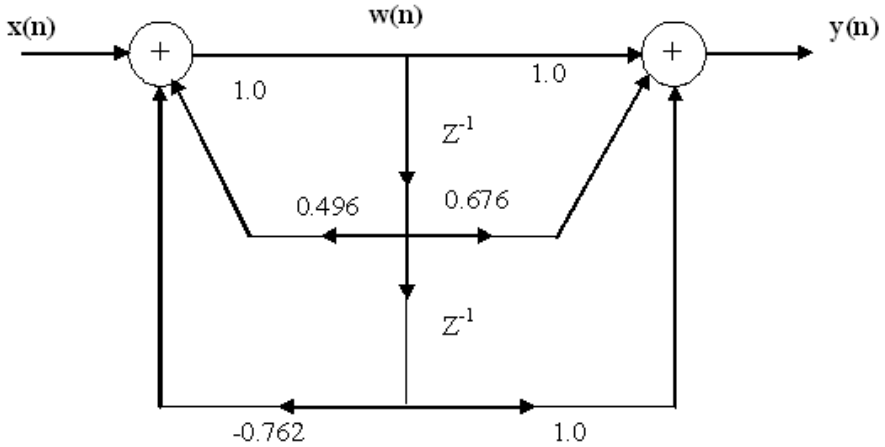


Fig.4.14

Assignment # 4

A4.1 Find and plot the z-transform of the following discrete-time sequences:

(i) $x(n) = \{0,1,3, \pm 5,-3,-1,0\}$

(ii) $x(n) = 1, \quad 0 \leq n \leq N-1$ (use N=10)

$= 0, \quad$ elsewhere

A4.2 Find, using the power series method the first five values of causal discrete-time sequence corresponding to the following z-transform:

$$X(z) = (z^3 - z^2 + z - 1)/(z + 0.9)^3$$

A4.3 Repeat problem (A4.2) using the partial fraction expansion method.

A4.4 Repeat problem (A4.2) using the residue method.

CHAPTER 5

DIGITAL FILTERS DESIGN

5.1 INTRODUCTION TO DIGITAL FILTERS

A filter is essentially a system or network that selectively changes the wave shape, amplitude-frequency and/or phase-frequency characteristics of a signal in a desired manner. Common filtering objectives are to improve the quality of a signal (for example, to remove or reduce noise), to extract information from signals or to separate two or more signals previously combined to make, for example, efficient use of an available communication channel.

A digital filter, as we shall see later, is a mathematical algorithm implemented in hardware and/or software that operates on a digital input signal to produce a digital output signal for the purpose of achieving a filtering objective. The term digital filter refers to the specific hardware or soft ware routine that performs the filtering algorithm. Digital filters often operate on digitized analogue signals or just numbers, representing some variable, stored in a computer memory.

A simplified block diagram of a real-time digital filter, with analogue input and output signals, is given in Fig. 5.1. The band limited analogue signal is sampled periodically and converted into a series of digital samples, x(n), n = 0, 1, …. The digital processor implements the filtering operation, mapping the input sequence, x(n), into the output sequence, y(n), in accordance with a computational algorithm for the filter. The DAC converts the digitally filtered output into analogue values which are then analogue filtered to smooth and remove unwanted high frequency components.

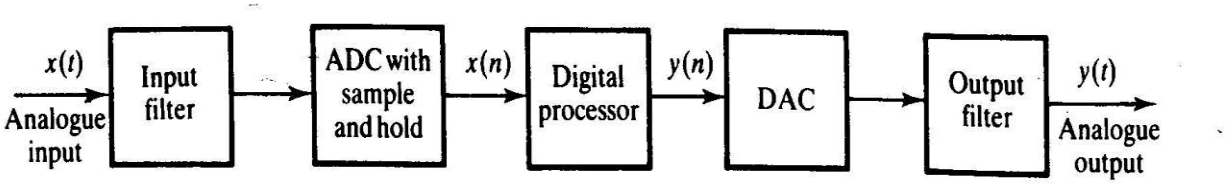


Fig. 5.1 A simplified block diagram of a real-time digital filter with analogue input and output signals.

Digital filters play very important roles in DSP. Compared with analogue filters they are preferred in a number of applications (for example: data compression, biomedical signal processing, speech processing, image processing, data transmission, digital audio, telephone echo cancellation) because of one or more of the following advantages:

1- Digital filters can have characteristics which are not possible with analogue filters, such as a truly linear phase response.

2- Unlike analogue filters, the performance of digital filters does not vary with environmental changes, e.g. thermal variations. This eliminates the need to calibrate periodically.

3- The frequency response of a digital filter can be automatically adjusted if it is implemented using a programmable processor, which is why they are widely used in adaptive filters.

4- Several input signals or channels can be filtered by one digital filter without the need to replicate the hardware.

5- Both filtered and unfiltered data can be saved for further use.

6- Advantage can be readily taken of the tremendous advancements in VLSI technology to fabricate digital filters and to make them small in size, to consume low power, and to keep the cost down.

7- In practice, the precision achievable with analogue filters is restricted; e.g., typically a maximum of only about 60 to 70 dB stopband attenuation is possible

with active filters designed with off-the-shelf components. With digital filters the precision is limited only by the word length used.

8- The performance of digital filters is repeatable from unit to unit.

9- Digital filters can be used at very low frequencies, found in many biomedical applications for example, where the use of analogue filters is impractical. Also, digital filters can be made to work over a wide range of frequencies by a mere change to the sampling frequency.

The following are the main disadvantages of digital filters compared with analogue filters:

1- **Speed limitation.**

The maximum bandwidth of signals that digital filters can handle, in real time, is much lower than for analogue filters. In real-time situations, the analogue-digital filter performance. The conversion time of the ADC and the settling time of the DAC limit the highest frequency that can be processed. Further, the speed of operation of a digital filter depends on the speed of the digital processor used and on the number of arithmetic operations that must be performed for the filtering algorithm, which increases the filter response is made tighter.

2- **Finite word length effects.**

Digital filters are subject to ADC noise resulting from quantizing a continuous signal, and to round off noise incurred during computation. With higher order recursive filters, the accumulation of round off noise could lead to instability.

3- **Long design and development times.**

The design and development times for digital filters, especially hardware development, can be much longer than for analogue filters. However, once developed the hardware and/or software can be used for other filtering or DSP tasks with little or no modifications. Good computer-aided design (CAD) support can make the design of digital filters an enjoyable task, but some expertise is required to make full and effective use of such design aids.

5.2 TYPES OF DIGITAL FILTERS: FIR AND IIR FILTERS

Digital filters are broadly divided into two classes, namely infinite impulse response (IIR) and finite impulse response (FIR) filters. Either type of filter, in its basic form, can be represented by its impulse response sequence, h(k) (k = 0, 1, …), as in Figure 5.2. The input and output signals to the filter are related by the convolution sum, which is given in Eqs. 5.1 for the IIR and in 5.2 for the FIR filter.

$$y(n) = \sum_{k=0}^{\infty} h(k)x(n-k) \tag{5.1}$$

$$y(n) = \sum_{k=0}^{N-1} h(k)x(n-k) \tag{5.2}$$

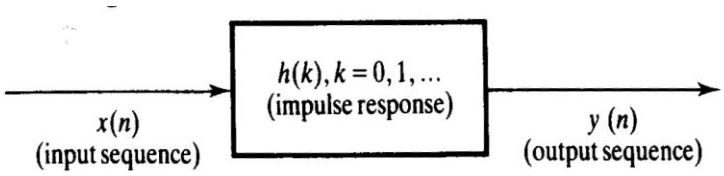


Fig. 5.2 A conceptual representation of a digital filter.

It is evident from these equations that, for IIR filters, the impulse response is of infinite duration whereas for FIR it is of finite duration, since h(k) for the FIR has only N values. In practice, it is not feasible to compute the output of the IIR filter using Eq. 5.1 because the length of its impulse response is too long (infinite in theory). Instead, the IIR filtering equation is expressed in a recursive form:

$$y(n) = \sum_{k=0}^{\infty} h(k)x(n-k) = \sum_{k=0}^{N} a_k x(n-k) - \sum_{k=1}^{M} b_k y(n-k) \tag{5.3}$$

where the a_k and b_k are the coefficients of the filter. Thus, Equations 5.2 and 5.3 are the difference equations for the FIR and IIR filters respectively. These equations, and in particular the values of h(k), for FIR , or a_k and b_k, for IIR, are often very important objectives of most filter design problems. We note that, in Eq. 5.3, the current output sample, y(n), is a function of past outputs as well as present and past input samples, that

is the IIR is a feedback system of some sort. This should be compared with the FIR equation in which the current output sample, y(n), is a function only of past and present values of the input.

Note, however, that when the b_k are set to zero, Eq. 5.3 reduces to the FIR Eq. 5.2.

Alternative representations for the FIR and IIR filters are given in Eqs. 5.4a and 5.4b respectively. These are the transfer functions for these filters and are very useful in evaluating their frequency responses.

$$H(z) = \sum_{k=0}^{N-1} h(k)z^{-k} \qquad (5.4a)$$

$$H(z) = \sum_{k=0}^{N} a_k z^{-k} \Big/ \left(1 + \sum_{k=1}^{M} b_k z^{-k}\right) \qquad (5.4b)$$

As will become clear in the next few sections, factors that influence the choice of options open to the digital filter designer at each stage of the design process are strongly linked to whether the filter in question is IIR or FIR. Thus, it is very important to appreciate the differences between IIR and FIR, their peculiar characteristics, and more importantly, how to choose between them.

5.3 CHOOSING BETWEEN FIR AND IIR FILTERS

The choice between FIR and IIR filters depends largely on the relative advantages of the two filter types.

1- FIR filters can have an exactly linear phase response. The implication of this is that no phase distortion is introduced into the signal by the filter. This is an important requirement in many applications, for example data transmission, biomedicine, digital audio and image processing. The phase responses of IIR filters are nonlinear, especially at the band edges.

2- FIR filters realized non-recursively, that is by direct evaluation of Eq. 5.2, are always stable. The stability of IIR filters cannot always be guaranteed.

3- The effects of using a limited number of bits to implement filters such as round off noise and coefficient quantization errors are much less sever in FIR than in IIR.

4- FIR requires more coefficients for sharp cutoff filters than IIR. Thus for a given amplitude response specification, more processing time and storage will be required for FIR implementation. However, one can readily take advantage of the computational speed of the FFT and multi-rate techniques to improve significantly the efficiency of FIR implementations.

5- Analogue filters can be readily transformed into equivalent IIR digital filters meeting similar specifications. This is not possible with FIR filters as they have no analogue counterpart. However, with FIR it is easier to synthesize filters of arbitrary frequency responses.

6- In general, FIR is algebraically more difficult to synthesize, if CAD support is not available.

From the above, a broad guideline on when to use FIR or IIR would be as follows:

- Use IIR when the only important requirements are sharp cutoff filters and high throughput, as IIR filters, especially those using elliptic characteristics, will give fewer coefficients than FIR.

- Use FIR if the number of filter coefficients is not too large and, is particular, if little or no phase distortion is desired. One might also add that newer DSP processors have architecture that are tailored to FIR filtering, and indeed some are designed specifically for FIRs.

Example 5.1

The following transfer functions represent two different filters meeting identical amplitude – frequency response specifications:

(1) $$H(z) = \frac{a_0 + a_1 z^{-1} + a_2 z^{-2}}{1 + b_1 z^{-1} + b_2 z^{-2}}$$

Where,

$a_0 = 0.4981819$ $a_1 = 0.9274777$ $a_2 = 0.4981819$

$b_1 = -0.6744878$ $b_2 = -0.3633482$

(2) $$H(z) = \sum_{k=0}^{11} h(k) z^{-k}$$

where,

$h(0) = 0.54603280 \times 10^{-2} = h(11)$

$h(1) = -0.45068750 \times 10^{-1} = h(10)$

$h(2) = 0.69169420 \times 10^{-1} = h(9)$

$h(3) = -0.55384370 \times 10^{-1} = h(8)$

$h(4) = -0.63428410 \times 10^{-1} = h(7)$

$h(5) = 0.57892400 \times 10^{0} = h(6)$

For each filter,

(a) State whether it is an FIR or IIR filter,

(b) Represent the filtering operation in a block diagram form and write down the difference equation, and

(c) Determine and comment on the computational and storage requirements.

Solution

(a) Filters (1) and (2) are IIR and FIR respectively.

(b) The block diagram for filter (1) is given in Fig. 5.3 (a).

The corresponding set of difference equations is:

$$w(n) = x(n) - b_1 w(n-1) - b_2 w(n-2)$$

$$y(n) = a_0 w(n) + a_1 w(n-1) + a_2 w(n-2)$$

The block diagram for filter (2) is given in Fig. 5.3(b). The corresponding difference eq. is:

$$y(n) = \sum_{k=0}^{11} h(k)x(n-k)$$

(c) From examination of the two difference equations the computational and storage requirements for both filters are summarized below:

	FIR	IIR
Number of multiplications	12	5
Number of additions	11	4
Storage locations (coefficients and data)	24	8

It is evident that the IIR filter is more economical in both computational and storage requirements than the FIR filter. However, we could have exploited the symmetry in the FIR coefficients to make the FIR filter more efficient, although at the expense of its obvious implementation simplicity. A point worth making is that, for the same amplitude response specifications, the number of FIR filter coefficients (12 in this example) is typically six times the order (the highest power of z in the denominator) of the IIR transfer function (2 in this case).

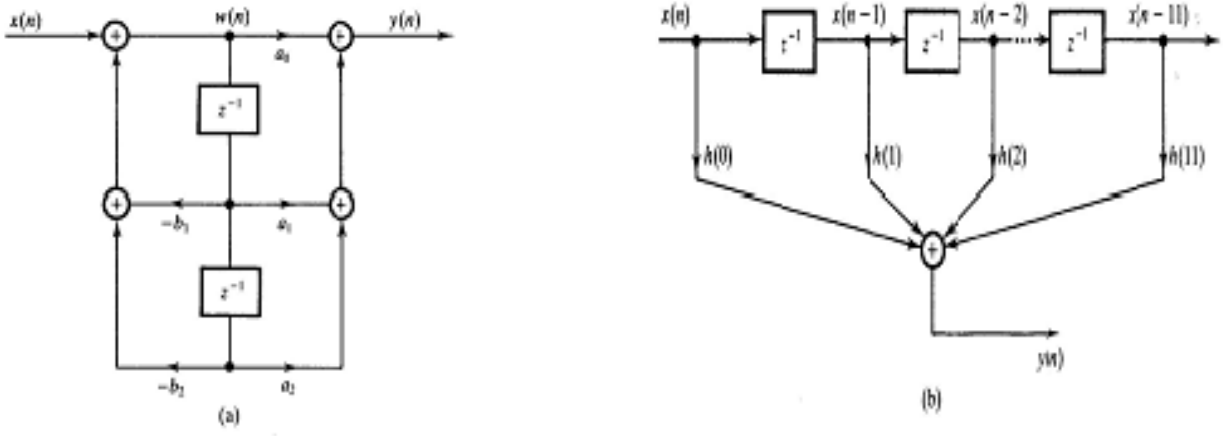


Fig. 5.3 (a) Block diagram representation of the IIR filter of Example 5.1. (b) Block diagram representation of the FIR filter of Example 5.1.

5.4 FILTER DESIGN STEPS

The design of a digital filter involves five steps:

- Specification of the filter requirements;
- Calculation of suitable filter coefficients;
- Representation of the filter by a suitable structure (realization);
- Analysis of the effects of finite word length on filter performance;
- Implementation of filter in software and/or hardware.

The five steps are not necessarily independent; nor are they always performed in the order given. In fact techniques are now available that combine the second and aspects of the third and fourth steps. However, the approach discussed here gives a simple step-by-step guide that will ensure a successful design. To arrive at an efficient filter, it may be necessary to iterate a few times between the steps, especially if the problem specification is not watertight, as is often the case, or if the designer wants to explore other possible designs. Detailed discussions of these steps now follow.

5.4.1 Specification of the filter requirements

Requirement specifications include specifying:

(i) signal characteristics (types of signal source and sink, I/O interface, data rates and width, and highest frequency of interest),

(ii) the characteristics of the filter (the desired amplitude and/or phase responses and their tolerances (if any), the speed of operation and modes of filtering (real time or batch),

(iii) the manner of implementation (for example, as a high level language routine in a computer or as a DSP processor-based system, choice of signal processor), and

(iv) other design constraints (for example, the cost of the filter).

The designer may not have enough information to specify the filter completely at the outset, but as many of the filter requirements as possible should be specified to simplify the design process.

Although the above requirements are application dependent it will be helpful to devote some time on some aspects of (ii). The characteristics of digital filters are often specified in the frequency domain. For frequency selective filters, such as lowpass and bandpass filters, the specifications are often in the form of tolerance schemes. Figure 5.4 depicts such a scheme for a lowpass filter. The shaded horizontal lines indicate the tolerance limits. In the passband, the magnitude response has a peak deviation of δ_P and, in the stopband, it has a maximum deviation of δ_s.

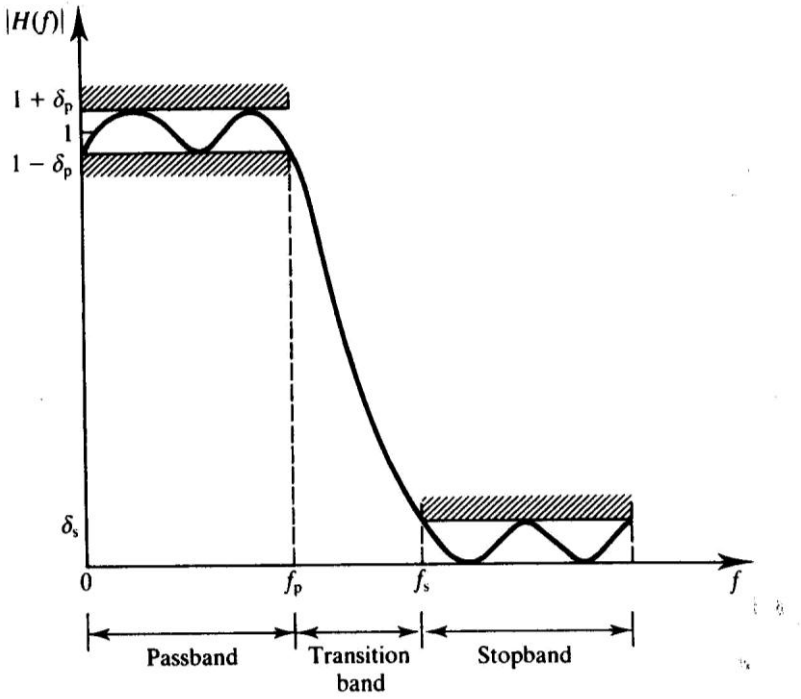


Fig. 5.4 Tolerance scheme for a lowpass filter.

The width of the transition band determines how sharp the filter is. The magnitude response decreases monotonically from the passband to the stopband in this region. The following are the key parameters of interest :

δ_P Passband deviation

δ_s stopband deviation

f_P passband edge frequency

f_s stopband edge frequency

The edge frequencies are often given in normalized form, that is as a fraction of the sampling frequency (f/F_s), but specifications using standard frequency units of hertz, kilohertz are valid and sometimes are more meaningful, especially to the inexperienced designer. Passband and stopband deviations may be expressed as ordinary numbers or in decibels when they specify the passband ripple and minimum stopband attenuation respectively. Thus the minimum stopband attenuation, A_s, and the peak passband ripple, A_P, in decibels are given as (for FIR filters)

$$A_s \text{ (stopband attenuation)} = -20\log_{10} \delta_s \qquad (5.5a)$$

$$A_p \text{ (passband ripple)} = 20 \log_{10}(1+\delta_p) \qquad (5.5b)$$

The phase response of digital filters is often not as meticulously specified as the amplitude response. In many cases it is sufficient to indicate that phase distortion is of concern or that linear phase response is desirable. However, in some applications where filters are used to equalize or compensate a system's phase response, for example, or as phase shifters, then the desired phase response will need to be specified.

Example 5.2

An FIR bandpass filter is to be designed to meet the following frequency response specifications:

passband	0.18 – 0.33 (normalized)
transition width	0.04 (normalized)
stopband deviation	0.001
passband deviation	0.05

1- Sketch the tolerance scheme for the filter.

2- Express the filter band edge frequencies in the standard unit of kilohertz, assuming a sampling frequency of 10kHz, and the stopband and passband deviations in decibels.

Solution

1- The tolerance scheme for the filter is given in Fig. 5.5.

2- The band edge frequencies, at a sampling frequency of 10kHz, and the stopband and passband deviations are given below :

passband	1.8 – 3.3 kHz
stopbands	0 – 1.4 kHz and 3.7 – 5 kHz
stopband attenuation	$-20\log_{10}(0.001) = 60$ dB
passband ripple	$20\log_{10}(1 + 0.05) = 0.42$ dB

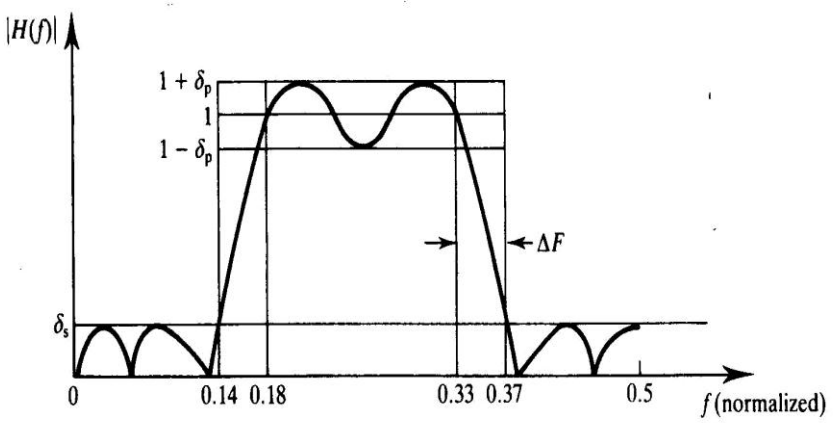


Fig. 5.5 Tolerance scheme for the bandpass filter of Example(2)

5.4.2 Calculation of suitable filter coefficients

In this step, we select one of a number of approximation methods and calculate the values of the coefficients, h(k), for FIR, or a_k and b_k, for IIR, such that the filter characteristics given in Section 5.4.1 are satisfied. The method used to calculate the filter coefficients depends on whether the filter is IIR of FIR type.

Calculations of IIR filter coefficients are traditionally based on the transformation of known analogue filter characteristics into equivalent digital filters. The two basic methods used are the impulse invariant and the bilinear transformation methods. With the impulse invariant method, after digitizing the analogue filter, the impulse response of the original analogue filter is preserved, but not its magnitude–frequency response. Because of inherent aliasing, the method is inappropriate for highpass or bandstop filters. The bilinear method, on the other hand, yields very efficient filters and is well suited to the calculation of coefficients of frequency selective filters. It allows the design of digital filters with known classical characteristics such as Butter-worth, Chebyshev and elliptic. Digital filters resulting from the bilinear transform method will, in general, preserve the magnitude response characteristics of the analogue filter but not the time domain properties. Efficient computer programs now exist for calculating filter coefficients, using the bilinear method, by merely specifying filter parameters of interest. The impulse invariant method is good for simulating analogue systems, but the bilinear method is best for frequency selective IIR filters.

The pole-zero placement method offers an alternative approach to calculating the coefficients of IIR filters. It is an easy way of calculating the coefficients of very simple filters. However, for filters with good amplitude response it is not recommended as it relies on ‘trial and error’ shuffling of the pole and zero positions.

As with IIR filters there are several methods of calculating the coefficients of FIR filters. The three methods discussed in this book in detail are the window, frequency

sampling, and the optimal (Parks – McClellan algorithm). The window method offers a very simple and flexible way of computing FIR filter coefficients, but it does not allow the designer adequate control over the filter parameters. The main attraction of the frequency sampling method is that it allows a recursive realization of FIR filters which can be computationally very efficient. However, it lacks flexibility of an efficient and easy-to-use program, the optimal method is now widely used in industry and, for most applications, will yield the desired FIR filters. Thus, for FIR filters, the optimal method should be the method of first choice unless the particular application dictates other-wise or a CAD facility is unavailable.

In summary, there are several methods of calculating filter coefficients of which the following are the most widely used:

- Impulse invariant (IIR);
- Bilinear transformation (IIR);
- Pole-zero placement (IIR);

- Window (FIR);
- Frequency sampling (FIR);
- Optimal (FIR);

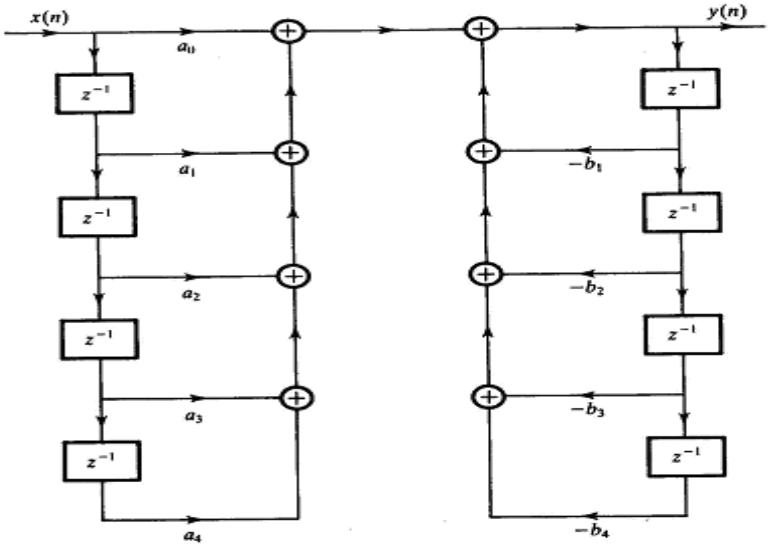
We choose the method that best suits our particular application. Our choice will be influenced by several factors, the most important of which are the critical requirements in the specifications. In general, the crucial choice is really between FIR and IIR. In most cases, if the FIR properties are vital then a good candidate is the optimal method, whereas, if IIR properties are desirable, then the bilinear method will in most cases suffice.

5.4.3 Representation of a filter by a suitable structure (realization)

Realization involves converting a given transfer function, H(z), into a suitable filter structure. Block or flow diagrams are often used to depict filter structures and they show the computational procedure for implementing the digital filter. The structure used depends on whether the filter is an IIR or FIR filter.

For IIR filters, three structures commonly used are the direct form, cascade and parallel forms. The direct form is simply a straightforward representation of the IIR transfer function. In the cascade form, the transfer function of the IIR filter, Equation 5.4b, is factored and expressed as the product of second-order sections. In the parallel form, H(z) is expanded, using partial fractions, as the sum of second-order sections. As an illustration and for comparison, Figure 5.6 shows the block diagrams for a fourth-order (that is, N=4) IIR filter represented in the direct, cascade and parallel structures. The corresponding set of transfer functions and the difference equations describing the filter structures are also given in the figure.

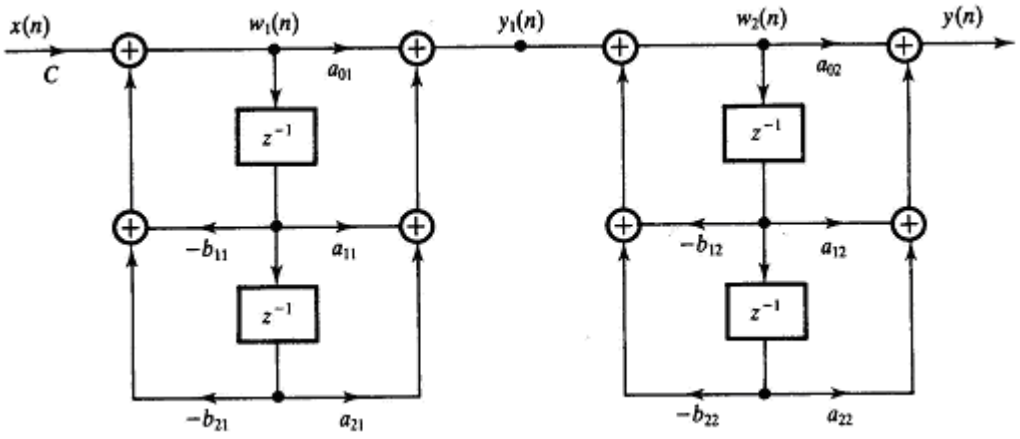
The parallel and cascade structures are the most widely used for IIR because they lead to simpler filtering algorithms and are far less sensitive to the effects of implementing the filter using a finite number of bits than the direct structure. The direct structure suffers sever coefficient sensitivity problems, especially for high order filters, and should be avoided in these cases.



$$y(n)=\sum_{k=0}^{4}a_k x(n-k)-\sum_{k=1}^{k=4}b_k y(n-k) \qquad \& \qquad H(z)=\frac{\sum_{K=0}^{4}a_k z^{-k}}{1+\sum_{k=0}^{4}bkz^{-k}}$$

(a)

Fig. 5.6 (a) Direct realization of a fourth-order IIR filter.

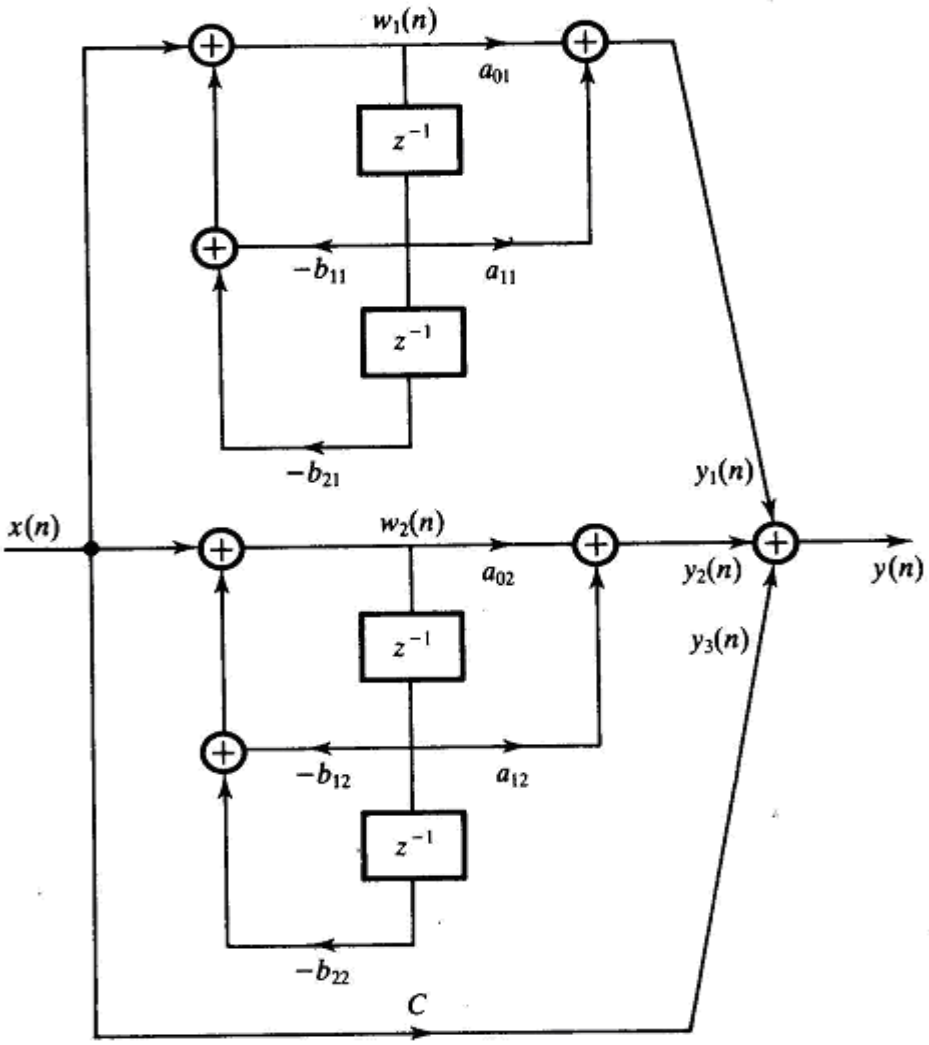


$$H(z)=C\prod_{k=1}^{2}\frac{1+a_{1k}z^{-1}+a_{2k}z^{-2}}{1+b_{1k}z^{-1}+b_{2k}z^{-2}}$$

$$w_1(n)=Cx(n)-b_{11}w_1(n)-b_{21}w_1(n-2)$$
$$y_1(n)=a_{01}w_1(n)+a_{11}w_1(n-1)+a_{21}w_1(n-2)$$
$$w_2(n)=y_1(n)-b_{12}w_2(n-1)-b_{22}w_2(n-2)$$
$$y(n)=a_{02}w_2(n)+a_{12}w_2(n-1)+a_{22}w_2(n-2)$$

(b)

Fig. 5.6 (b) Cascade realization of a fourth-order IIR filter.



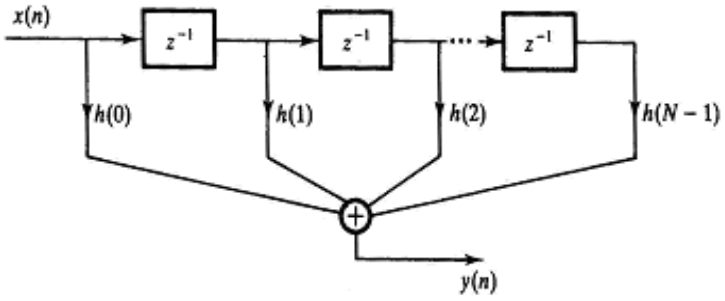
$$H(z) = C + \sum_{k=1}^{2} \frac{a_{0k} + a_{1k} z^{-1}}{1 + b_{1k} z^{-1} + b_{2k} z^{-2}}$$

$$w_1(n) = x(n) - b_{11} w_1(n-1) - b_{21} w_1(n-2)$$
$$w_2(n) = x(n) - b_{12} w_2(n-1) - b_{22} w_2(n-2)$$
$$y_1(n) = a_{01} w_1(n) + a_{11} w_1(n-1)$$
$$y_2(n) = a_{02} w_2(n) + a_{12} w_2(n-2)$$
$$y_3(n) = Cx(n)$$
$$y(n) = y_1(n) + y_2(n) + y_3(n)$$

(c)

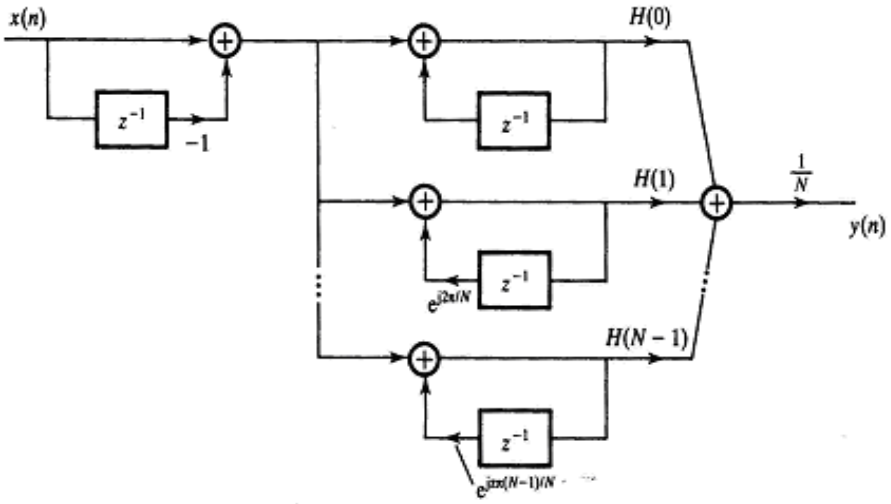
Fig. 5.6 (c) Parallel realization of a fourth-order IIR filter.

The most widely used structure for FIR is the direct form, Fig. 5.7(a), because it is particularly simple to implement. In this form, the FIR is sometimes called a tapped delay line (because it resembles a tapped delay line) or transversal filter. Two other FIR structures that are also used are the frequency sampling structure and the fast convolution technique, Figure 5.7(b) and 5.7(c). compared with the transversal structure, the frequency sampling structure can be computationally more efficient as it leads to fewer coefficients, but it may not be as simple to implement and would require more storage. The fast convolution uses the computational advantage of the fast Fourier transform (FFT) and is particularly attractive in situations where the power spectrum of the signal is also required.



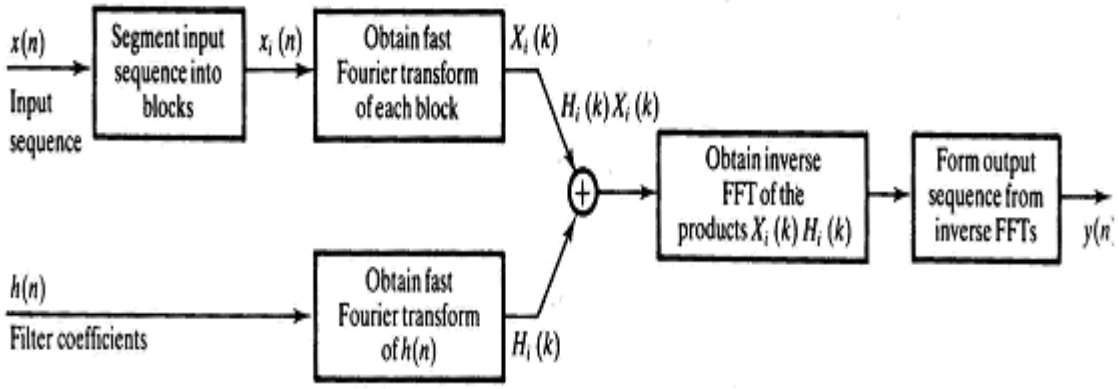
(a)

Fig. 5.7(a) transversal filter (or direct) realization for FIR filters.



(b)

Fig. 5.7(b) frequency sampling realization for FIR filters.

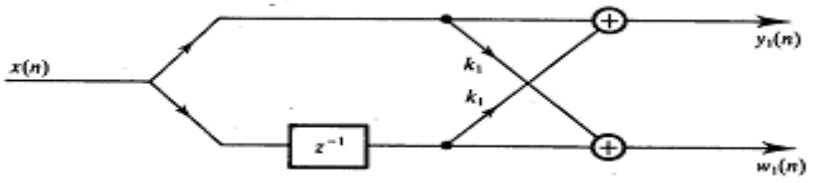


(c)

Fig. 5.7(c) fast convolution realization for FIR filters.

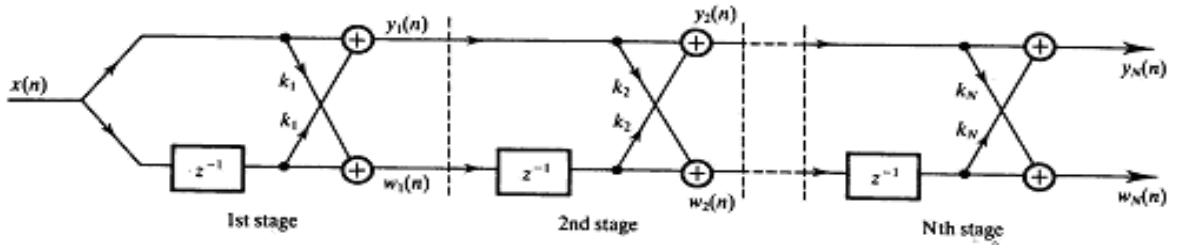
Fig. 5.7 Realization structures for FIR filters: (a) transversal filter (or direct); (b) frequency sampling structure; (c) fast convolution.

There are many other practical structures for digital filters, but most of these are popular only in specific application areas. An example is the lattice structure, which finds use in speech processing and linear prediction applications. The lattice structure may be used to represent FIR as well as IIR filters. The basic lattice structure is characterized by a single input and a pair of outputs, as shown in Figure 5.8(a). A lattice structure, derived from the basic structure in Fig. 5.8(a), for an N-point FIR filter is shown in Fig. 5.8(b), and that for a second-order all-pole IIR filter (that is, with only denominator coefficients) is given in Fig. 5.8(c). Further details of the lattice structure are given in Example (5) section (5.5).



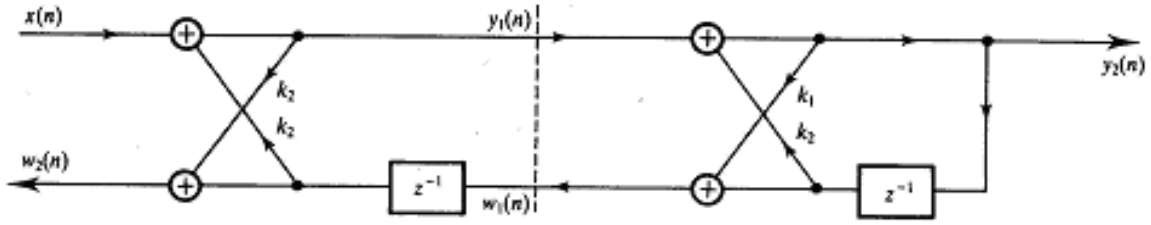
(a)

Fig. 5.8 (a) The basic lattice structure.



(b)

Fig. 5.8 (b) An N-stage FIR lattice structure.



(c)

Fig. 5.8 (c) A 2-stage all-pole IIR lattice structure.

Fig. 5.8 (a) The basic lattice structure. (b) An N-stage FIR lattice structure. (c) A 2-stage all-pole IIR lattice structure.

In summary, the following are the commonly used realization structures for FIR and IIR filters:

FIR	IIR
transversal (direct)	direct form
frequency sampling	cascade
fast convolution	parallel
lattice	

For a given filter the choice between structures depends largely on

(i) whether it is FIR or IIR,

(ii) the ease of implementation and

(iii) how sensitive the structure is to the effects of finite word length.

5.4.4 Analysis of finite word length effects

The approximation and realization steps assume infinite or very high precision. However, in actual implementation it is often necessary to represent the filter coefficients using limited number of bits, typically 8 to 16 bits, and the arithmetic operations indicated in the difference equation are performed using finite precision arithmetic.

The effects of using a finite number of bits are to degrade the performance of the filter and in some cases to make it unusable. The designer must analyze these effects and choose suitable word lengths (that is, numbers of bits), for the filter coefficients, filter variables, that is the input and output samples, and for the arithmetic operations within the filter.

The main sources of performance degradation in digital filters are as follows .

1- Input/output signal quantization.

In particular, the ADC noise due to quantizing of the input signal samples is significant.

2- Coefficient quantization.

This leads to deviations in the frequency response of both FIR and IIR filters, and possibly to instability in IIR filters.

3- Arithmetic round off errors.

The use of finite precision arithmetic to perform filtering operations yields results that require additional bits to represent. When these are quantized to the permissible word length, often by rounding, round off noise is the result. This can cause undesirable effects such as instability in an IIR filter.

4- Overflow.

This occurs when the result of an addition exceeds permissible word length. It leads to wrong output samples and to possible instability in IIR filters.

The extent of filter degradation depends on

(i) the word length and type of arithmetic used to perform the filtering operation,

(ii) the method used to quantize filter coefficients and variables to the chosen word length, and

(iii) the filter structure. From a knowledge of these factors, the designer can assess the effects of finite word length on the filter performance and take remedial action if necessary.

Depending on how the filter is to be implemented, some of the effects may be insignificant. For example, when implemented as a high level language program on most large computers, coefficient quantization and round off errors are not important. For real-time processing, finite word lengths (typically 8 bits, 12 bits, and 16 bits) are used to represent the input and output signals, filter coefficients and the results of arithmetic operations. In these cases, it is nearly always necessary to analyze the effects of quantization on the filter performance.

5.4.5 Implementation of a filter

Having calculated the filter coefficients, chosen a suitable structure, and verified that the filter degradation, after quantizing the coefficients and filter variables to the selected word lengths, is acceptable, the difference equation must be implemented as a soft ware routine or in hardware. Whatever the method of implementation, the output of the filter must be computed, for each sample, in accordance with the difference equation (assuming a time domain implementation).

As the examination of any difference equation will show (Eqs. 5.2 and 5.3), the computation of y(n) (the filter output) involves only multiplications, additions/subtractions, and delays. Thus to implement a filter, we need the following basic building blocks:

- memory (for example ROM) for storing filter coefficients;
- memory (such as RAM) for storing the present and past inputs and outputs, that is {x(n), x(n – 1), ….} and {y(n), y(n – 1),…..};
- hardware or software multiplier(s);
- adder or arithmetic logic unit.

The designer provides these basic blocks and also ensures that they are suitably configured for the application. The manner in which the components are configured depends to a large extent on whether batch (that is, non-real-time) or real-time processing is required. In batch processing, the entire data is already available in some memory device. Such is the case in applications where, for example, experimental data is acquired from elsewhere for later analysis. In this case, the filter is often implemented in a high level language and runs in a general-purpose computer, such as a personal computer or a mainframe computer, where all the basic blocks are already configured. Thus, batch processing may be described as a purely software implementation (although the designer may wish to incorporate additional hardware to increase the speed of processing).

In real-time processing, the filter is required either

(i) to operate on the present input sample, x(n), to produce the current output sample, y(n), before the next input sample arrives, that is within the inter-sample period, or

(ii) to operate on an input block of data, using an FFT algorithm for example, to produce an output block of data within a period proportional to the block length. Real-time filtering may require fast and dedicated hardware if the sample rate is very high or if the filter is of a high order. For most audio frequency work, DSP

processors such as the DSP56000 (by Motorola) and the TMS320C25 (by Texas Instruments) will be adequate and offer considerable flexibility. These processors have all the basic blocks on board, including built-in hardware multiplier(s).

In some applications, standard 8-bit or 16-bit microprocessors such as the Motorola 6800 or 68000 families offer attractive alternative implementations. In addition to the signal processing hardware, the designer must also provide suitable input-output (for example, analogue-digital conversion) interfaces to the digital hardware, depending on the type of data source and sink.

5.5 ILLUSTRATIVE EXAMPLES

Example 5.3

Discuss the five main steps involved in the design of digital filters, using the following design problem to illustrate your answer.

A digital filter is required for real-time physiological noise reduction. The filter should meet the following amplitude response specifications:

passband	0 – 10Hz
stopband	20 – 64Hz
sampling frequency	128Hz
maximum passband deviation	< 0.036 dB
stopband attenuation	> 30dB

Other important requirements are that

1- minimal distortion of the harmonic relationships between the components of the in-band signals is highly desirable,

2- the time available for filtering is limited, the filter being part of a larger process, and

3- the filter will be implemented using the Texas Instruments TMS32010 DSP processor with the analogue input digitized to 12 bits.

Solution

This filter was designed and used in a certain biomedical signal processing project. We shall give here only an outline discussion of the design, postponing detailed discussion to Chapter 6 where FIR filter design methods are fully covered.

1- Requirement specification. As discussed previously, the designer must give the exact role and performance requirements for the filter together with any important constraints. These have already been given for the example.

2- Calculation of suitable coefficients. The requirements of minimal distortion and limited processing time are best achieved with a linear phase FIR filter, with coefficients obtained using the optimal method.

3- Selection of filter structure. The transversal structure will lead to the most efficient implementation using the TMS32010 processor.

4- Analysis of finite word length effects. Since the TMS32010 processor will be used, fixed point arithmetic should be used with each coefficient represented by 16 bits (after rounding) for efficiency. FIR filter degradation may result from input signal quantization, coefficient quantization, round off and overflow errors. A check should be made to ensure that the word lengths are sufficiently long. Analysis of finite word length effects for this case showed that the input quantization noise and deviation in the frequency response due to coefficient quantization are both insignificant. The use of the TMS32010's 32-bit accumulator to sum the coefficient-data products, rounding only the final sum, would reduce round off errors to negligible levels. To avoid overflow, each coefficient should be divided by $\sum_{k=0}^{N-1}|h(k)|$ before quantizing to 16 bits.

5- Implementation. Design and configure the TMS32010-based hardware (if it does not already exist) with the necessary input/output interfaces. Then write a TMS32010 program to handle the I/O protocols and calculate filter output,

$$y(n)= \sum_{k=0}^{N-1} h(k)\, x(n-k), \text{ for each new input, } x(n).$$

Example 5.4

An analogue filter is to be converted into an equivalent digital filter that will operate at a sampling frequency of 256 Hz. The analogue filter has the transfer function:

$$H(s) = \frac{1}{s^3 + 2s^2 + 2s + 1}$$

1- Obtain suitable coefficients for the digital filter.

2- Assuming that the digital filter is to be realized using the cascade structure, draw a suitable realization block diagram and develop the difference equations.

3- Repeat (2) for the parallel structure.

Solution

1- To preserve the amplitude response of the analogue function, the bilinear method was used to obtain the filter coefficients. The application of the bilinear transformation approach to the analogue transfer function yields the following transfer functions:

$$H(z) = \frac{0.1432(1 + 3z^{-1} + 3z^{-2} + z^{-3})}{1 - 0.1801z^{-1} + 0.3419z^{-2} - 0.0165z^{-3}}$$

2- For the cascade realization, H(z) is factorized using partial fractions :

$$H(z) = 0.1432 \frac{1 + 2z^{-1} + z^{-2}}{1 - 0.1307z^{-1} + 0.3355z^{-2}} \frac{1 + z^{-1}}{1 - 0.0490z^{-1}}$$

The block diagram representation and the corresponding set of difference equations are given in Figure 5.9 and as follows:

$w_1(n) = 0.1432x(n) + 0.1307w_1(n-1) - 0.3355w_1(n-2)$
$y_1(n) = w_1(n) + 2w_1(n-1) + w_1(n-2)$
$w_2(n) = y_1(n) + 0.049w_2(n-1)$
$y_2(n) = w_2(n) + w_2(n-1)$

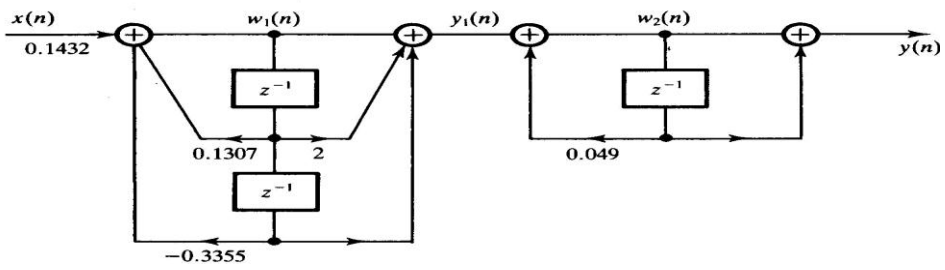


Fig. 5.9

3- For the parallel realization, H(z) is expressed using partial fractions as:

$$H(z) = \frac{1.2916 - 0.08407z^{-1}}{1 - 0.131z^{-1} + 0.3355z^{-2}} + \frac{7.5268}{1 - 0.049z^{-1}} - 8.675$$

The parallel realization diagram and its corresponding set of difference equation are given in Fig. 5.10 and as follows:

$w_1(n) = x(n) + 0.131w_1(n-1) - 0.3355w_1(n-2)$
$y_1(n) = 1.2916w_1(n) - 0.08407w_1(n-1)$
$w_2(n) = x(n) + 0.049w_2(n-1)$
$y_2(n) = 7.5268w_2(n)$
$y_3(n) = -8.6753x(n)$
$y(n) = y_1(n) + y_2(n) + y_3(n)$

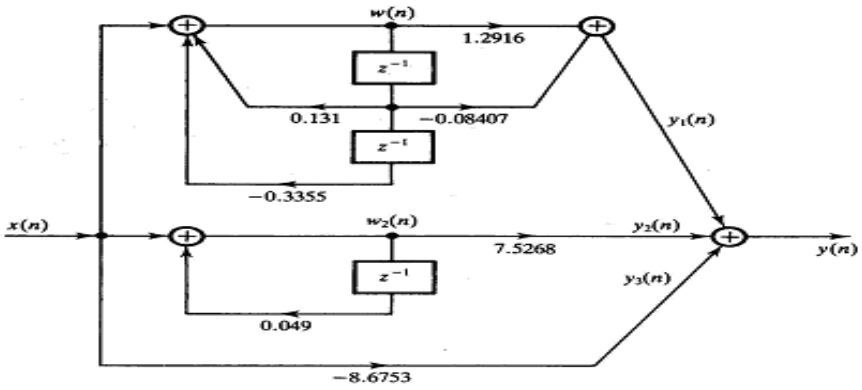


Fig. 5.10

Example 5.5

The transfer function for an FIR filter is given by

$$H(z) = 1 - 1.3435z^{-1} + 0.9025z^{-2}$$

Draw the realization block diagram for each of the following cases:

1- Transversal structure;

2- A two-stage lattice structure.

Calculate the values of the coefficients for the lattice structure.

Solution

1- From the transfer function, the diagram for the transversal structure is given in Figure 5.11. The input and output of the transversal structure are given by

$$y(n) = x(n) + h(1)x(n-1) + h(2)x(n-2) \qquad (5.6)$$

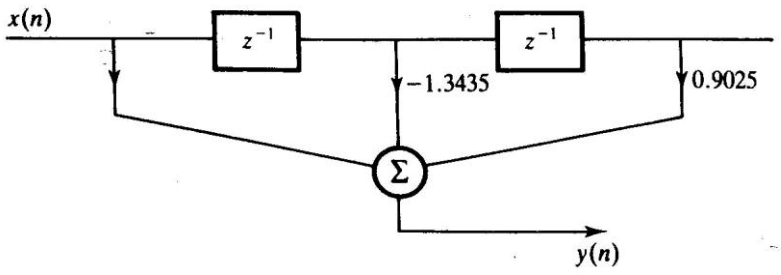


Fig. 5.11

2- A two-stage lattice structure for the filter is given in Figure 5.12. The outputs of the structure are related to the input as:

$$y_2(n) = y_1(n) + K_2w_1(n-1)$$
$$= x(n) + K_1(1 + K_2)x(n-1) + K_2x(n-2) \qquad (5.7a)$$

$$w_2(n) = K_2x(n) + K_1(1 + K_2)x(n-1) + x(n-2) \qquad (5.7b)$$

Comparing Equation 5.6 and 5.7a, and equating coefficients, we have

$$K_1 = \frac{h(1)}{1+h(2)} \qquad \text{and} \qquad K_2 = h(2)$$

from which $K_2 = 0.9025$ and

$K_1 = - 1.3435/ (1 + 0.9025) = - 0.7062.$

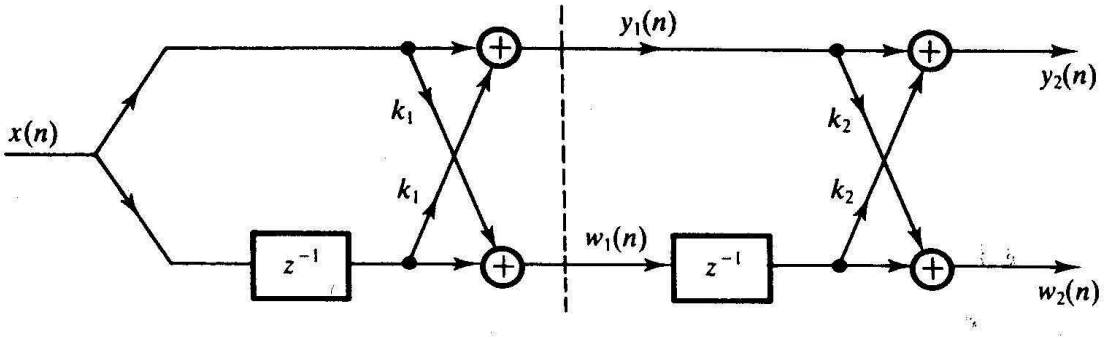


Fig. 5.12

Notice that the coefficients of $y_2(n)$ and $w_2(n)$ (Eqs. 5.7a and 5.7b) are identical except that one is written in reverse order. This is a characteristic feature of the FIR lattice structure. Further details of the lattice structure including recursive techniques for converting the coefficients of an FIR or IIR filter to those of equivalent lattice structures are available in several books (see, for example, Proakis and Manolakis, 1992).

SHEET # 5

5.1 The following transfer functions represent three different filters meeting identical amplitude frequency response specifications:

$$H_1(Z) = \frac{a_o + a_1 Z^{-1} + a_2 Z^{-2}}{1 + b_1 Z^{-1} + b_2 Z^{-2}} \qquad H_2(Z) = \frac{a_3 + a_4 Z^{-1} + a_5 Z^{-2}}{1 + b_3 Z^{-1} + b_4 Z^{-2}}$$, and

$$H_3(Z) = \sum_{k=0}^{22} h_k Z^{-k}$$

Where:

$a_o = 3.136362 \times 10^{-1}$ $a_1 = -5.456657 \times 10^{-2}$ $a_2 = 4.635728 \times 10^{-1}$
$a_3 = -5.456657 \times 10^{-2}$ $a_4 = 3.136362 \times 10^{-1}$
$b_1 = -8.118702 \times 10^{-1}$ $b_2 = 3.339288 \times 10^{-1}$ $b_3 = -2.794577 \times 10^{-1}$
$b_4 = 3.030631 \times 10^{-1}$
and

$h_o = 0.39826480 \times 10^{-1} = h_{22}$
$h_1 = -0.16874380 \times 10^{-1} = h_{21}$
$h_2 = 0.34781130 \times 10^{-1} = h_{20}$
$h_3 = 0.12052890 \times 10^{-1} = h_{19}$
$h_4 = -0.44731860 \times 10^{-1} = h_{18}$
$h_5 = 0.27894610 \times 10^{-1} = h_{17}$
$h_6 = -0.87573360 \times 10^{-1} = h_{16}$
$h_7 = -0.90972060 \times 10^{-1} = h_{15}$
$h_8 = -0.15667550 \times 10^{-1} = h_{14}$
$h_9 = -0.28499560 \times 10^{0} = h_{13}$
$h_{10} = 0.74035030 \times 10^{-1} = h_{12}$
$h_{11} = 0.62349560 \times 10^{-10}$

(a) $H(Z) = H_1(Z) \times H_2(Z)$ (b) $H(Z) = H_1(Z) + H_2(Z)$ (c) $H(Z) = H_3(Z)$
(d) $H(Z) = [\, H_1(Z) \times H_2(Z) \,] + H_3(Z)$ (e) $H(Z) = H_1(Z) + H_2(Z) + H_3(Z)$

For each of the above filters find the following:

1- State whether it is an FIR or IIR filter or neither.

2- Represent the filter operation in a block diagram form an write down the difference equation.

3- Determine and comment on the computational and storage requirements.

5.2 The coefficients of Lattice filter are $K_1 = -0.266$ and $K_2 = 0.69$. Draw the realization diagram for the Lattice filter. Compute the impulse response coefficients for the filter and draw the equivalent transversal function.

5.3 A second-order IIR digital filter is characterized by the following transfer function:

$$H(Z) = \frac{1}{1 - 0.9Z^{-1} + 0.81Z^{-2}}$$

Draw the realization diagram for each of the following structure:

(a) Direct.

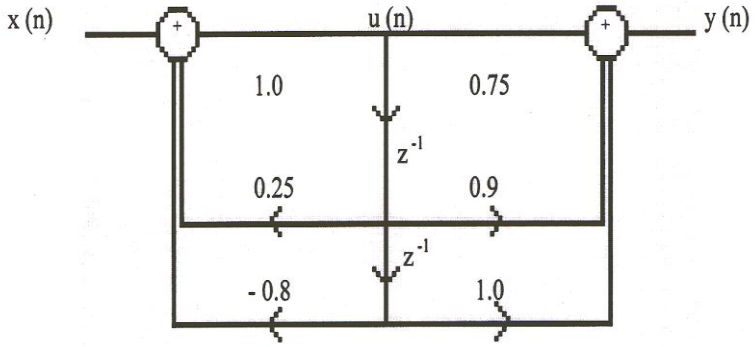
(b) Lattice.

Then, find the coefficients of the Lattice structure from the given transfer function.

5.4 The coefficients of transversal filter are $h_0 = h_1 = -0.266$ and $h_2 = 0.69$. Draw the realization structure for the transversal filter. Find the impulse response and the difference equation of the filter.

Assignment # 5

Give the realization structure of an IIR filter as shown in the following figure. Find the two-step difference equation relating the output, y (n), and input, x (n). Draw the equivalent Lattice filter's structure, and then compute its coefficients.



EXAMS

Modern Academy
for Engineering and Technology in Maadi
Electronic Engineering and Communications Technology Department

Academic Year	2015 / 2016
Semester	First
Exam Date	18/11/2015

QUESTIONS FOR THE Mid Term EXAM

Subject:	(E501) Digital Signal Processing	**Spec.:**	5th Elec. Eng. & Comm. Tech. Dep.
Examiner:	Dr. Samir Kamal	**Time:**	75 Mints

Number of Pages: one | Number of Questions: **2** | Attempt all questions

Model-A

Question 1 [20 Points]

(a) In A/D conversion, in detail explain the coding process.

(b) Find the relationship between the frequencies, f, F & F_S, where these frequencies respectively are: frequency of analog signal, frequency of discrete-time signal and sampling frequency.

(c) For the signal $x(t) = \cos(300\pi t)$ to be sampled with sampling frequency F_S = 300Hz.

(i) Is F_S is suitable for sampling the given signal, $x(t)$, if it is suitable, find an expression for its corresponding sequence, x(n). Then calculate its fundamental period N_P.

(ii) Find the samples values of the first cycle in x(n), then plot $x(t)$ and x(n) on the same figure.

Question 2 [20 Points]

(a) Show that the DFT, X(K), of a sequence x(n) is periodic.

(b) For the following sequence: x(n) = {1, 0, 1, 0}:

(i) Using the direct method, DFT, find the frequency components of this sequence and check your results by calculating x(n) using the IFFT algorithm.

(ii) Using your results, verify Parseval's theorem.

(iii) If we use FFT algorithm instead of the DFT algorithm, what is the computational saving?

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Examiner:	Dr. Samir Kamal	**Time:**	75 Mints

Number of Pages: one | Number of Questions: **2** | Attempt all questions

Model-B

Question 1 [20 Points]

(a) In A/D conversion, in detail explain the quantization process.

(b) Given two signals: $x_1(t) = 5\cos(200\pi t)$ and $x_2(t) = 2\sin(500\pi t)$ to be sampled with sampling frequencies F_{S1} = 400Hz and F_{S2} = 1KHz respectively.

(i) Are the sampling frequencies given above are suitable for sampling the given signals, $x_1(t)$ and $x_2(t)$

(ii) If they are suitable, use them to find expressions for $x_1(n)$ and $x_2(n)$.

Question 2 [20 Points]

(a) Show that the DFT, X(K), of a sequence x(n) is periodic.

(b) For the following sequence: x(n) = {0, 1, 0, 1}:

(i) Using the direct method, DFT, find the frequency components of this sequence and check your results by calculating this frequency components using FFT algorithm.

(ii) Using your results, verify Parseval's theorem.

(iii) Plot the amplitude and the phase spectrums.

G O O D L U C K

Modern Academy
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Electronic Engineering and
Communications Technology Department

Academic Year	2015-2016
Semester	First
Exam Date	17/1/ 2016

QUESTIONS FOR THE FINAL WRITTEN EXAM

Subject: (E501) Digital Signal Processing **Spec.:** 5th Elec. Eng. & Comm. Tech. Dep.

Examiners: Dr. Samir Kamal and Prof. Mostafa Afifi **Time:** 3 hours

Number of Pages: 2 Number of Questions: 5 Attempt all questions

Question 1 **(20 points)**

a - In detail, what are the main characteristics of the continuous time sinusoidal signals?

b- Check the periodicity of the following sequences, showing their fundamental periods.

(i) y (t) = $7\cos(400\pi t)$

(ii) x (n) = $\sin(2n + \pi/4)$

(c) For the signal $x(t) = 2\cos(3000\pi t)$ to be sampled with sampling frequency F_S = 3 KHz.

(i) Is F_S suitable for sampling the given signal, $x(t)$,

(ii) If it is suitable, use it to find an expression for the corresponding sequence, x(n),

(iii) Calculate the fundamental period, N_P, of the obtained x(n),

(iv) Find the values of x(n) in the first cycle and

(v) Plot (showing the steps) the given $x(t)$ and the obtained x(n) on the same figure.

Question 2 **(25 points)**

a- In A/D conversion, in detail explain the sampling process.

b - For the following sequence: x(n) = {1, 1, 0, 1}:

(i) Using the FFT algorithm, find the frequency components of this sequence and check your results by calculating x(n) using the IDFT method.

(ii) Using your results, verify Parseval's theorem.

(iii) Plot the amplitude and the phase spectrums.

c- The transfer function for an FIR filter is given by: $H(z) = 1 - 1.4z^{-1} + 0.9z^{-2}$. Design and realize the block diagram for this filter in the following cases:

(i) Transversal structure.

(ii) A two-stage lattice structure.

Question 3 **(15 points)**

a- In correlation operation define the following items: Correlation operation, correlation coefficient and the end effect problem.

b - Given two sets of samples $x_1(n)$ = {1, 2.5, 1} and $x_2(n)$ = {0, 2, 1.5, -1}.

(i) Find the cross- correlation between $x_1(n)$ and $x_2(n)$.

(ii) Plot the correlation function obtained in part (i) against the lag number, j.

(iii) Find the autocorrelation function of the sequence $x_1(n)$.

c- Find the output of an electrical system, y(t), with impulse response h(t) shown in Fig. 1a when the input signal, x(t) is the signal shown in Fig. 1b. Plot the obtained y(t) v.s t,

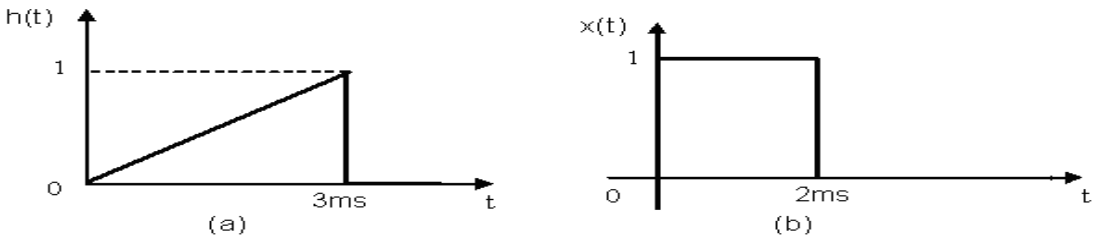


Fig. 1

Please Turn Over the Page

Question 4 **(20 points)**

a- For LTIS define: Causality, linearity and the difference equation.

b- By applying a unit impulse to the discrete-time system characterized by the following z-transfer function, find its impulse response, h(n), (calculate the first 4-values, assuming an initial condition of y(–1) = 0).

$$H(Z) = \frac{1 + 2z^{-1} + z^{-2}}{1 - z^{-1} - 2z^{-2}}$$

c- Using the residue method, find the impulse response, h(n), of the system characterized by the following transfer function, $H(z)$:

$$H(z) = \frac{z^2(z+1)}{(z-0.5)(z-1)^2}$$

For this transfer function:

(i) Plot its pole-zero diagram,
(ii) Find its region of convergence,
(iii) Discuss the stability of this system,
(iv) Find the difference equation of this system,
(v) Draw its filtering operation in a block diagram form and
(vi) Determine the computational and storage requirements.

Question 5 **(10 points)**

Check (✔) for the right statements and (X) for the wrong statements and correct the mistakes.

(1) In quantization process, when the quantization step (Δ) increases, the accuracy of the Quantizer increases. ( )
(2) The smaller the quantization step the longer will be the required codeword length. ( )
(3) The spectrum of a continuous periodic signal could be obtained using Fourier transform whereas the spectrum of discrete time signal could be obtained using the FFT algorithm.( )
(4) FFT is calculated for a signal that has any number of samples. ( )
(5) Correlation coefficients never exceed one. ( )
(6) We can calculate the normalized energy of any signal performing convolution at lag=1( )
(7) A discrete time system is said to be time invariant if its output is independent of the time the input is applied. ( )
(8) The impulse response of a discrete- time system may be defined as the z- transform of the system's transfer function H (z). ( )
(9) IIR requires more coefficients for sharp cut off filters than FIR. ( )
(10) For a given amplitude response specification more processing time and storage will be required for IIR filter implementation. ( )

G O O D L U C K

REFERENCES

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